

Cancer

CHAPTER 20

About one in five of us will die of cancer, but that is not why we devote a chapter to this disease. Cancer cells break the most basic rules of cell behavior by which multicellular organisms are built and maintained, and they exploit every kind of opportunity to do so. These transgressions, while often tragic, help to reveal what the normal rules are and how they are enforced. As a result, cancer research helps to illuminate the fundamentals of cell biology—especially cell signaling (Chapter 15), the cell cycle and cell growth (Chapter 17), apoptosis (Chapter 18), and the control of tissue architecture (Chapters 19 and 22). Of course, with a deeper understanding of these normal processes, we also gain a deeper understanding of the disease and better tools to treat it.

In this chapter, we first consider what cancer is and describe the natural history of the disease from a cellular standpoint. We then discuss the molecular changes that make a cell cancerous. And we end the chapter by considering how our enhanced understanding of the molecular basis of cancer is leading to improved methods for its prevention and treatment.

CANCER AS A MICROEVOLUTIONARY PROCESS

The body of an animal operates as a society or ecosystem, whose individual members are cells that reproduce by cell division and organize themselves into collaborative assemblies called *tissues*. This ecosystem is very peculiar, however, because self-sacrifice—as opposed to survival of the fittest—is the rule. Ultimately, all of the somatic cell lineages in animals dedicate their existence to the support of the germ cells, which alone have a chance of continued survival (discussed in Chapter 21). Because the genome of the somatic cells is the same as that of the germ-cell lineage that gives rise to sperm or eggs, through their self-sacrifice the somatic cells help to propagate copies of their own genes.

Unlike free-living cells such as bacteria or yeast, which compete to survive, the cells of a multicellular organism must be committed to collaboration. To coordinate their behavior, the cells send, receive, and interpret an elaborate set of extracellular signals that serve as *social controls*, directing cells how to act (discussed in Chapter 15). As a result, each cell normally behaves in a socially responsible manner—resting, growing, dividing, differentiating, or dying—as needed for the good of the organism.

Molecular disturbances that upset this harmony mean trouble for a multicellular society. In a human body with more than 10^{13} cells, billions of cells experience mutations every day, potentially disrupting the social controls. Most dangerously, a mutation may give one cell a selective advantage, allowing it to grow and divide slightly more vigorously and survive more readily than its neighbors. In this way, a mutated cell can become a founder of a growing mutant clone. Over time, repeated rounds of mutation, competition, and natural selection operating within the population of somatic cells can cause matters to go from bad to worse, jeopardizing the future of the multicellular organism. These are the basic ingredients of cancer: it is a disease in which an individual mutant clone of cells begins by prospering at the expense of its neighbors. In the end—as the

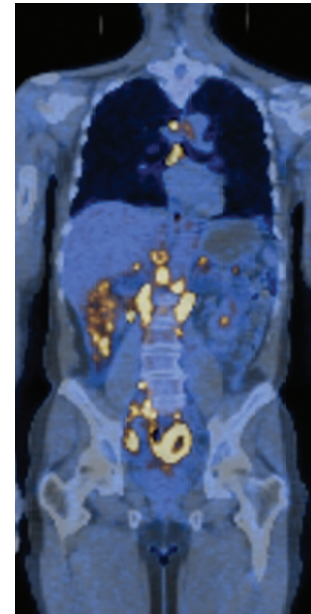
IN THIS CHAPTER

Cancer as a Microevolutionary Process

Cancer-critical Genes: How They Are Found and What They Do

Cancer Prevention and Treatment: Present and Future

Figure 20–1 Metastasis. Malignant tumors typically give rise to metastases, making the cancer hard to eradicate. Shown in this fusion image is a whole-body scan of a patient with metastatic non-Hodgkin's lymphoma (NHL). The background image of the body's tissues was obtained by CT (computed x-ray tomography) scanning. Overlaid on this image, a PET (positron emission tomography) scan reveals the tumor tissue (yellow), detected by its unusually high uptake of radioactively labeled fluorodeoxyglucose (FDG). High FDG uptake occurs in cells with unusually active glucose uptake and metabolism, a characteristic of cancer cells (see Figure 20–18). The yellow spots in the abdominal region reveal multiple metastases. (Courtesy of S.S. Gambhir.)



clone evolves over time and spreads—it can destroy the entire cellular society (**Movie 20.1**).

In this section, we discuss the development of cancer as a microevolutionary process that takes place within the course of a human life span in a subpopulation of cells in the body. As we shall see, the process depends on the same principles of mutation and natural selection that have driven the evolution of living organisms on Earth for billions of years.

Cancer Cells Bypass Normal Proliferation Controls and Colonize Other Tissues

Cancer cells are defined by two heritable properties: (1) they reproduce in defiance of the normal restraints on cell growth and division, and (2) they invade and colonize territories normally reserved for other cells. It is the combination of these properties that makes cancers particularly dangerous. An abnormal cell that goes through successive and inappropriate rounds of growth and division to proliferate out of control will give rise to a tumor, or *neoplasm*—literally, a new growth. As long as the *neoplastic* cells have not yet become invasive, however, the tumor is said to be **benign**. For most types of such neoplasms, removing or destroying the mass locally usually achieves a complete cure.

A tumor is considered a true cancer only if it is **malignant**; that is, when its cells have acquired the ability to invade surrounding tissue. Invasiveness is an essential characteristic of cancer cells. It allows them to break loose, enter blood or lymphatic vessels, and form secondary tumors called **metastases** at other sites in the body (**Figure 20–1**). In general, the more widely a cancer spreads, the harder it becomes to eradicate. It is metastases that usually kill the cancer patient by causing failure of a vital organ.

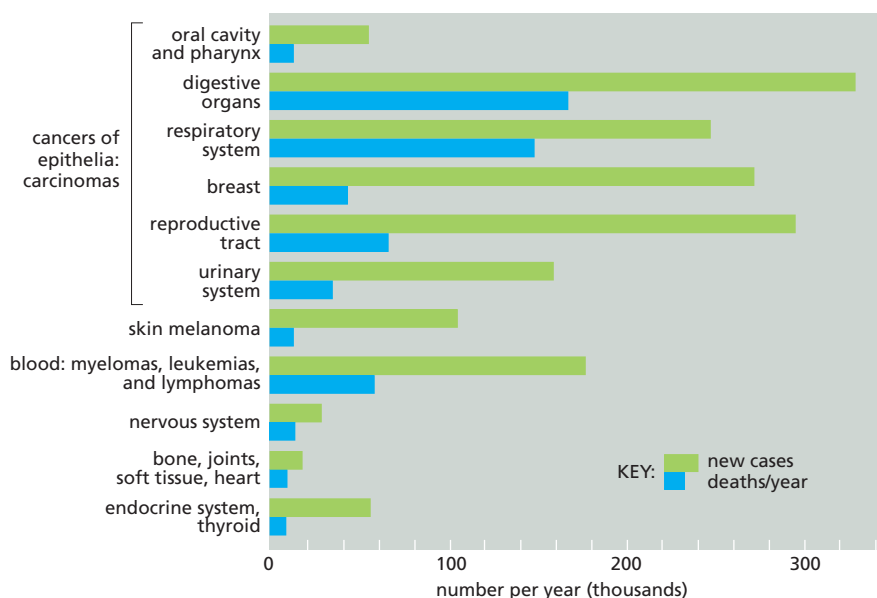


Figure 20–2 Cancer incidence and mortality in the United States. The estimated total number of new cases diagnosed in 2018 in the United States was 1,762,450, and total cancer deaths were 606,880. Note that deaths reflect cases diagnosed at many different stages and that well under half of the people who develop cancer die of it. The most common cancers are those of the digestive organs (including colon, pancreas, and liver), respiratory system (primarily lung and bronchus), reproductive tract (prostate and uterine), and breast. Skin cancers other than melanomas are not included in these figures, as almost all are cured easily and many are unrecorded. Each broad category has many subdivisions according to the specific cell type, the location in the body, and the microscopic appearance of the tumor. The data for the United Kingdom are similar. However, incidences are different in some other parts of the world, reflecting widespread exposures to different infectious agents and environmental toxins. (Data from American Cancer Society, *Cancer Facts & Figures 2019*.)

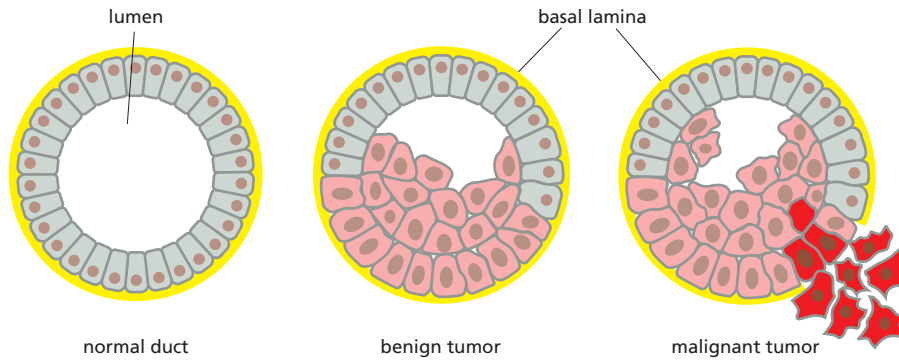


Figure 20-3 Benign versus malignant tumors. A benign glandular tumor (*pink cells*; an adenoma) remains inside the basal lamina (*yellow*) that marks the boundary of the normal structure (a duct, in this example). In contrast, a malignant glandular tumor (*red cells*; an adenocarcinoma) can develop from a benign tumor cell, and it destroys the integrity of the tissue, as shown. There are many different forms that such tumors may take.

Cancers are traditionally classified according to the tissue and cell type from which they arise. **Carcinoma** is the name given to cancers arising from epithelial cells, and they are by far the most common cancers in humans, accounting for about 85% of cancers. Most of the normal cell turnover by proliferation and death in adults occurs in epithelia, and the cell types that undergo the most cell division cycles have the greatest probability of accumulating the multiple mutations needed to become cancerous. In addition, epithelial tissues are the most likely to be exposed to the various forms of physical and chemical damage that favor the development of cancer. **Figure 20-2** shows the types of cancers diagnosed in the United States, together with their incidence and death rates. After carcinomas, the next most common cancer types include the various **myelomas**, **leukemias**, and **lymphomas**, derived from white blood cells and their precursors (hemopoietic cells). **Sarcomas** that arise from connective tissue or muscle cells and cancers derived from cells of the nervous system are much less common.

In parallel with the set of names for malignant tumors, there is a related set of names for benign tumors: an *adenoma*, for example, is a benign epithelial tumor with a glandular organization; the corresponding type of malignant tumor is an *adenocarcinoma* (**Figure 20-3**).

Most cancers have characteristics that reflect their origin. Thus, for example, the cells of a *basal-cell carcinoma*, derived from a keratinocyte stem cell in the skin, generally continue to synthesize cytokeratin intermediate filaments, whereas the cells of a *melanoma*, derived from a pigment cell in the skin, will often (but not always) continue to make pigment granules. Cancers originating from different cell types are, in general, very different diseases. Basal-cell carcinomas of the skin, for example, are only locally invasive and rarely metastasize, whereas melanomas can become much more malignant and often form metastases. Basal-cell carcinomas are readily cured by surgery or local irradiation, whereas malignant melanomas, once they have metastasized widely, are frequently fatal.

Later, we shall see that there is also a different, newer way to classify cancers, one that cuts across the traditional classification by site of origin: we can now classify many of them in terms of the mutations that make the particular tumor cells cancerous. The final section of the chapter will show how this information can be crucial to the design and choice of treatments.

Most Cancers Derive from a Single Abnormal Cell

Even after a cancer has metastasized, we can usually trace its origins to a single **primary tumor**, arising in a specific organ. The primary tumor is thought to arise by cell division from a single cell that initially experienced some heritable change. Subsequently, additional changes have accumulated in some of the descendants of this cell, allowing them to outgrow, out-divide, and often outlive their neighbors.

By the time it is first detected, a typical human cancer will have been developing for many years and will already contain a billion cancer cells or more (**Figure 20-4**). Tumors will usually also contain a variety of other cell types and

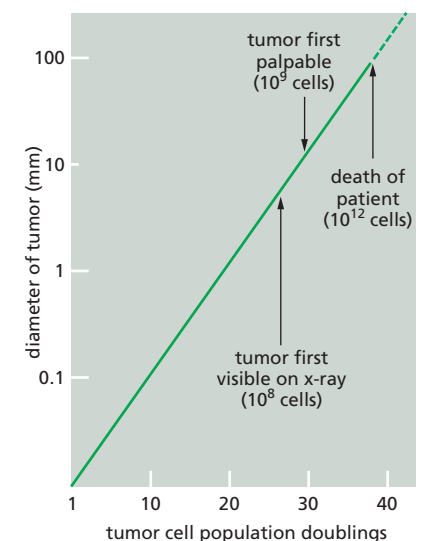


Figure 20-4 The growth of a typical human tumor, such as a tumor of the breast. The diameter of the tumor is plotted on a logarithmic scale. Years may elapse before the tumor becomes noticeable. The doubling time for a typical breast tumor, for example, is about 100 days. However, particularly aggressive tumors may grow much more rapidly.

associated extracellular matrix, termed the *tumor stroma*. For example, fibroblasts will be present in the supporting connective tissue associated with a carcinoma, in addition to immune cells and vascular endothelial cells. How can we be sure that the cancer cells are the clonal descendants of a single abnormal cell?

One way of proving clonal origin is through molecular analysis of the chromosomes in tumor cells. In almost all individuals with *chronic myelogenous leukemia* (CML), for example, we can distinguish the leukemic white blood cells from the individual's normal cells by a specific chromosomal abnormality: the so-called *Philadelphia chromosome*, created by a translocation between the long arms of chromosomes 9 and 22 (Figure 20-5). When the DNA at the site of translocation is cloned and sequenced, it is found that the site of breakage and rejoining of the translocated fragments is identical in all the leukemic cells in any given individual, but that this site differs slightly (by a few hundred or thousand base pairs) from one individual to another. This is the expected result if, and only if, the cancer in each individual arises from a unique accident occurring in a single cell. (We will see later how this particular translocation promotes the development of CML by creating a novel hybrid gene encoding a protein that promotes cell proliferation.)

Many other lines of evidence, from a variety of different cancers, point to the same conclusion: most cancers originate from a single aberrant cell.

Cancer Cells Contain Somatic Mutations

If a single abnormal cell is to give rise to a tumor, it must pass on its abnormality to its progeny: the aberration has to be heritable. Thus, the development of a clone of cancer cells depends on genetic changes. The tumor cells contain **somatic mutations**: they have one or more shared detectable abnormalities in their DNA sequence that distinguish them from the normal cells surrounding the tumor, as in the example of CML just described. (The mutations are called *somatic* because they occur in the soma, or body cells, not in the germ line.) Cancers are also driven by *epigenetic changes*—persistent, heritable changes in gene expression that result from modifications of chromatin structure without alteration of the cell's DNA sequence. But somatic mutations that alter a DNA sequence appear to be a fundamental and universal feature, and cancer is in this sense a genetic disease.

Factors that cause genetic changes tend to provoke the development of cancer. Thus, **carcinogenesis** (the generation of cancer) can be linked to *mutagenesis* (the production of a change in the DNA sequence). This correlation is particularly clear for two classes of external agents: (1) *chemical carcinogens* (which typically cause simple changes in the nucleotide sequence), and (2) *radiation*, such as x-rays (which typically cause chromosome breaks and translocations) or ultraviolet (UV) light (which causes specific DNA base alterations).

As would be expected, people who have inherited a genetic defect in one of several DNA repair mechanisms, causing their cells to accumulate mutations at an elevated rate, run a heightened risk of cancer. Those with the disease *xeroderma pigmentosum*, for example, have defects in the system that repairs DNA damage induced by UV light, and they have a greatly increased incidence of skin cancers. Overall, inherited mutations are thought to play a role in 5–10% of all cancers, whereas somatic mutations and epigenetic changes are much more prevalent.

A Single Mutation Is Not Enough to Change a Normal Cell into a Cancer Cell

It is estimated that there are 3.7×10^{13} cells in an average-sized human (not counting bacteria), and that 10^{16} cell divisions occur within the body over the course of a typical lifetime. This cell proliferation occurs primarily in epithelia and the hemopoietic system and is balanced by cell death to maintain normal tissue homeostasis. Even in an environment that is free of mutagens, mutations would

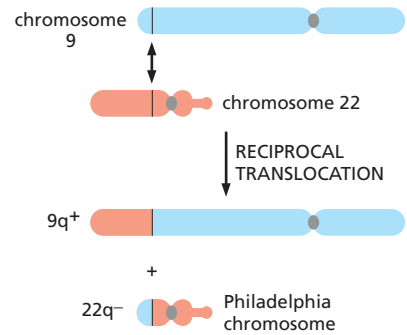


Figure 20-5 The translocation between chromosomes 9 and 22 responsible for chronic myelogenous leukemia. The normal structures of chromosomes 9 and 22 are shown at the top. When a reciprocal translocation occurs between them at the indicated site, the result is the abnormal pair at the bottom. The smaller of the two resulting abnormal chromosomes (22q⁻) is called the Philadelphia chromosome, after the city where the abnormality was first recorded.

occur spontaneously at an estimated rate of three per cell division, corresponding to about 10^{-6} mutations per gene per cell division. This unavoidable error rate is set by fundamental limitations on the accuracy of DNA replication and repair (see pp. 253–254). This means that, over the course of a typical lifetime, every single human gene will have undergone mutation somewhere in the body on roughly 10^{10} separate occasions. Among the resulting mutant cells, there will be many that have sustained deleterious mutations in genes that regulate cell growth and division, causing the cells to disobey the normal rules controlling cell turnover. Given the huge number of unavoidable mutations in a large organism like ourselves, the problem of cancer might seem to be not why it occurs, but why it occurs so infrequently.

If a mutation in a single gene were enough to convert a typical healthy cell into a cancer cell, we would not be viable organisms. Many lines of evidence indicate that the development of a cancer typically requires that a substantial number of independent, rare genetic and epigenetic accidents occur in the lineage that emanates from a single cell. One such indication comes from epidemiological studies of the incidence of cancer as a function of age (Figure 20–6). If a single mutation were responsible for cancer, occurring with a fixed probability per year, the chance of developing cancer in any given year of life should be independent of age. In fact, for most types of cancer, the incidence rises steeply with age—as would be expected if cancer is caused by a progressive, random accumulation of a set of mutations in a single lineage of cells. Notably, however, cancer incidence declines markedly among the very elderly. One interpretation of this phenomenon is that decreased cell proliferation, which is characteristic of declining stem cell function in octogenarians, provides fewer opportunities for mutation.

As discussed later, these indirect arguments linking the number of accumulated mutations to cancer development have now been confirmed by systematically sequencing the genomes of the tumor cells from individual cancer patients and cataloging the mutations that they contain.

Many Cancers Develop Gradually Through Successive Rounds of Random Inherited Change Followed by Natural Selection

For those cancers known to have a specific external cause, the disease does not usually become apparent until long after exposure to the causal agent. The incidence of lung cancer, for example, does not begin to rise steeply until after decades of heavy smoking (Figure 20–7). Similarly, the incidence of leukemias in those exposed to intense radiation in Hiroshima and Nagasaki did not show a marked rise until about 5 years after the explosion of the atomic bombs. And industrial workers exposed for a limited period to chemical carcinogens do not usually develop the cancers characteristic of their occupation until 10, 20, or even more years after the exposure. During this long incubation period, the prospective cancer cells undergo a succession of changes, and the same presumably applies to cancers where the initial genetic lesion has no such obvious external cause.

The fact that the development of a cancer requires a gradual accumulation of mutations in a number of different genes within a cell helps to explain the well-known phenomenon of **tumor progression**, whereby an initial mild disorder of cell behavior evolves gradually into a full-blown cancer (Figure 20–8).

At each stage of progression, some individual cell acquires an additional mutation or epigenetic change that gives it a selective advantage over its neighbors, making it better able to thrive in its environment—an environment that, inside a tumor, may be harsh, with low levels of oxygen, scarce nutrients, and the natural barriers to growth presented by the surrounding normal tissues. The larger the number of tumor cells, the higher the chance that at least one of them will undergo a change that favors it over its neighbors. Thus, as the tumor grows, progression accelerates. The offspring of the best-adapted cells continue to divide, eventually producing the dominant clones in a developing lesion

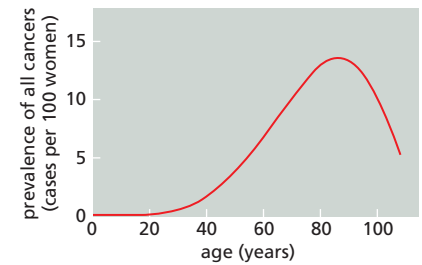


Figure 20–6 Cancer prevalence as a function of age. Age versus malignant cancer prevalence is plotted for women in the Surveillance, Epidemiology, and End Results (SEER) 9 cancer registries for the year 2000. The prevalence of cancer rises steeply as a function of age. If only a single mutation were required to trigger a cancer and the mutation had an equal chance of occurring at any time, the prevalence of cancer would be the same at all ages. (Data from C. Harding et al., *Cancer* 118:1371–1386, 2012, doi 10.1002/cnrc.26376.)

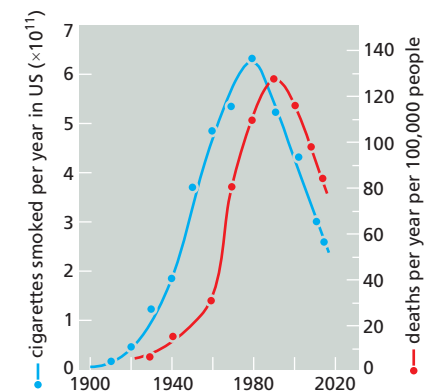


Figure 20–7 Smoking and lung cancer. A major increase in cigarette smoking that started in the early 1900s (blue line) caused a dramatic rise in lung cancer deaths (red line) after a lag time of about 20 years. Because cigarette smoking peaked in 1980, lung cancer deaths are now declining after a similar lag. (Data from National Center for Health Statistics, Centers for Disease Control and Prevention, 2017.)

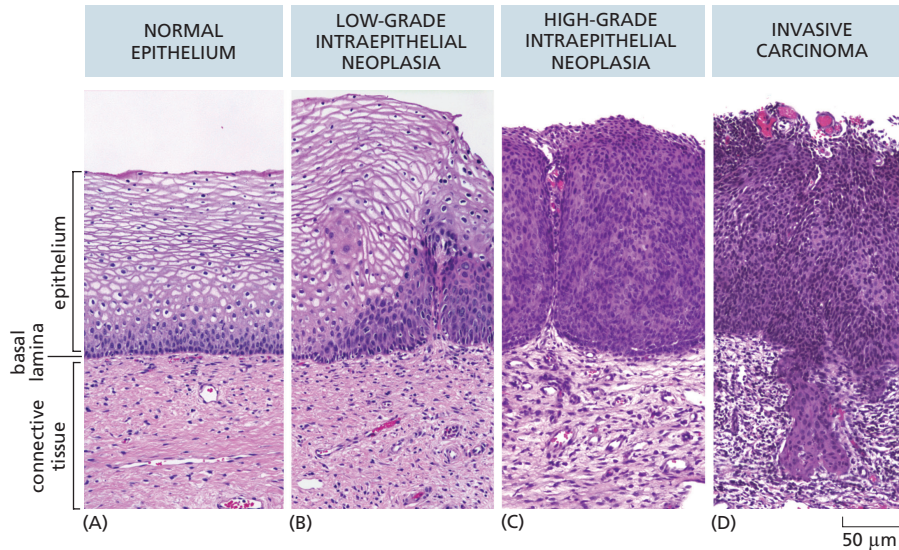


Figure 20–8 Stages of progression in the development of cancer of the epithelium of the uterine cervix. Pathologists use standardized terminology to classify the types of disorders they see in tissue slices like those shown here. (A) In a stratified squamous epithelium, dividing cells are confined to the basal layer. (B) In this low-grade intraepithelial neoplasia (right half of image), dividing cells can be found throughout the lower third of the epithelium; the superficial cells are still flattened and show signs of differentiation. (C) In high-grade intraepithelial neoplasia, cells in all the epithelial layers are proliferating and exhibit defective differentiation. (D) True malignancy begins when the cells move through or destroy the basal lamina that underlies the epithelium and invade the underlying connective tissue. (Courtesy of Andrew J. Connolly.)

(Figure 20–9). Thus, tumor progression involves a large element of chance and usually takes many years, which may be why the majority of us will die of causes other than cancer.

Just as in the evolution of plants and animals, a kind of speciation often occurs: the original cancer cell lineage can diversify to give many genetically different subclones of cells. These may coexist in the same mass of tumor tissue; or they may migrate and colonize separate environments suited to their individual quirks, where they settle, thrive, and progress as independently evolving metastases. As new mutations arise within each tumor mass, different subclones may gain an advantage and come to predominate, only to be overtaken by others or outgrown by their own sub-subclones. The large amount of genetic diversity in most tumors is one of the chief factors that make cancer cures difficult; it also increases the importance of detecting a tumor as early as possible.

Cancers Can Evolve Abruptly Due to Genetic Instability

If cancer cells evolved exclusively through the gradual accumulation of single deleterious mutations, then the time scale of transformation from premalignant disease to metastatic cancer should be predictable for a particular cancer type. However, this is not the case. As in the evolution of species, cancer evolution may proceed through long periods with no discernible change punctuated by the sudden generation of new phenotypes. The “Big Bang” theory of cancer evolution posits that in addition to gradual mutagenesis and selection for the fittest cells, periodic, cataclysmic genome disruption can promote rapid steps in the evolution of a cancer cell toward malignancy.

Unlike normal dividing cells, most human cancer cells accumulate genetic changes at an abnormally rapid rate and are said to be **genetically unstable**. This instability provides a selective advantage by speeding the process of tumor progression—allowing the subsequent accumulation of many additional mutations

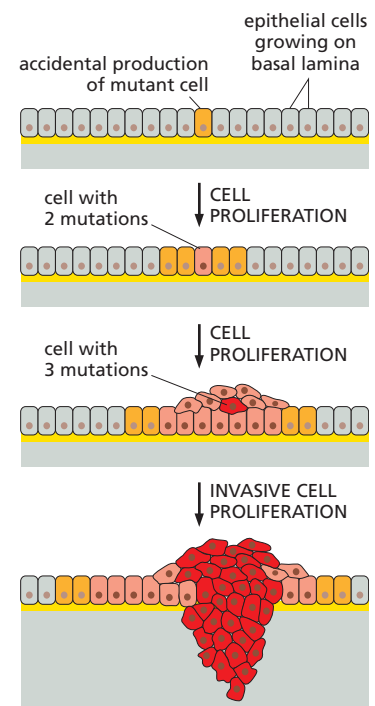


Figure 20–9 Clonal evolution during tumor progression. In this schematic diagram, a tumor develops through repeated rounds of mutation and proliferation, giving rise to a clone of fully malignant cancer cells. At each step, a single cell undergoes a mutation that either enhances cell proliferation or decreases cell death, so that its progeny become the dominant clone in the tumor. Proliferation of each clone hastens the occurrence of the next step of tumor progression by increasing the size of the cell population that is at risk of undergoing an additional mutation. The final step depicted here is invasion through the basal lamina, an initial step in metastasis. In reality, there may be more than the three steps shown here, depending on the tumor type, and a combination of genetic and epigenetic changes is involved. Not shown here is the fact that, over time, a variety of competing subclones will often arise in a tumor. As we will discuss later, this heterogeneity complicates cancer therapies.

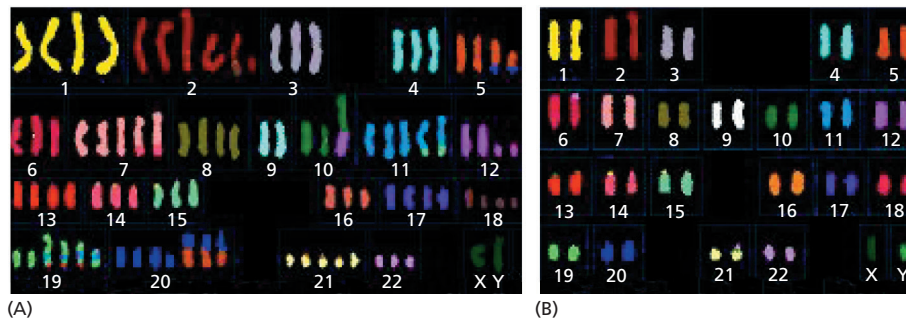


Figure 20-10 Chromosome complements (karyotypes) of colon cancers showing different kinds of genetic instability. (A) The karyotype of a typical cancer shows many gross abnormalities in chromosome number and structure. Considerable variation can also exist from cell to cell (not shown). (B) The karyotype of a tumor that has a stable chromosome complement with few chromosomal anomalies; the genetic abnormalities in these tumors are mostly invisible, having been created by defects in DNA repair. All of the chromosomes in this figure were stained as in Figure 4-11, the DNA of each human chromosome being marked with a different combination of fluorescent dyes. (Courtesy of Wael Abdel-Rahman and Paul Edwards.)

required to produce a cancer. But the extent of the instability and its molecular origins differ from cancer to cancer and from individual to individual, both in severity and in character. In some cases, the karyotype—the set of chromosomes as they appear at mitosis—is normal or nearly so, but many point mutations are detected in individual genes, suggesting a failure of the repair mechanisms that normally correct errors in the replication or maintenance of DNA sequences. Often, however, the cancer cell karyotype is severely disordered, with many chromosomal breaks and rearrangements, resulting in many deletions, duplications, and amplifications of parts of the genome (Figure 20-10). Such highly disrupted genomes indicate that catastrophic events have occurred, likely due to defects in chromosome duplication or segregation during mitosis. For example, a common feature of many cancer cells is a failure to correctly attach all of the chromosomes to the mitotic spindle, which can result in chromosome breakage or aneuploidy, the gain or loss of individual chromosomes. More dramatically, mitotic defects can also lead to the isolation of a single chromosome in one of the daughter cells within a “micronucleus,” where it is prone to massive DNA damage and chromosomal rearrangement, termed “chromothripsis” (Figure 20-11).

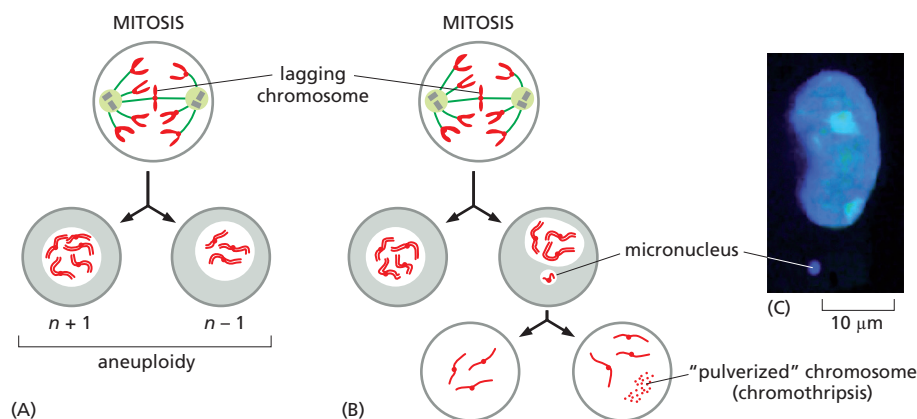


Figure 20-11 Chromosome segregation defects can give rise to aneuploidy and/or chromothripsis. Correct chromosome segregation requires that replicated copies of each chromosome, termed sister chromatids, are attached to opposite spindle poles. A mistake in this process is depicted in which one sister of a chromatid pair is attached to both spindle poles and is therefore pulled in opposite directions during anaphase, causing it to lag behind in the center of the cell, resulting in a *lagging chromosome*. (A) A lagging chromosome may end up in the same daughter cell as its sister, resulting in aneuploidy with $n+1$ and $n-1$ karyotypes (n = chromosome number). (B) Another possible outcome is that the lagging chromosome remains separated from the rest of the chromosomes in the daughter cell, forming its own “micronucleus” in the following interphase. DNA replication in micronuclei is frequently incomplete, and DNA damage accumulates, but this does not delay the cell cycle. In the next mitosis, the isolated chromosome is prone to undergo fragmentation during chromosome condensation, in a process called chromothripsis. (C) Micronuclei can persist over several generations of cell division, undergoing rounds of chromosome fragmentation and reassembly, or they can be reincorporated into a daughter cell nucleus (Movie 20.2). The fluorescence micrograph shows a primary nucleus and a micronucleus present in the same cell. Nuclear DNA staining (blue) appears uneven due to normal nuclear substructures. (C, from M.L. Leibowitz et al., *Annu. Rev. Genet.* 49:183–211, 2015. With permission from Annual Reviews.)

From an evolutionary perspective, none of this should be a surprise: anything that increases the probability of random changes in gene function that are heritable from one cell generation to the next—and that are not too deleterious—will likely speed the evolution of a clone of cells toward malignancy, thereby causing this property to be selected for during tumor progression. Genome instability likely also contributes to the heterogeneity of cancer cells frequently observed within an individual tumor. By leading to multiple aberrant karyotypes, chromosome mis-segregation, aneuploidy—and more rarely chromothripsis—all act to shuffle the genetic cards, allowing cancer cells to sample a variety of different phenotypes and for different clones to coexist within a population.

Some Cancers May Harbor a Small Population of Stem Cells

Self-renewing tissues, where cell division continues throughout life, are the breeding ground for the great majority of human cancers. They include the epidermis (the outer epithelial layer of the skin), the epithelial lining of the digestive and reproductive tracts, and the bone marrow, where blood cells are generated (see Chapter 22). With each cell division cycle comes the chance of mutation, which can be amplified dramatically by environmental factors. As a result, there is a strong correlation between the frequency of cell division and the incidence of cancer in a particular tissue (Figure 20–12).

In almost all proliferating tissues, renewal depends on the presence of stem cells, which divide to give rise to terminally differentiated cells, which do not divide. This creates a mixture of cells that are genetically identical and closely related by lineage but are in different states of differentiation. Many tumors similarly appear to consist of populations of cells in various states of differentiation. A comparison of tumor development to the normal homeostasis of stem cell-derived tissues may help us better understand the origin of some cancers and also why some tumors are so resistant to treatment.

To consider the implications, it is helpful to consider how normal stem-cell systems operate. When a normal stem cell divides, each daughter cell has a choice—it can remain a stem cell or it can commit to a pathway leading to differentiation. A stem-cell daughter remains in place to generate more cells in the future. A committed daughter typically undergoes some rounds of cell proliferation (as a so-called *transit amplifying cell*), but it then stops dividing, terminally differentiates, and eventually is discarded and replaced (it may die by apoptosis, with recycling of its materials, or be shed from the body). Therefore, stem cells tend to be vastly outnumbered by the cells that are committed to terminal differentiation. However, though few and far between and often relatively slowly dividing, stem cells carry the entire responsibility for maintenance of the tissue

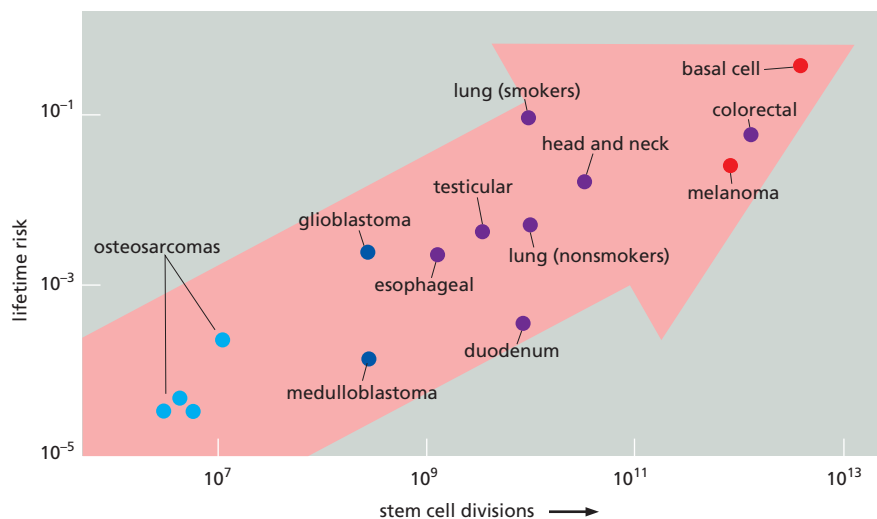


Figure 20–12 The lifetime cancer risk is correlated with the division rate of the cell of origin of the cancer. This plot shows the relationship between the number of stem cell divisions in a given tissue and the lifetime risk of cancer in that tissue. At the lower extreme are osteosarcomas (cancers that originate in bone), which are derived from mesenchymal cells that divide infrequently and rarely give rise to cancer. In contrast, epithelial cells of the skin and digestive tract are highly proliferative and give rise to malignancies much more frequently. Dots represent specific cancer types and are colored to indicate their classification: sarcoma (*light blue*); neuronal (*dark blue*); carcinoma (*purple*); and skin (*red*). Note that environmental factors strongly amplify the risk of many cancers, such as for lung cancer between smokers and nonsmokers. (Data from C. Tomasetti and B. Vogelstein, *Science* 347:78–81, 2015, doi 10.1126/science.1260825.)

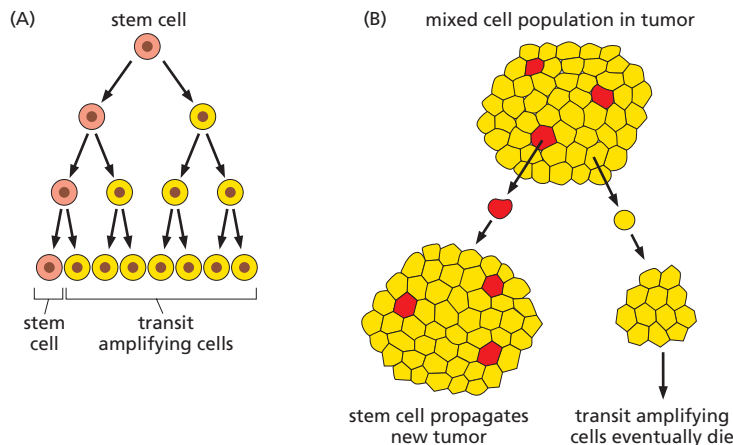


Figure 20-13 Cancer stem cells can be responsible for a tumor's growth and yet remain only a small part of the tumor-cell population. (A) Lineage of stem cells producing transit amplifying cells. (B) How a small proportion of cancer stem cells can maintain a tumor. Although the much more abundant transit amplifying cells will eventually die, the number of cancer stem cells will increase slowly but steadily to produce a growing tumor.

over the long term. In a healthy body, feedback controls regulate the process, adjusting the balance of cell-fate choices, cell proliferation, and cell death in a way that corrects for any departure from proper cell population numbers.

During the development of cancer, mutations have the potential to subvert the normal cellular differentiation program in multiple ways; for example, by leading to an overproliferation of transit amplifying cells or to an inhibition of their terminal differentiation or death. More insidious, however, are mutations that lead to the generation of cancer stem cells. Some tumors appear to exhibit this etiology: they consist of rare **cancer stem cells** capable of self-renewal, together with much larger numbers of dividing transit amplifying cells that are derived from the cancer stem cells but have a limited capacity for self-renewal (**Figure 20-13**).

Evidence for the existence of cancer stem cells comes from experiments in which individual cells from a cancer are tested for their ability to give rise to fresh tumors when implanted into a mouse. It has been known for more than half a century that there is usually only a small chance—typically much less than 1%—that a tumor cell chosen at random and tested in this way will generate a new tumor. This by itself does not prove that the tumor cells are heterogeneous: like seeds scattered on difficult ground, each of them may have only a small chance of finding a spot where it can survive and grow. Modern technologies for sorting cells have shown, however, that subpopulations of cancer cells expressing markers typically found on the surface of stem cells have a greatly enhanced ability to found new tumors. Moreover, the new tumors consist of mixtures of cells that express the stem cell markers and cells that do not, all generated from the same founder cell that expressed the markers. The cancer stem-cell phenomenon, whatever its basis, implies that even when the tumor cells are genetically similar, they may be phenotypically diverse. A treatment that wipes out tumor cells in one state will often allow survival of others that remain a danger. Radiotherapy or a cytotoxic drug, for example, may selectively kill off the rapidly dividing cells, reducing the tumor volume to almost nothing, and yet spare a few slowly dividing cells that go on to resurrect the disease. This greatly adds to the difficulty of cancer therapy, and it is part of the reason why treatments that seem at first to succeed often end in relapse and disappointment.

A Common Set of Hallmarks Typically Characterizes Cancerous Growth

Clearly, to produce a cancer, a cell must acquire a range of aberrant properties—a collection of subversive new skills—as it evolves. Different cancers require different combinations of these properties. Nevertheless, cancers all share some common hallmarks. These defining properties are commonly combined with other features, such as genetic instability, that help the miscreants to arise

and thrive. A list of the key attributes of cancer cells in general would include the following:

1. Altered homeostasis that results in cells growing and dividing at a faster rate than they die
2. Bypass of normal limits to cell proliferation
3. Evasion of cell-death signals
4. Altered cellular metabolism
5. Manipulation of the tissue environment to support cell survival and to evade a deleterious immune response
6. Escape of cells from their home tissues and proliferation in foreign sites (metastasis)

Below we discuss these key features in more detail. In the next section of the chapter, we examine the mutations and molecular mechanisms that underlie these and other properties of cancer cells.

Cancer Cells Display an Altered Control of Growth and Homeostasis

Mutability and large cell population numbers create opportunities for mutations to occur, but the driving force for development of a cancer has to come from some sort of *selective advantage* possessed by the mutant cells. Most obviously, a mutation or epigenetic change can confer such an advantage by increasing the rate at which a clone of cells proliferates or by enabling it to continue proliferating when normal cells would stop or die. One of the most important properties of many types of cancer cells is that they fail to undergo apoptosis when a normal cell would do so (Figure 20–14).

Cancer cells that can be grown in culture or cultured cells artificially engineered to contain the types of mutations encountered in cancers typically show a **transformed** phenotype. They are abnormal in their shape, their motility, their responses to growth factors in the culture medium, and, most characteristically, in the way they react to contact with the culture dish and with one another. Whereas most normal cells will not divide unless they are attached to the surface, transformed cells will often divide even if held in suspension. More generally, transformed cells no longer require all of the positive signals from their surroundings that normal cells require. In addition, transformed cells fail to recognize some negative influences. Thus, for example, normal cells become inhibited from moving and dividing when the culture reaches confluence (where the cells are touching one another), while transformed cells continue moving and dividing even after confluence, and so pile up in layer upon layer in the culture dish (Figure 20–15).

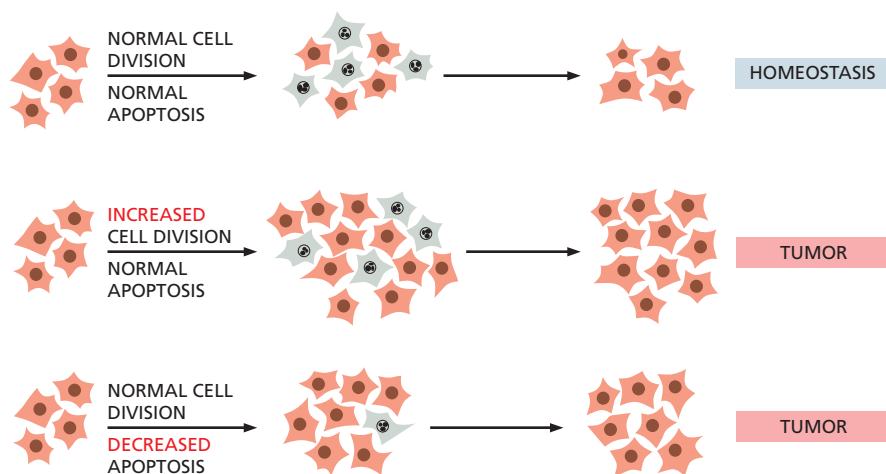


Figure 20–14 Both increased cell division and decreased apoptosis can contribute to tumorigenesis. In normal tissues, apoptosis balances cell division to maintain homeostasis (see Movie 18.1). During the development of cancer, either an increase in cell division or an inhibition of apoptosis can lead to the increased cell numbers important for tumorigenesis. The cells fated to undergo apoptosis are gray in this diagram. Both an increase in cell division and a decrease in apoptosis commonly contribute to tumor growth.

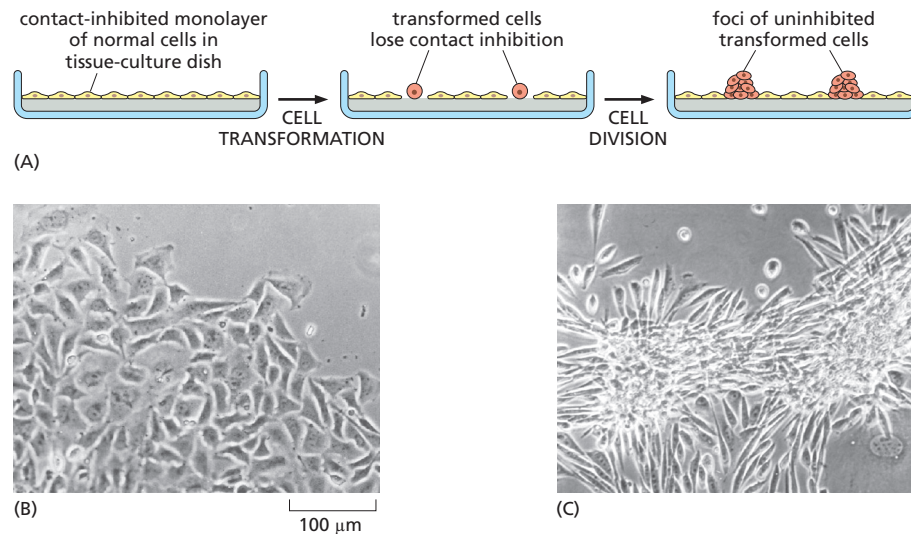


Figure 20-15 Loss of contact inhibition by cancer cells in cell culture. Most normal cells stop proliferating once they have carpeted the dish with a single layer of cells: proliferation seems to depend on contact with the dish and to be inhibited by contacts with other cells—a phenomenon known as *contact inhibition*. Cancer cells, in contrast, usually disregard these restraints and continue to grow, so that they pile on top of one another (**Movie 20.3**). (A) Schematic drawing. (B and C) Light micrographs of normal (B) and transformed (C) fibroblasts. (B and C, courtesy of Lan Bo Chen.)

Cancer cells also misbehave in their natural environment, embedded in a tissue. For example, normal cells of the gut epithelium are constantly turning over through rounds of cell division and death, but maintain their barrier function by seamlessly extruding old or damaged cells into the gut lumen. Once detached from the matrix and matrix-associated survival signals, these cells die through a form of apoptosis. By overriding normal death signals, transformed cells may be able to survive after their ejection from the epithelium. However, because the normal direction of cell extrusion is into the lumen of the gut, they would nevertheless be swept away through the waste canal. However, some of the mutations selected for during cancer progression can change the direction of extrusion, thereby enabling the tumor cells to cross the basement membrane and invade the surrounding tissue, potentially leading to metastasis (**Figure 20-16**).

In conclusion, by disobeying the normal etiquette of where to live and when to die, cancer cells both evade growth suppression and explore new opportunities to prosper and multiply.

Human Cancer Cells Escape a Built-in Limit to Cell Proliferation

Many normal human cells have a built-in limit to the number of times that they can divide when stimulated to proliferate in culture: they permanently stop dividing after a certain number of population doublings (25–50 for human fibroblasts, for example). This cell-division-counting mechanism is termed **replicative cell senescence**, and it generally depends on the progressive shortening of the telomeres at the ends of chromosomes, a process that eventually changes their structure (discussed in Chapter 17). As discussed in Chapter 5, the maintenance

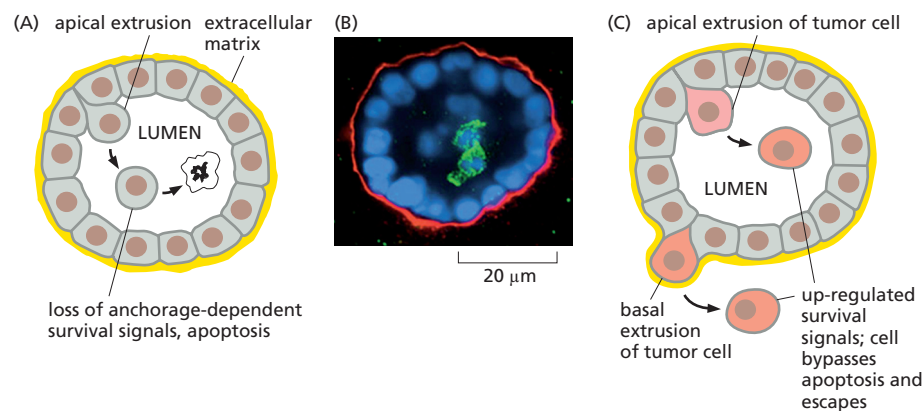


Figure 20-16 The direction in which a cell extrudes from an epithelium has important consequences for its fate. (A) Under normal conditions, as cells proliferate in a tissue such as the gut or mammary epithelium and become more crowded, they are extruded into the lumen to maintain cell number. Extruded cells generally undergo apoptosis due to loss of survival signals. (B) Fluorescence micrograph of a cluster of mammary epithelial cells grown in three-dimensional (3D) culture and stained for its basal boundary (red) and nuclei (blue). As the sac of tissue grows, the central cells occupying the lumen undergo apoptosis as shown by staining with caspase-3 (green). (C) Tumor cells with up-regulated survival signals that have been extruded apically might survive but are likely to be eliminated nevertheless; for example, by excretion through the digestive tract or by secretion from the mammary gland. In contrast, tumor cells that have been extruded basally can more readily initiate invasion. (B, courtesy of J. Debnath.)

of telomere DNA during S phase depends on the enzyme *telomerase*, which maintains a special telomeric DNA sequence that promotes the formation of protein cap structures to protect chromosome ends. Because many proliferating human cells (stem cells being an exception) are deficient in telomerase, their telomeres shorten with every division, and their protective caps deteriorate, creating a DNA damage signal because the unprotected chromosome ends resemble double-strand breaks. Eventually, the altered chromosome ends trigger a permanent cell-cycle arrest or cause the cell to die.

Human cancer cells avoid replicative cell senescence in one of two ways. Most often, they reactivate the telomerase gene as they proliferate, so that their telomeres do not shorten or become uncapped; alternatively, they can evoke a mechanism based on a DNA repair process that depends on homologous recombination for elongating their chromosome ends (called ALT). Regardless of the strategy used, the result is that the cancer cells continue to proliferate under conditions when normal cells would stop due to telomere erosion.

Cancer Cells Have an Abnormal Ability to Bypass Death Signals

A large multicellular organism requires powerful safety mechanisms that guard against the trouble caused by damaged and deranged cells. These mechanisms are essential because, as previously explained, a large number of mutated cells will inevitably be produced. Normally, internal disorder gives rise to danger signals in the faulty cell, activating protective measures to reverse and cure this disorder or, failing that, activating decisions that lead to cell death by apoptosis (see Chapter 18). To survive, cancer cells require mutations to elude or break through these defenses designed to eliminate defective cells.

Cancer cells generally contain mutations that drive the cell into an abnormal state, where metabolic processes may be unbalanced and essential cell components may be produced in ill-matched proportions. States of this type, where the cell's homeostatic mechanisms are inadequate to cope with an imposed disturbance, are loosely referred to as states of *cell stress*. As one example, chromosome breakage and other forms of DNA damage are commonly observed during the development of cancer, reflecting the genetic instability that cancer cells display. Thus, to survive and divide without limit, a prospective cancer cell must accumulate changes that disable the normal safety mechanisms that would otherwise cause a cell that is stressed to commit suicide by apoptosis.

While cancer cells tend to avoid apoptosis, this does not mean that they rarely die. On the contrary, in the interior of a large solid tumor, cell death often occurs on a massive scale: living conditions are difficult, with severe competition among the cancer cells for oxygen and nutrients. Most of these cells die by an alternative cell-death mechanism termed necrosis ([Figure 20-17](#)).

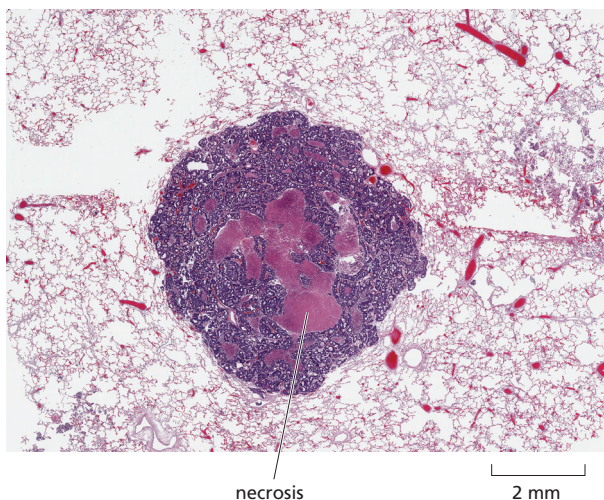


Figure 20-17 The interior of a large tumor is deprived of oxygen and nutrients. This cross section of a colon adenocarcinoma that has metastasized to the lung shows colorectal cancer cells that have formed a cohesive nodule (*dark-staining region*). The metastasis has central *pink areas* of necrosis where dying cancer cells have outgrown their blood supply and burst open, releasing their contents. Such anoxic regions are common in the interior of tumors. (Courtesy of Andrew J. Connolly.)

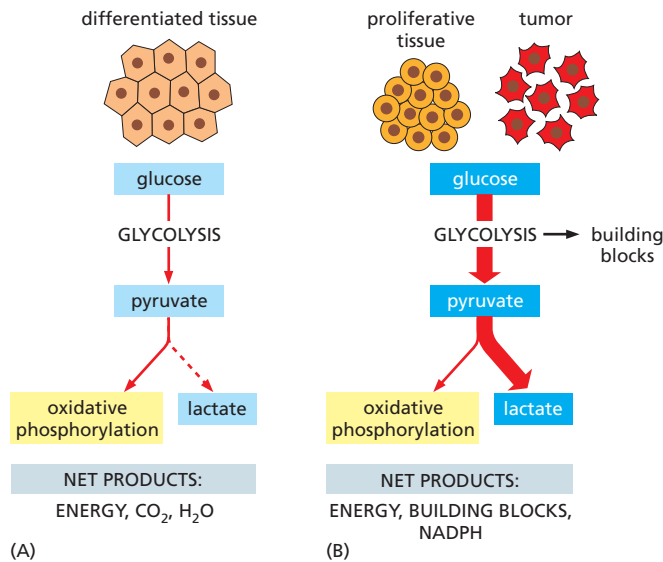


Figure 20-18 The Warburg effect in tumor cells reflects a dramatic change in glucose uptake and sugar metabolism.

(A) Cells that are not proliferating will normally oxidize nearly all of the glucose that they import from the blood to produce ATP through oxidative phosphorylation that takes place in their mitochondria. Only when deprived of oxygen will these cells generate most of their ATP from glycolysis, converting the pyruvate produced to lactate in order to regenerate the NAD⁺ that they need to keep glycolysis going (see Figure 2-50). (B) Tumor cells, by contrast, will generally produce abundant lactate even in the presence of oxygen. This results from a greatly increased rate of glycolysis that is fed by a very large increase in the rate of glucose import. In this way, tumor cells resemble the rapidly proliferating cells in embryos, which likewise require for biosynthesis a large supply of the small-molecule building blocks that can be produced from imported glucose (see also Figure 2-60).

The tumor grows because the cell birthrate outpaces the cell death rate, but often by only a small margin. For this reason, the time that a tumor takes to double in size can be far longer than the cell-cycle time of the tumor cells.

Cancer Cells Have Altered Sugar Metabolism

Given sufficient oxygen, differentiated cells in most adult tissues will fully oxidize almost all the carbon in the glucose they take up to CO₂, which is eventually exhaled by the lungs as a waste product. A growing tumor needs nutrients in abundance to provide the building blocks to make new macromolecules. Correspondingly, most tumors are metabolically more similar to a growing embryo than to normal adult tissue. Tumor cells consume glucose avidly, importing it from the blood at a rate that can be as much as 100 times higher than that for neighboring normal cells. Moreover, only a fraction of this imported glucose is fully oxidized to CO₂ by mitochondrial oxidative phosphorylation, which normally enables highly efficient production of ATP. Instead, the metabolism of carbon atoms from glucose is rewired to support the production of raw materials for the synthesis of the proteins, nucleic acids, and lipids that enable cellular proliferation (Figure 20-18). In other words, even though glycolysis is a much less efficient mode of ATP production than oxidative phosphorylation, it can continue unabated in cancer cells that find themselves in an oxygen-poor environment, and it has the critical added benefit of producing abundant cellular building blocks.

This tendency of tumor cells to de-emphasize oxidative phosphorylation even when oxygen is plentiful, while at the same time taking up large quantities of glucose, is necessary for the rapid proliferation of many cancer cells and is called the *Warburg effect*—so named because Otto Warburg first noticed the phenomenon in the early twentieth century. It is this abnormally high glucose uptake that allows tumors to be selectively imaged in whole-body scans (see Figure 20-1), thereby providing one way to monitor cancer progression and responses to treatment.

The Tumor Microenvironment Influences Cancer Development

While the cancer cells in a tumor are the bearers of dangerous mutations and are often grossly abnormal, the other cells in the tumor—especially those of the supporting connective tissue, or **stroma**—are far from passive bystanders. The development of a tumor relies on a two-way communication between the cancer cells and tumor stroma, just as the normal development of epithelial organs relies on communication between epithelial cells and mesenchymal cells (discussed in Chapter 22).

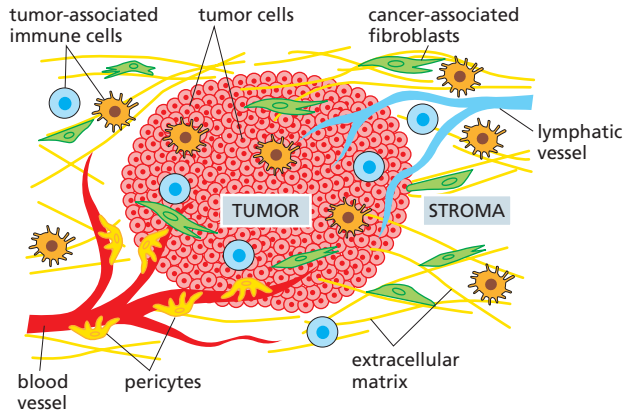


Figure 20–19 The tumor microenvironment plays a role in tumorigenesis. Tumors consist of many cell types, including cancer cells, endothelial cells, pericytes (vascular smooth muscle cells), fibroblasts, and inflammatory white blood cells. Communication among these and other cell types plays an important part in tumor development. Note, however, that only the cancer cells in a tumor are thought to contain mutations that make them genetically abnormal.

The stroma provides a framework for the tumor. As for normal connective tissue, the stroma is composed of fibroblasts and inflammatory white blood cells and of the endothelial cells that form blood and lymphatic vessels with their attendant smooth muscle cells (Figure 20–19). As a carcinoma progresses, the cancer cells induce changes in this stroma by secreting signal proteins that alter the behavior of the stromal cells and proteolytic enzymes that modify the extracellular matrix. The stromal cells in turn act back on the tumor cells by secreting signal proteins that stimulate cancer cell growth and division and proteases that further remodel the extracellular matrix. In these ways, the tumor and its stroma evolve together, like weeds and the ecosystem that they invade, and the tumor can become dependent on its particular stromal cells. Experiments using mice indicate that the growth of some transplanted carcinomas depends on the tumor-associated fibroblasts and that normal fibroblasts will not suffice.

Cancer cells have a complex interaction with the cells of the immune system that are present in the stroma, which have the potential to destroy the tumor if it is recognized as aberrant tissue but can also promote tumor growth by providing signals that stimulate cancer cell proliferation. The tumor can manipulate the immune system to its advantage in at least two ways. First, tumors may invoke an inflammatory reaction similar to what occurs when normal tissue is damaged, which helps them acquire the stroma they need for survival and growth. Tumors have therefore been likened to unhealed wounds, eliciting some of the same responses, including an increase in the permeability of nearby blood vessels, which allows the flow of signaling molecules in and out of the vessels, and the deposition of extracellular matrix. Tumors also stimulate the formation of new blood vessels, a process termed *angiogenesis*, which promotes survival of a tumor as it grows larger and becomes hypoxic in its interior. Second, and of equal importance, the tumor establishes an immunosuppressive microenvironment by blocking the activation of white blood cells that could lead to its destruction. Strategies that override this ability of a tumor to suppress an immune response have recently emerged as powerful tools in cancer therapy, and they are described in the last section of this chapter.

Cancer Cells Must Survive and Proliferate in a Foreign Environment

In order to kill us, cancer cells generally need to spread and multiply at new sites in the body through a process called metastasis. This is the most deadly—and least understood—aspect of cancer, being responsible for 90% of cancer-associated deaths. By spreading through the body, a cancer becomes almost impossible to eradicate by either surgery or local irradiation. **Metastasis** is itself a multistep process: the cancer cells first have to invade local tissues and vessels, move through the circulation, leave the vessels, and then establish new cellular colonies

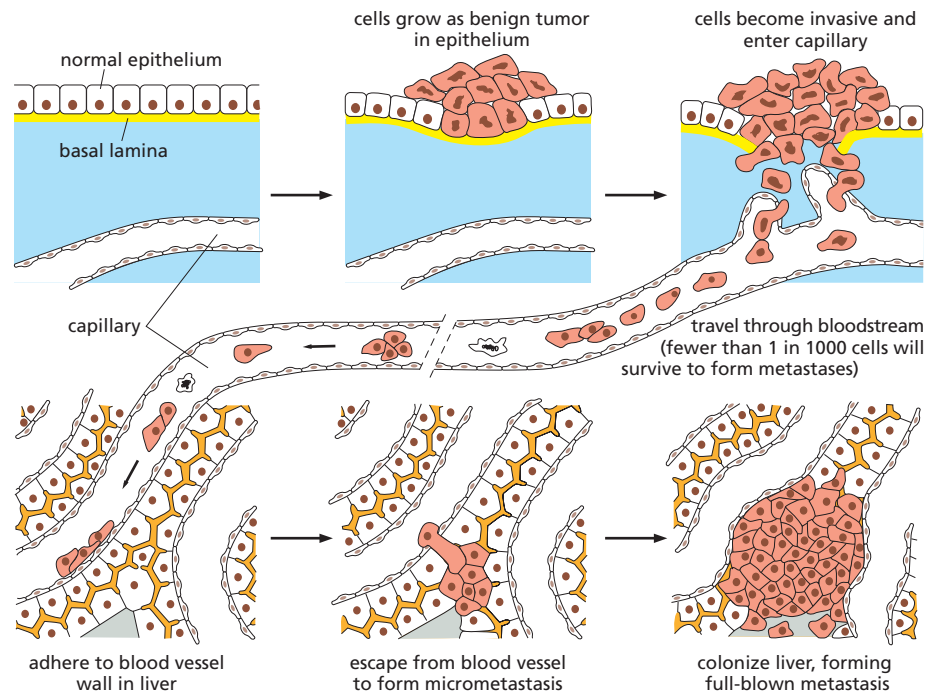


Figure 20-20 Steps in the process of metastasis. This example illustrates the spread of a tumor from its site of origin (such as the bladder) to another organ (such as the liver). Tumor cells may enter the bloodstream directly by crossing the wall of a blood vessel, as diagrammed here, or, more commonly perhaps, by crossing the wall of a lymphatic vessel that ultimately discharges its contents (lymph) into the bloodstream. Tumor cells that have entered a lymphatic vessel often become trapped in lymph nodes along the way, giving rise to lymph-node metastases. Studies in animals show that typically fewer than one in every thousand malignant tumor cells introduced into the bloodstream is viable after 24 hours, and that less than 0.1% of these surviving circulating tumor cells (CTCs) will colonize a new tissue so as to produce a detectable tumor at a new site.

at distant sites (**Figure 20-20**). Each of these events is complex, and most of the molecular mechanisms involved are not yet clear, but the last step—colonization of distant sites—is rate limiting.

For a cancer cell to become malignant, it must break free of constraints that keep normal cells in their proper places and prevent them from invading neighboring tissues. Invasiveness is thus one of the defining properties of malignant tumors, which show a disorganized pattern of growth and ragged borders, with extensions into the surrounding tissue (see, for example, **Figure 20-8**). Although the underlying molecular changes are not well understood, invasiveness almost certainly requires a disruption of the adhesive mechanisms that normally keep cells tethered to their proper neighbors and to the extracellular matrix.

The next step in metastasis—the establishment of colonies in distant organs—begins with entry into the circulatory system: the invasive cancer cells must penetrate the wall of a blood or lymphatic vessel. Lymphatic vessels, being larger and having more flimsy walls than blood vessels, allow cancer cells to enter in small clumps; such clumps may then become trapped in lymph nodes, giving rise to lymph-node metastases. The cancer cells that enter blood vessels, in contrast, do so singly or, more rarely, in small clusters. With modern techniques for sorting cells according to their surface properties, it has become possible to detect these *circulating tumor cells (CTCs)* in samples of blood from cancer patients, even though they are only a minute fraction of the total blood-cell population. Notably, in a mouse model, CTC clusters gave rise to metastases at a significantly higher rate than single CTCs did. Presumably the adhesion of epithelial-derived cancer cells to one another helps to override death signals and suppress apoptosis.

Of the cancer cells that enter the lymphatics or bloodstream, only a tiny proportion succeed in making their exit, settling in new sites, and surviving and proliferating there as founders of metastases. To discover which of the later steps in metastasis present cancer cells with the greatest difficulties, one can label the cells with a fluorescent dye or green fluorescent protein (GFP), inject them into the bloodstream of a mouse, and then monitor their fate (**Movie 20.4**). In such experiments, one observes that many cells survive in the circulation, lodge in small vessels, and exit into the surrounding tissue, regardless of whether they come from a tumor that metastasizes or one that does not. Some cells die immediately after they enter foreign tissue; others survive entry into the foreign tissue but

fail to proliferate. Still others divide a few times and then stop, forming micrometastases containing ten to several thousand cells. Very few establish full-blown metastases. Experiments show that fewer than one in thousands, perhaps one in millions, manages this feat. The final step of colonization seems to be the most difficult: like the Vikings who landed on the inhospitable shores of Greenland, the migrant cells may fail to survive in the alien environment or they may only thrive there for a short while to found a little colony—a *micrometastasis*—that then dies out (**Movie 20.5**).

Many cancers are discovered before they have managed to found metastatic colonies, and these can be cured by destruction of the primary tumor. But on occasion, an undetected, distant micrometastasis will remain dormant for many years, only to reveal its presence by erupting into growth to form a large secondary tumor long after the primary tumor has been removed.

Summary

Cancer cells, by definition, grow and proliferate in defiance of normal controls (that is, they are neoplastic) and gain the ability to invade surrounding tissues and colonize distant organs (that is, they are malignant). By giving rise to secondary tumors, or metastases, they become difficult to eradicate by surgery or local irradiation. Cancers are thought to originate from a single cell that has experienced an initial mutation, but the progeny of this cell must undergo many further changes to become cancerous. Tumor progression usually takes many years and reflects the operation of a Darwinian-like process of evolution, in which somatic cells undergo mutation and epigenetic changes accompanied by natural selection and occasional bursts of genomic chaos that give rise to heterogeneity.

Cancer cells acquire a variety of special properties as they evolve, multiply, and spread. Their mutant genomes enable them to grow and divide in defiance of the signals that normally keep cell proliferation under tight control. As part of the evolutionary process of tumor progression, cancer cells acquire a collection of additional abnormalities, including defects in the controls that permanently stop cell division or induce apoptosis in response to cell stress or DNA damage and defects in the mechanisms that normally keep cells from straying from their proper place. All of these changes increase the ability of cancer cells to survive, grow, and divide in their original tissue and then to metastasize, founding new colonies in foreign environments. The evolution of a tumor also depends on other cells present in the tumor microenvironment, collectively called stromal cells, that the cancer attracts and manipulates.

Because many changes are needed to confer this collection of asocial behaviors, it is not surprising that most cancer cells are genetically and/or epigenetically unstable. This instability is thought to be selected for in the clones of aberrant cells that are able to produce tumors, because it greatly accelerates the accumulation of the further genetic and epigenetic changes that are required for tumor progression.

CANCER-CRITICAL GENES: HOW THEY ARE FOUND AND WHAT THEY DO

As we have seen, cancer depends on the accumulation of heritable changes in somatic cells; that is, changes that are transmitted by a cell to its progeny. To understand it at a molecular level we need to identify the mutations and epigenetic alterations involved and to discover how they give rise to cancerous cell behavior. Finding the relevant cells is often easy; they are favored by natural selection and call attention to themselves by giving rise to tumors. But how do we identify those genes that have undergone cancer-promoting changes among all the other genes in the cancerous cells? A typical cancer depends on a whole set of mutations and epigenetic changes that are never exactly the same in two different patients, although there are commonalities among certain tumor types. In addition, a given cancer cell will also contain a large number of somatic mutations that are accidental by-products—so-called *passengers* rather than *drivers*—of its genetic instability, and it can be difficult to distinguish these incidental changes

from those changes that have a causative role in the disease. Despite these difficulties, many of the genes that are repeatedly altered in human cancers have been identified over the past 40 years. We will call such genes, for want of a better term, *cancer-critical genes*, meaning all genes whose alteration can contribute to the causation or evolution of cancer.

In this section, we shall first discuss how cancer-critical genes are identified. We shall then examine their functions and the parts they play in conferring on cancer cells the properties outlined in the first part of the chapter. We shall end the section by discussing colon cancer as an extended example, showing how a succession of changes in cancer-critical genes enables a tumor to evolve from one pattern of bad behavior to another that is worse.

The Identification of Gain-of-Function and Loss-of-Function Cancer Mutations Has Traditionally Required Different Methods

Cancer-critical genes are grouped into two broad classes, according to whether the cancer risk arises from too much activity of the gene product or too little. Genes of the first class, in which a gain-of-function mutation can drive a cell toward cancer, are called **proto-oncogenes**; their mutant, overactive or overexpressed forms are called **oncogenes**. Genes of the second class, in which a loss-of-function mutation can contribute to cancer, are called **tumor suppressor genes**. In either case, the mutation may lead toward cancer directly (by causing cells to proliferate when they should not) or indirectly; for example, by causing genetic or epigenetic instability and so hastening the occurrence of other inherited changes that directly stimulate tumor growth. Those genes whose alteration results in genomic instability represent a subclass of cancer-critical genes that are sometimes called *genome maintenance genes*.

As we shall see, mutations in oncogenes and tumor suppressor genes can have similar effects in promoting the development of cancer; overproduction of a signal for cell proliferation, for example, can result from either kind of mutation. Thus, from the point of view of a cancer cell, oncogenes and tumor suppressor genes—and the mutations that affect them—are flip sides of the same coin. The techniques that led to the discovery of these two categories of genes, however, are quite different.

The mutation of a single copy of a proto-oncogene that converts it to an oncogene has a dominant, growth-promoting effect on a cell (**Figure 20-21A**). Thus,

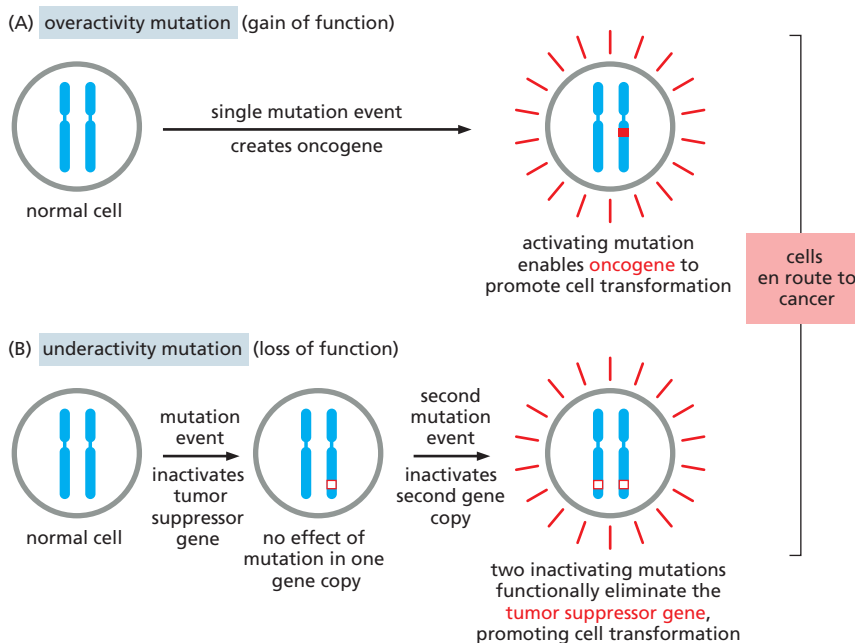


Figure 20-21 Cancer-critical mutations fall into two readily distinguishable categories, dominant and recessive.

In this diagram, activating mutations are represented by *solid red boxes*, inactivating mutations by *hollow red boxes*. (A) Oncogenes act in a dominant manner: a gain-of-function mutation in a single copy of the cancer-critical gene can drive a cell toward cancer. (B) Mutations in tumor suppressor genes, on the other hand, generally act in a recessive manner: the function of both alleles of the cancer-critical gene must be lost to drive a cell toward cancer. Although in this diagram the second allele of the tumor suppressor gene is inactivated by mutation, it is often inactivated instead by loss of the second chromosome. Not shown is the fact that mutation of some tumor suppressor genes can have an effect even when only one of the two gene copies is damaged.

we can identify the oncogene by its effect when it is *added*—by DNA transfection, for example, or through transduction with a viral vector—to the genome of a suitable type of tester cell or experimental animal. In the case of the tumor suppressor gene, on the other hand, the cancer-causing alleles produced by the change are generally recessive: often (but not always) both copies of the normal gene must be removed or inactivated in the diploid somatic cell before an effect is seen (Figure 20-21B). This calls for a different experimental approach, one focusing on discovering what is *missing* in the cancer cell.

We begin by discussing a few examples of each class of cancer-critical genes to illustrate basic principles. These examples are chosen also for their historical importance: the experiments that led to their discovery—at different times and by different methods—marked turning points in the understanding of cancer.

Retroviruses Led to the Identification of Oncogenes

The search for the genetic causes of human cancer took a circuitous route, beginning with clues that came from the study of **tumor viruses**. Although viruses are involved only in a minority of human cancers, a set of viruses that infect animals provided critical early tools for studying cancer.

One of the first animal viruses to be implicated in cancer was discovered in chickens more than 100 years ago, when an infectious agent that causes connective-tissue tumors, or sarcomas, was identified as a virus—the *Rous sarcoma virus*. Like all the other *RNA tumor viruses* discovered since, it is a **retrovirus**. When it infects a cell, its RNA genome is copied into DNA by reverse transcription, and the DNA is inserted into the host genome, where it can persist and be inherited by subsequent generations of cells. Something in the DNA inserted by the Rous sarcoma virus made the host cells cancerous, but what was it? The answer was a surprise. It turned out to be a piece of DNA that was unnecessary for the virus's own survival or reproduction; instead, it was a passenger, a gene called *v-Src*, that the virus had picked up on its travels. *v-Src* was unmistakably similar, but not identical, to a gene—*c-Src*—that is found in all vertebrate genomes and encodes a protein tyrosine kinase. *c-Src* had evidently been taken up accidentally by the retrovirus from the genome of a previously infected host cell, and it had undergone mutation in the process to become an oncogene (*v-Src*).

This Nobel Prize-winning finding was followed by a flood of discoveries of other viral oncogenes carried by retroviruses that cause cancer in nonhuman animals. Each such oncogene turned out to have a counterpart proto-oncogene in the normal vertebrate genome. As was the case for *Src*, these other oncogenes generally differed from their normal counterparts, either in structure or in level of expression.

But how did this relate to typical human cancers, in which retroviruses were not known to play a role? In an assay to identify human oncogenes, DNA was extracted from human tumor cells, broken into fragments, and introduced into cultured mouse cells. Occasional colonies of abnormally proliferating cells began to appear in the culture dish that showed a transformed phenotype, outgrowing the untransformed cells in the culture and piling up in layer upon layer (see Figure 20-15). Each colony was a clone originating from a single cell that had incorporated a DNA fragment that drove cancerous behavior. Once isolated and sequenced, the DNA fragments were found to contain a human version of a gene already known from study of a retrovirus that caused tumors in rats—an oncogene called *v-Ras*. The newly discovered oncogene was clearly derived by mutation from a normal human gene, one of a small family of proto-oncogenes called **Ras**. This discovery in the early 1980s of the same oncogene in human tumor cells and in an animal tumor virus was electrifying. The implication that cancers are caused by mutations in a limited number of cancer-critical genes transformed our understanding of the molecular biology of cancer.

As discussed in Chapter 15, normal Ras proteins are monomeric GTPases that help transmit signals from cell-surface receptors to the cell interior (see Movie 15.7). The *Ras* oncogenes isolated from human tumors contain point

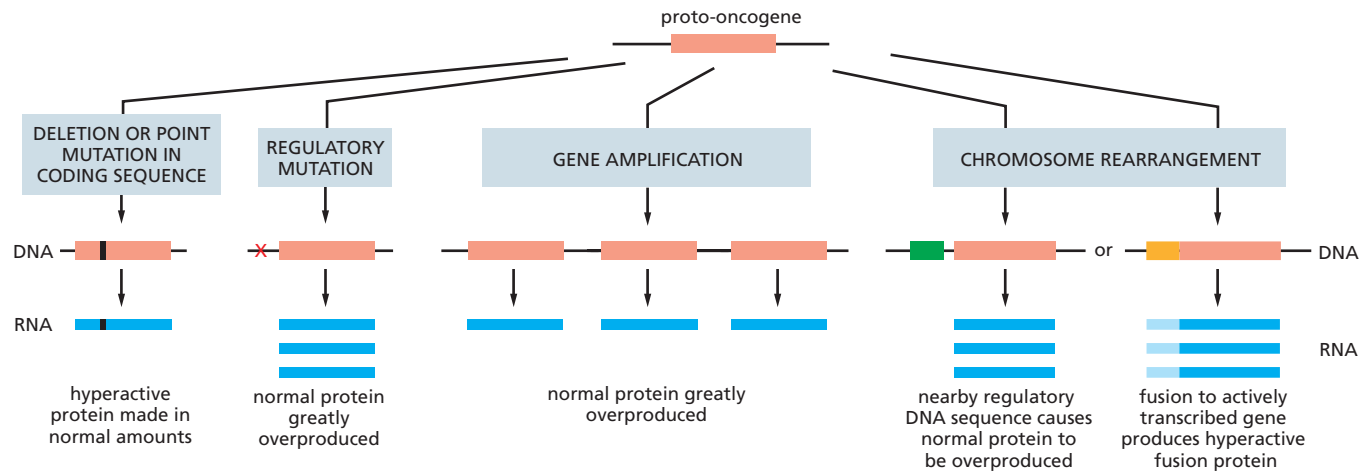


Figure 20–22 The types of accidents that can convert a proto-oncogene into an oncogene.

mutations that create a hyperactive Ras protein that cannot shut itself off by hydrolyzing its bound GTP to GDP. Because this makes the protein hyperactive, its effect is dominant; that is, only one of the cell's two gene copies needs to change to have an effect. One or another of the three human *Ras* family members is mutated in about 30% of all human cancers. *Ras* genes are thus among the most important of all cancer-critical genes.

Genes Mutated in Cancer Can Be Made Overactive in Many Ways

Figure 20–22 summarizes the types of accidents that can convert a proto-oncogene into an oncogene. (1) A small change in DNA sequence such as a point mutation or deletion may produce a hyperactive protein when it occurs within a protein-coding sequence or lead to protein overproduction when it occurs within a regulatory region for that gene. (2) Gene amplification events, such as those that can be caused by errors in DNA replication, may produce extra gene copies; this can lead to overproduction of the protein. (3) A chromosomal rearrangement—involving the breakage and rejoining of the DNA helix—may either change the protein-coding region, resulting in a hyperactive fusion protein, or alter the control regions for a gene so that a normal protein is overproduced.

As one example, the receptor for the extracellular signal protein *epidermal growth factor* (*EGF*) can be activated by a deletion that removes part of its extracellular domain, causing it to be active even in the absence of EGF (**Figure 20–23**). The mutant EGF receptor thus produces an inappropriate stimulatory signal, like a faulty doorbell that rings even when nobody is pressing the button. Mutations

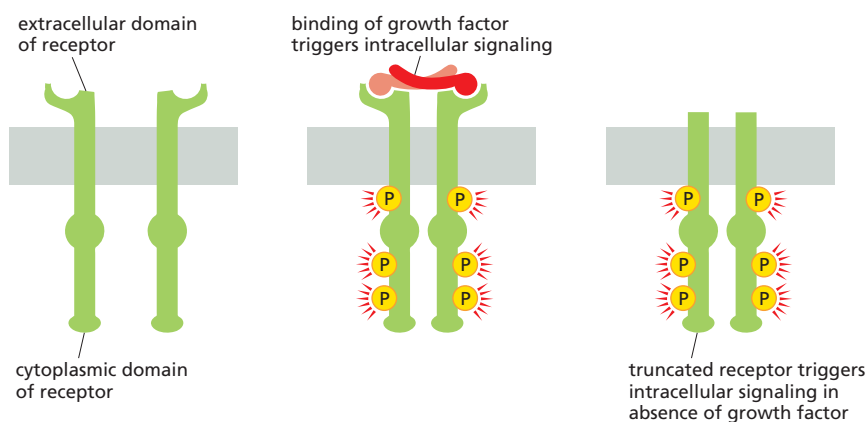


Figure 20–23 Mutation of the epidermal growth factor (EGF) receptor can make it active even in the absence of EGF, and consequently oncogenic. Normally, binding of EGF to the receptor's extracellular domain leads to phosphorylation of the intracellular domain, which activates signaling. Truncation of the extracellular domain leads to hyperphosphorylation and inappropriate activation. Other types of activating mutations are observed in different cancers.

of this type are frequently found in the most common type of human brain tumor, called glioblastoma.

As another example, the *Myc protein*, which acts in the nucleus to stimulate cell growth and division (see Chapter 17), generally contributes to cancer by being overproduced in its normal form. In some cases, the gene is amplified; that is, errors of DNA replication lead to the creation of large numbers of gene copies in a single cell. Or a point mutation can stabilize the protein, which normally turns over very rapidly. More commonly, the overproduction appears to be due to a change in a regulatory element that acts on the gene. For example, a chromosomal translocation can inappropriately bring powerful gene regulatory sequences next to the *Myc* protein-coding sequence, so as to produce unusually large amounts of *Myc* mRNA. Thus, in Burkitt's lymphoma, a translocation brings the *Myc* gene under the control of sequences that normally drive the expression of antibody genes in B lymphocytes. As a result, the mutant B cells tend to proliferate excessively and form a tumor. Different specific chromosome translocations are common in other cancers.

Studies of Rare Hereditary Cancer Syndromes First Identified Tumor Suppressor Genes

Identifying a gene that has been inactivated in the genome of a cancer cell requires a different strategy from finding a gene that has become hyperactive: one cannot, for example, use a cell transformation assay to identify something that simply is not there. The key insight that led to the discovery of the first tumor suppressor gene came from studies of a rare type of human cancer, **retinoblastoma**, which arises from cells in the retina of the eye that are converted to a cancerous state by an unusually small number of mutations. As often happens in biology, the discovery arose from examination of a special case, but it turned out to reveal a gene of widespread importance.

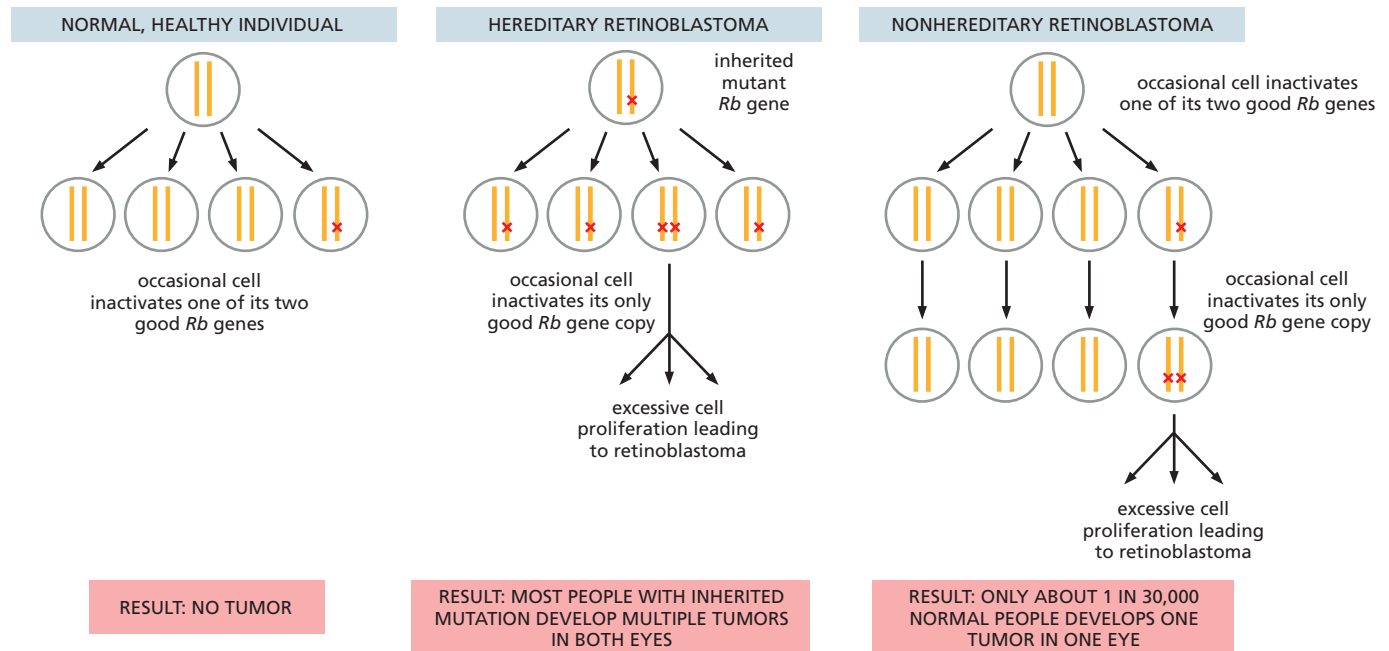
Retinoblastoma occurs in childhood, and tumors develop from neural precursor cells in the immature retina. About one child in 20,000 is afflicted. One form of the disease is hereditary, and the other is not. In the hereditary form, multiple tumors usually arise independently, affecting both eyes; in the nonhereditary form, only one eye is affected, and by only one tumor. A few individuals with retinoblastoma have a visibly abnormal karyotype, with a deletion of a specific band on chromosome 13 that, if inherited, predisposes an individual to the disease. Deletions of this same region are also encountered in tumor cells from some patients with the nonhereditary disease, which suggested that the cancer was caused by loss of a critical gene in that location.

By mapping the location of this chromosomal deletion, it was possible to identify the *Rb* gene. It was then discovered that those who suffer from the hereditary form of the disease have a deletion or loss-of-function mutation present in one copy of the *Rb* gene in every somatic cell. These cells are predisposed to becoming cancerous but do not do so if they retain one good copy of the gene. The retinal cells that are cancerous are defective in both copies of *Rb* because of a somatic event that has eliminated the function of the previously good copy.

In individuals with the nonhereditary form of the disease, by contrast, the noncancerous cells show no defect in either copy of *Rb*, while the cancerous cells have become defective in both copies. These nonhereditary retinoblastomas are very rare because they require two independent events that inactivate the same gene on two chromosomes in a single retinal cell lineage (**Figure 20–24**).

The *Rb* gene is also missing in several common types of sporadic cancer, including carcinomas of lung, breast, and bladder. These more common cancers arise by a more complex series of genetic changes than does retinoblastoma, and they make their appearance much later in life. But in all of them, it seems, loss of *Rb* function is frequently a major step in the progression toward malignancy.

The *Rb* gene encodes the **Rb** protein, which is a universal regulator of the cell cycle in almost all cells of the body (see Figure 17–59). It acts as one of the main brakes on progress through the cell-division cycle, and its loss can allow cells to enter the cell cycle inappropriately, as we discuss later.



Both Genetic and Epigenetic Mechanisms Can Inactivate Tumor Suppressor Genes

For tumor suppressor genes, it is their inactivation that is dangerous. This inactivation can occur in many ways, with different combinations of mishaps serving to eliminate or cripple both gene copies. The first copy may, for example, be lost by a small chromosomal deletion or inactivated by a point mutation due to a random error in DNA replication. The second copy is more commonly eliminated by a less specific mechanism that is likely to occur in cells progressing toward cancer that have become genetically unstable. For example, the chromosome carrying the remaining normal copy may be lost from the cell or damaged due to errors in chromosome segregation (see Figure 20–11) or the normal gene, along with neighboring genetic material, may be replaced by a mutant version through either a *mitotic recombination* event or a *gene conversion* that accompanies it (see pp. 305–306). Epigenetic changes provide another important way to permanently inactivate a tumor suppressor gene. Most commonly, the C nucleotides in CG sequences in its promoter may become methylated in a heritable manner, which can irreversibly silence the gene in a cell and in all of its progeny (see pp. 435–436). **Figure 20–25**

Figure 20–24 The genetic mechanisms that cause retinoblastoma. In the hereditary form, all cells in the body lack one of the normal two functional copies of the *Rb* tumor suppressor gene, and tumors arise from a clone of cells where the remaining copy is lost or inactivated by a somatic event (either mutation or epigenetic silencing). In the nonhereditary form, all cells initially contain two functional copies of the gene, and the tumor arises because both copies are lost or inactivated through the coincidence of two somatic events in a single line of cells.

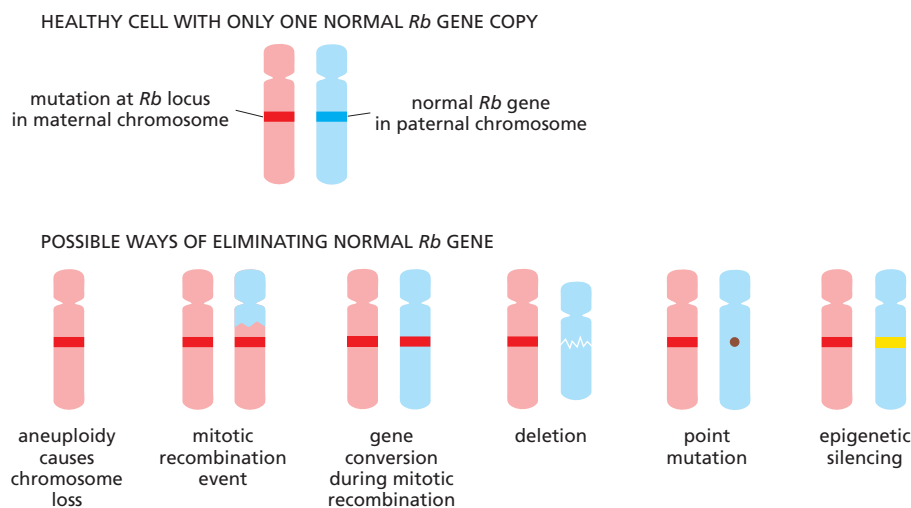


Figure 20–25 Six ways of inactivating the remaining good copy of a tumor suppressor gene through changes in DNA sequence or an epigenetic mechanism. A cell that is defective in only one of its two copies of a tumor suppressor gene—for example, the *Rb* gene—usually behaves as a normal, healthy cell; the diagrams show how this cell may lose the function of the other gene copy as well and thereby progress toward cancer.

summarizes the range of ways in which the remaining good copy of a tumor suppressor gene can be lost through a DNA sequence or epigenetic change, using the *Rb* gene as an example.

Systematic Sequencing of Cancer Cell Genomes Has Transformed Our Understanding of the Disease

Methods such as those we have described above shone a spotlight on a set of cancer-critical genes that were identified in a piecemeal fashion. Meanwhile, the rest of the cancer cell genome remained in darkness: it was a mystery how many other mutations might lurk there, of what types, in which varieties of cancer, at what frequencies, with what variations from individual to individual, and with what consequences. With the sequencing of the human genome and the dramatic advances in DNA sequencing technology (see Figure 8–44), it has become possible to see the whole picture—to view cancer cell genomes in their entirety. This transforms our understanding of the disease.

Cancer cell genomes can be scanned systematically in several different ways. At one extreme—the most costly, but no longer prohibitively so—one can determine a tumor’s complete genome sequence. More cheaply, one can focus just on the 21,000 or so genes in the human genome that code for protein (the so-called *exome*), looking for mutations in the cancer cell DNA that alter the amino acid sequence of the product or prevent its synthesis (Figure 20–26). In addition, DNA sequencing enables a survey of the genome for regions that have undergone deletion or duplication to reveal copy number variations and the loss or gain of chromosomes (aneuploidy). The genome can also be scanned for epigenetic changes to reveal changes in methylation patterns of DNA or other changes that accompany transcriptional silencing or activation without affecting DNA sequence. Systematic sequencing of RNAs can reveal alterations in levels of gene expression by analysis of mRNAs (see Figure 7–5), as well as levels of regulatory noncoding RNAs. Finally, to measure the expression of known **cancer-critical genes** directly by detecting their protein products, tumor cell lysates can be surveyed quantitatively using mass spectrometry. These approaches involve comparing cancer cells with normal controls—ideally, noncancerous cells originating in the same tissue and from the same individual.

A combination of the high-throughput methods described above has been applied to more than 10,000 tumors spanning 33 cancer types through a consortium called The Cancer Genome Atlas, which is coordinated by the National Institutes of Health. Other international and freely available resources, including COSMIC, the Catalog of Somatic Mutations in Cancer, are continually compiling new cancer data. As we discuss below, these large-scale projects have shed important new light on what types of alterations are present in a cancer genome, what kinds of genes or pathways are altered across and within cancer types, how

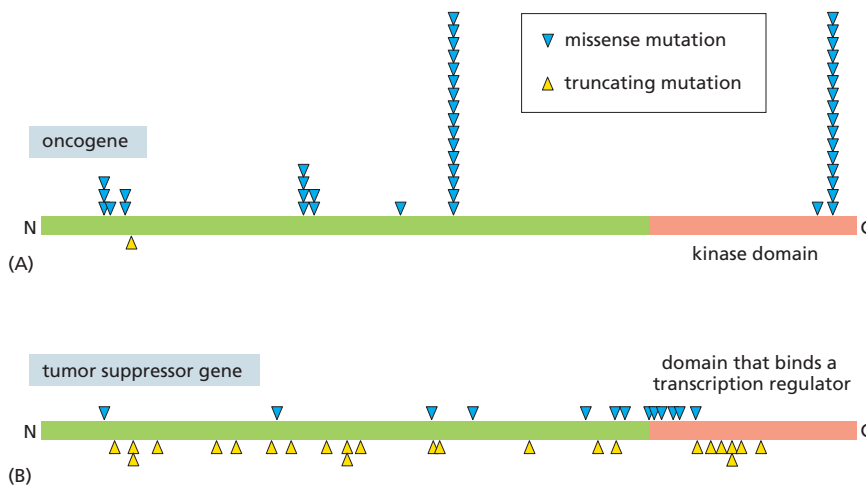


Figure 20–26 The distinct types of DNA sequence changes found in oncogenes compared to those in tumor suppressor genes. In this diagram, mutations that change an amino acid are denoted by blue arrowheads, whereas mutations that truncate the polypeptide chain are marked by yellow arrowheads. (A) As in this example (the *PIK3CA* gene, which produces a phosphoinositide 3-kinase subunit), oncogene mutations can be detected by the fact that the same nucleotide change is repeatedly found among the missense mutations in a gene. (B) As in this example (the *Rb* gene), for tumor suppressor genes missense mutations that abort protein synthesis by creating stop codons predominate. Note that only a few of the possible mutations in a protein-coding sequence are likely to be activating, while inactivation can be a consequence of missense, nonsense, and frameshift mutations. (Adapted from B. Vogelstein et al., *Science* 339:1546–1558, 2013.)

heterogeneous the cancer cells within a tumor are, and how the patterns of alterations evolve over time during disease progression and treatment.

Many Cancers Have an Extraordinarily Disrupted Genome

Cancer genome analysis reveals, first of all, the scale of gross genetic disruption in cancer cells. This varies greatly among cancer types and from one cancer patient to another, both in severity and in character. As discussed previously (see p. 1169) the chromosomal karyotype may appear normal, yet is riddled with numerous point mutations in individual genes due to a failure of the repair mechanisms that normally correct errors in the replication or maintenance of DNA sequences. Frequently, however, the karyotype is extremely disrupted, with many chromosomal breaks and rearrangements and chromosome sequences that are completely scrambled, indicating that extensive fragmentation of the DNA occurred followed by random re-assembly (as in chromothripsis; see Figure 20–11). From the pattern of changes, one can infer that disruptive events have occurred repeatedly during the evolution of the tumor, with a progressive increase in genetic disorder.

A common feature of many human cancers is aneuploidy—a change in chromosome karyotype from the normal number (46). A study of chromosome-level gain or loss across a panel of more than 10,000 cancer genomes revealed that most cancer types harbor a characteristic pattern of chromosomal abnormalities, with different arms or whole chromosomes altered at different frequencies. In many cases, the entire set of chromosomes was doubled or quadrupled because of complete failures in cell division, generating polyploid cells that subsequently experience additional chromosome gain and loss. Almost 90% of all the cancers surveyed showed some level of aneuploidy. **Figure 20–27** depicts a subset of the data showing selected tumor types, each of which displays a unique pattern of aneuploidy. Thus, for example, whereas 74% of thyroid carcinomas displayed a normal karyotype, less than 5% of glioblastoma and cervical carcinoma tumors possessed the normal chromosome number. Although the degree of aneuploidy is often characteristic of a cancer class, heterogeneity is also apparent. For example, one subset of glioblastomas is characterized by gain of chromosome 7 and loss of chromosome 10, but this combination of defects is not present in all cases, indicating that there are multiple pathways of genome disorder that can lead to different cancer subtypes.

Cancer genome studies also illuminate the underlying defects that bring about genome disruption. In ovarian cancers, for example, chromosome breaks, translocations, and deletions are very common, and these aberrations correlate with a high frequency of mutations and epigenetic silencing in the genes needed for repair of DNA double-strand breaks by homologous recombination, especially *Brca1* and *Brca2* (see pp. 300–301). In a subset of endometrial cancers, on the other hand, one instead finds many point mutations scattered throughout the genome, which could for example be caused by mutation of enzymes required for proofreading during DNA replication (see pp. 267–269). Thus, different cancers and cancer subtypes possess highly variable mutation rates, ranging from one base substitution per exome in some pediatric cancers to thousands of mutations per exome in mutagen-induced malignancies such as lung cancer and melanoma.

Epigenetic and Chromatin Changes Contribute to Most Cancers

So far we have focused primarily on the mutations in cancer-critical genes that contribute directly to the hallmarks of cancer. Equally important to the genesis of cancer, however, are changes that modify gene expression epigenetically, without altering the DNA sequence. As discussed previously, increased methylation of DNA in the promoter region of a gene such as *Rb* can permanently silence it. In addition, reversible covalent modifications such as methylation and acetylation also occur on the histone proteins that package DNA into chromatin. These

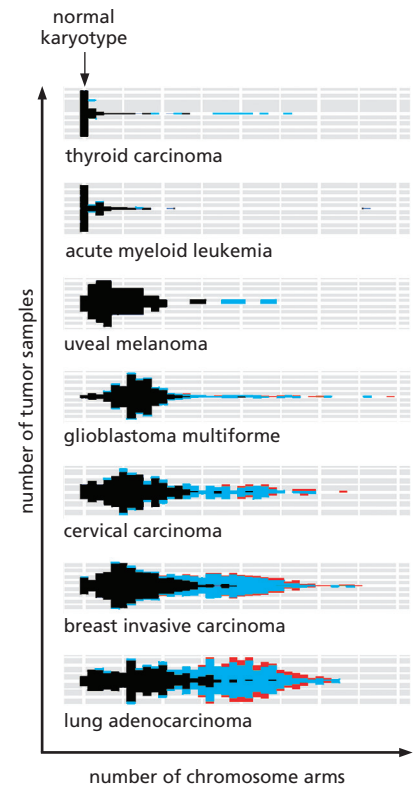


Figure 20–27 The prevalence of aneuploidy among different tumor types. For each tumor type, the total number of chromosome arms detected is plotted on the X axis. The number of chromosome arms in a normal karyotype is indicated. Genome doubling status is also shown (black, not double; blue, one genome doubling; red, two or more genome doublings). The number of tumor samples possessing the respective arm numbers is represented by the length of the bar on the Y axis for each tumor type. Note that samples from some tumor types predominantly possess a normal karyotype, while samples from other tumor types display extreme heterogeneity with a dramatic increase in chromosome arm numbers because of extensive aneuploidy. (Adapted from A.M. Taylor et al., *Cancer Cell* 33:676–689, 2018.)

histone marks modulate gene expression by altering the conformation and accessibility of chromatin to regulatory factors such as transcription factors and chromatin remodeling complexes (see Chapter 7). An important finding of large-scale genome sequencing projects is that roughly 50% of human cancers harbor mutations in chromatin proteins, which includes mutations in enzymes that add, remove, or recognize covalent marks on histones and DNA. Many of the resulting defects have the potential to cause heritable, epigenetic changes that modulate gene expression and contribute to tumorigenesis.

Frequently, genetic and epigenetic changes cooperate to cause cancer (Figure 20–28). In some cases, epigenetic changes precede characteristic oncogenic mutations and can even lead to them, such as when epigenetic silencing of genes required for the repair of DNA damage acts to increase mutation rates. In other cases, an accidental change in DNA sequence can disrupt chromatin and epigenetic regulation. One example is a gain-of-function mutation in the gene encoding isocitrate dehydrogenase (IDH), which is known to be a frequent initiating event in several cancers including glioma, leukemia, and other tumors. The mutant IDH produces elevated levels of a metabolite, 2-hydroxyglutarate, which acts to inhibit enzymes that demethylate DNA. Therefore, DNA in IDH-mutant cells becomes hypermethylated at random sites in the genome. Just as for DNA mutations, those epigenetic changes that provide a selective advantage in growth or survival will allow the affected cell and its descendants to persist and accumulate further changes, thereby contributing to cancer progression.

The indirect effect of mutant IDH on DNA hypermethylation illustrates another important point: metabolic conditions affect the status of chromatin, which in turn can influence gene expression. Notably, only a minority of cancers with aberrant DNA methylation can be explained by an underlying genetic event. These observations suggest that environmental conditions, which may not themselves be mutagenic, can provide a source of epigenetic changes. Many DNA- and histone-modifying enzymes require metabolites as cofactors, providing a potential link between known risk factors for human cancer, such as diet and inflammation, and cancer development.

In addition to modulating the expression of individual genes, epigenetic changes can also have more global effects on the chromatin state, thereby influencing gene networks that underlie programs of cellular behavior. For example, different patterns of gene expression determined by epigenetic marks could restrict the ability of a cell to activate apoptosis or prevent it from exiting the cell cycle and undergoing differentiation. Evidence for such changes in cancer comes from characterization of rare pediatric brain tumors that lack somatic mutations and instead display aberrant DNA methylation profiles.

It is crucial to remember that altered patterns of gene activation and silencing due to epigenetic regulation are a normal feature of cellular differentiation, enabling the same genome to produce and maintain a multitude of different cell types through precise developmental programs (see Chapter 21). Thus, as for other properties of cancer cells, epigenetic changes do not represent new biology, but rather the subversion of existing cellular mechanisms.

Hundreds of Human Genes Contribute to Cancer

Among the billions of somatic mutations now identified in cancer cells, how can we discover which of them are **drivers** of cancer; that is, causal factors in the development of the disease? Clearly, most will be merely **passengers**—mutations that happen to have occurred in the same cell as the driver mutations, thanks to genetic instability, but are irrelevant to the development of the cancer. One criterion is frequency of occurrence. Driver mutations affecting a gene that play a part in the disease will be seen repeatedly, in many different individuals with a particular type of cancer. In contrast, a passenger mutation that confers no selective advantage on the cancer cell is likely to be encountered only rarely.

By compiling the genome sequence data for different types of cancer, each with its own set of identified driver mutations, we can develop a comprehensive

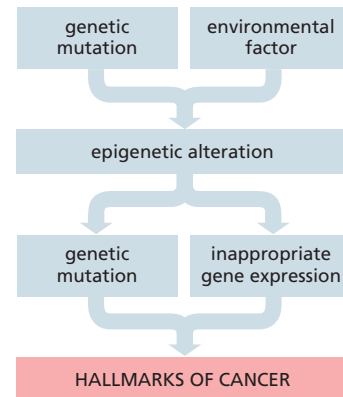


Figure 20–28 Epigenetic and genetic mechanisms can cooperate to promote the evolution of cancer. Both genetic mutations and environmental factors (such as metabolic state) can lead to epigenetic changes. As indicated, these epigenetic changes can in turn increase the rate at which genetic as well as further epigenetic changes accumulate—thereby speeding the process of cancer progression by altering gene expression in ways that promote cancer cell proliferation and are inherited in clones of cells.

catalog of those genes that are strongly suspected to be cancer-critical for at least one type of tumor. Current estimates put the total number of such genes at around 300, about 1% of the genes in the human genome. These cancer-critical genes are amazingly diverse, revealing an unexpected breadth of mechanisms. Whereas some of the new cancer genes encode classical signaling and cell-cycle proteins that might be anticipated to become mutated in cancer cells, others populate new and sometimes surprising categories. These include functions as diverse as metabolism, chromatin biology, RNA splicing, protein homeostasis, and cell differentiation. It is clear that alterations in any of these processes can contribute, in one tissue or another, to the evolution of cells with the cancerous properties that were listed on page 1172.

Clearly, the molecular changes that cause cancer are complex. As we now explain, however, the complexity is not quite as daunting as it might initially seem.

Disruptions in a Handful of Key Pathways Are Common to Many Cancers

Some genes, like *Rb* and *Ras*, are mutated in many cases of cancer and in cancers of many different types. The involvement of genes such as *Rb* and *Ras* in cancer is no surprise, now that we understand their normal functions: they control fundamental processes of cell division and growth. But even these common culprits feature in considerably less than half of individual cases. What is happening to the control of these processes in the many cases of cancer where, for example, *Rb* is intact or *Ras* is not mutated? What part do mutations in the hundreds of other cancer-critical genes play in the development of the disease? With our increasing knowledge of the normal functions of the genes in the human genome, it is becoming easier to detect patterns in the cataloged driver mutations and to give some simplifying answers to these questions.

In order to clarify the type of patterns revealed, consider the deadly disease *glioblastoma*—the most common type of human brain cancer. Analysis of the genomes of tumor cells from 91 individuals identified a total of at least 79 genes that were mutated in more than one individual. The normal functions of most of these genes were known or could be guessed, allowing them to be assigned to specific biochemical or regulatory pathways. Three functional groupings stood out, accounting for 21 of the recurrently mutated genes. One group consists of genes in the *Rb pathway* (that is, *Rb* itself, along with genes that directly regulate *Rb*); this pathway governs initiation of the cell-division cycle. Another consists of genes in the same regulatory subnetwork as *Ras*—referred to as the *RTK/Ras/PI 3-kinase pathway*, after three of its core components; this pathway serves to transmit signals for cell growth and cell division from outside the cell to the cell interior. A third grouping consists of genes in a pathway regulating responses to stress and DNA damage—the *p53 pathway*. We shall have more to say about these pathways below.

For the 91 glioblastomas, 74% had identifiable mutations in all three pathways. If one were to trace these three pathways further upstream and include all the components, known and unknown, on which they depend, this percentage would almost certainly be even higher. In other words, in almost every case of glioblastoma, there are mutations that disrupt each of three fundamental controls: the control of cell growth, the control of cell division, and the control of responses to stress and DNA damage.

Strikingly, in any given tumor-cell clone, there is a strong tendency for no more than one gene to be mutated in each pathway. Evidently, what matters for tumor evolution is the disruption of the control mechanism, and not the genetic means by which that is achieved. Thus, for example, in an individual whose tumor cells have no mutation in *Rb* itself, there is generally a mutation in some other component of the *Rb pathway*, producing a similar biological effect. This indicates that there is no further selection for a mutant in a pathway if the pathway has been disabled by a mutation elsewhere.

Similar patterns are seen in other types of cancers. A survey of many specimens of the major variety of ovarian cancer, for example, identified 67% of individuals

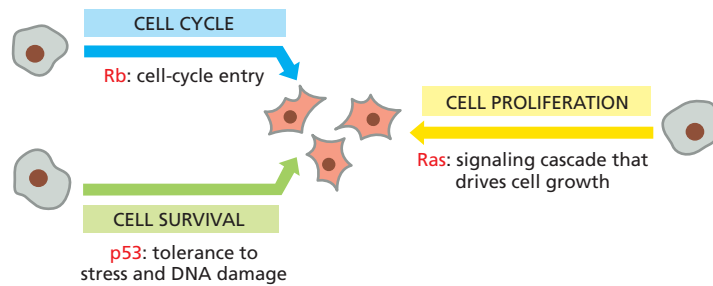


Figure 20–29 The three major cellular pathways that contribute to tumorigenesis. The specific examples listed here are described in this chapter. Cell-cycle events controlled by Rb are described in detail in Chapter 17.

as having mutations in the Rb pathway, 45% in the Ras/PI 3-kinase pathway (defined more narrowly than in the glioblastoma study), and more than 96% in the p53 pathway. Allowing for additional pathway components not included in the analysis, it seems that most cases of this type of cancer, too, have mutations disrupting the same three controls, leading to misregulated cell growth, misregulated cell proliferation, and an abnormal disregard of stress and DNA damage.

It seems that these three fundamental controls are subverted in one way or another in virtually every type of cancer. However, because specialized tissues can depend on different mechanisms to relay environmental signals to the core control machinery, these controls are open to subversion in a different set of ways in different types of cancers. In fact, one can find examples of driver mutations in practically all the major signaling pathways through which cells communicate during development and tissue maintenance (discussed in Chapters 15, 21, and 22).

Figure 20–29 outlines the three central pathways just described, abbreviating them as cell cycle, cell proliferation, and cell survival. We have previously discussed Rb (see pp. 1182–1183) and devoted an entire chapter to cell-cycle controls (Chapter 17). Some important details of the other two control pathways in Figure 20–29 are reviewed next.

Mutations in the PI 3-kinase/Akt/mTOR Pathway Drive Cancer Cells to Grow

Cell proliferation is not simply a matter of progression through the cell cycle; it also requires coordinated cell growth, which involves complex anabolic processes through which the cell synthesizes all the necessary macromolecules from small-molecule precursors (Figure 20–30). Cancer depends, therefore, not only on a loss of restraints on cell-cycle progression but also on a disrupted control of cell growth.

Downstream of RTK/Ras activation, the PI 3-kinase/Akt/mTOR intracellular signaling pathway is critical for cell growth control. As described in Chapter 15, various extracellular signal proteins, including insulin and insulin-like growth factors, normally activate this pathway. In cancer cells, however, the pathway is activated by mutation so that the cell can grow in the absence of such signals. The resulting abnormal activation of the protein kinases Akt and mTOR not only stimulates protein synthesis but also greatly increases both glucose uptake and the production of the acetyl CoA in the cytosol required for cell lipid synthesis, as outlined in Figure 20–30B.

The abnormal activation of the PI 3-kinase/Akt/mTOR pathway, which normally occurs early in the process of tumor progression, helps to explain the excessive rate of glycolysis that is observed in tumor cells, known as the Warburg effect, as discussed earlier (see Figure 20–18). As expected from our previous discussion, cancers can activate this pathway in many different ways. Thus, for example, a growth factor receptor can become abnormally activated, as in Figure 20–23. Also very common in cancers is the loss of the phosphatase and tensin homolog (PTEN) phosphatase that normally functions to counteract this pathway. PTEN suppresses the PI 3-kinase/Akt/mTOR pathway by dephosphorylating the phosphatidylinositol 3,4,5-trisphosphate [PI(3,4,5)P₃]

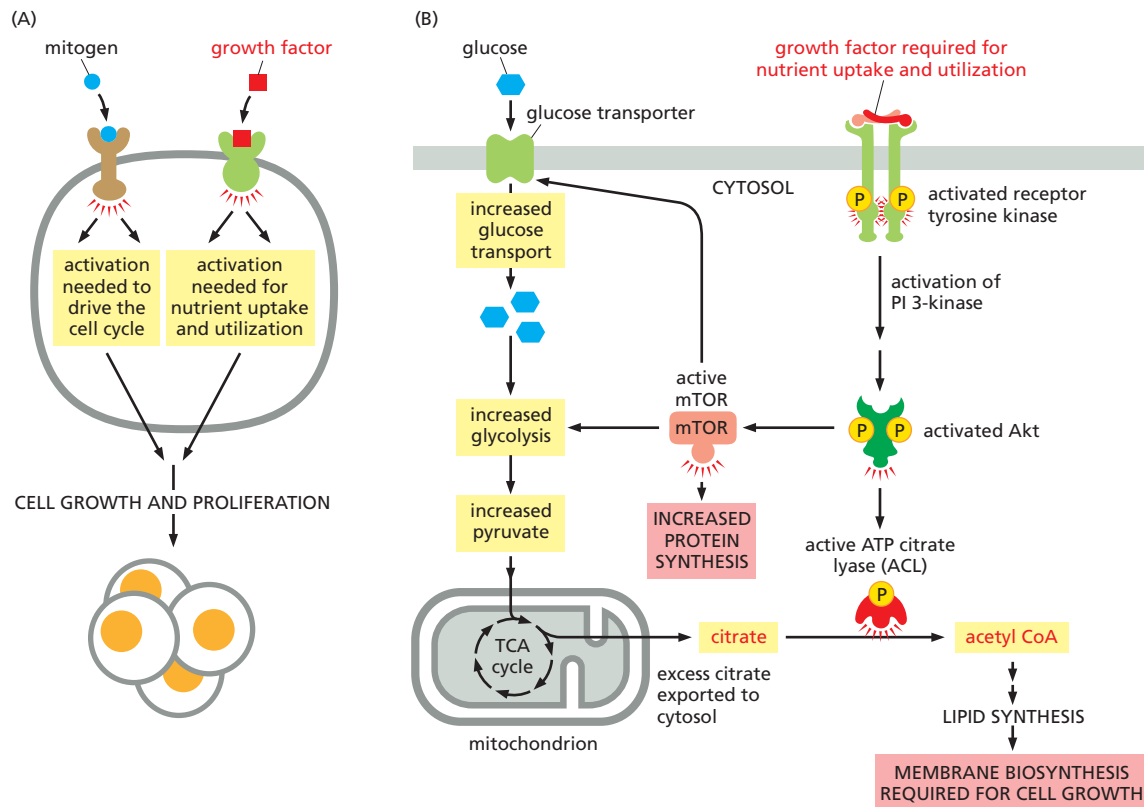


Figure 20–30 Cells seem to require two types of signals to proliferate. (A) In order to multiply successfully, most normal cells require both extracellular signals that drive cell-cycle progression (shown here as *blue* mitogen) and extracellular signals that drive cell growth (shown here as *red* growth factor). How mitogens activate signaling through the Rb pathway to drive entry into the cell cycle is described in Figure 17–59. (B) Diagram of the signaling system involving activation of Akt and mTOR that drives cell growth through greatly stimulating glucose uptake and utilization, including a conversion of the excess citric acid produced from sugar intermediates into the acetyl CoA that is needed in the cytosol for lipid synthesis and new membrane production. As indicated, protein synthesis is also increased. This system becomes abnormally activated early in tumor progression. TCA cycle indicates the tricarboxylic acid cycle (citric acid cycle).

molecules that the PI 3-kinase generates (see p. 920). *PTEN* is commonly mutated in tumors.

Mutations in the p53 Pathway Enable Cancer Cells to Survive and Proliferate Despite Stress and DNA Damage

That cancer cells must break the normal rules governing cell growth and cell division is obvious: that is part of the definition of cancer. It is not so obvious why cancer cells should also be abnormal in their response to stress and DNA damage, and yet this too is an almost universal feature. The gene that lies at the center of this response, the *p53* gene, is mutated in about 50% of all cases of cancer—a higher proportion than for any other known cancer-critical gene. When we include with *p53* the other genes that are closely involved in its function, we find that most cases of cancer harbor mutations in the *p53* pathway. Why should this be? To answer, we must first consider the normal function of this pathway.

In contrast to Rb, most cells in the body contain very little **p53** protein under normal conditions: although the protein is synthesized, it is rapidly degraded. Mice in which both copies of the gene have been deleted or inactivated typically appear normal in all respects except one—they universally develop cancer before 10 months of age. These observations suggest that *p53* has a function that is required only in special circumstances. In fact, cells raise their concentration of *p53* protein in response to a whole range of conditions that have only one obvious thing in common: they are, from the cell's point of view, pathological, putting the cell in danger of death or serious injury. These conditions include DNA damage,

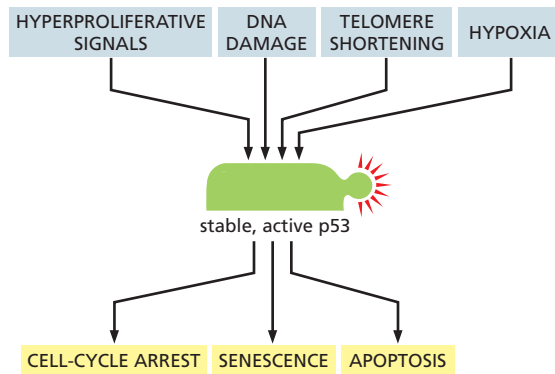


Figure 20–31 Modes of action of the p53 tumor suppressor. The p53 protein is a cellular stress sensor. In response to hyperproliferative signals, DNA damage, hypoxia, telomere shortening, and various other stresses, the p53 levels in the cell rise. As indicated, this may either arrest cell cycling in a way that allows the cell to adjust and survive, trigger cell suicide by apoptosis, or cause cell *senescence*—an irreversible cell-cycle arrest that stops damaged cells from dividing.

which puts the cell at risk from a faulty genome; telomere loss or shortening, also dangerous to the integrity of the genome; hypoxia, which deprives the cell of the oxygen it needs to maintain mitochondrial respiration; osmotic stress, which causes the cell to swell or shrivel; and oxidative stress, which generates dangerous levels of highly reactive free radicals.

Yet another form of stress that can activate the p53 pathway arises, it seems, when regulatory signals are so intense or uncoordinated as to drive the cell beyond its normal limits and into a danger zone where its mechanisms of control and coordination break down, as in an engine driven too fast. The p53 concentration rises, for example, when *Myc* is overexpressed to oncogenic levels.

All these circumstances call for desperate action, which may take either of two forms: the cell can block any further progress through the division cycle in order to take time out to repair or recover from the pathological condition or it can accept that it must die, and do so in a way that minimizes damage to the organism. A good death, from this point of view, is a death by apoptosis. In apoptosis, the cell is phagocytosed by its neighbors and its contents are efficiently recycled. A bad death is a death by necrosis. In necrosis, the cell bursts or disintegrates and its contents are spilled into the extracellular space, inducing inflammation.

The p53 pathway, therefore, behaves as a sort of antenna, sensing the presence of a wide range of dangerous conditions and, when any are detected, triggering appropriate action—either a temporary or permanent arrest of cell cycling or suicide by apoptosis (Figure 20–31). These responses serve to prevent deranged cells from proliferating. Cancer cells are indeed generally deranged, and their survival and proliferation thus depend on inactivation of the p53 pathway. If the p53 pathway were active in them, they would be halted in their tracks or die (Movie 20.6). For example, if the p53 pathway is functional, a cell with unrepaired DNA damage will stop dividing or die; it cannot proliferate.

The p53 protein performs its job mainly by acting as a transcription regulator (see Movie 17.8). Indeed, the most common mutations observed in p53 in human tumors are in its DNA-binding domain, where they cripple the ability of p53 to bind to its DNA target sequences. As discussed in Chapter 17, the p53 protein exerts its inhibitory effects on the cell cycle, in part at least, by inducing the transcription of *p21*, which encodes a protein that binds to and inhibits the cyclin-dependent kinase (Cdk) complexes required for progression through the cell cycle. By blocking the kinase activity of these Cdk complexes, the p21 protein prevents the cell from progressing through S phase and replicating its DNA.

The mechanism by which p53 induces apoptosis includes stimulation of the expression of many pro-apoptotic genes, as described in Chapter 18.

Studies Using Mice Help to Define the Functions of Cancer-critical Genes

The ultimate test of a gene's role in cancer has to come from investigations in the intact, mature organism. The most favored organism for experimental studies is the mouse. To explore the function of a candidate oncogene or tumor suppressor

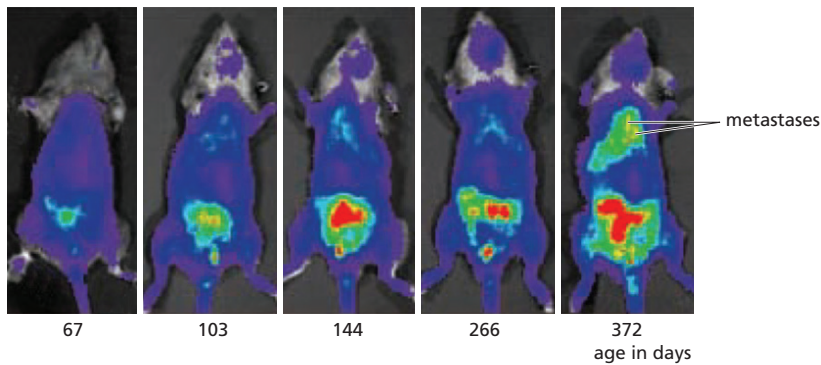
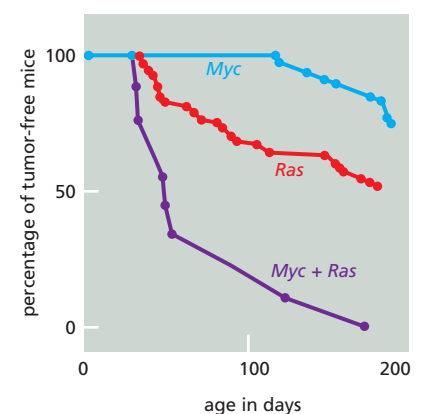


Figure 20-32 Monitoring tumor growth and metastasis in a mouse with a luminescent reporter. A mouse was genetically engineered in a way that allows both copies of its *PTEN* tumor suppressor gene to be inactivated in the prostate gland, simultaneously with the prostate-specific activation of a gene engineered to produce the enzyme luciferase (derived from fireflies). After an injection of luciferin (the substrate molecule for luciferase) into the mouse's bloodstream, the cells in the prostate emit light and can be detected by their bioluminescence in a live mouse, as seen in the 67-day-old animal at the left. Cells lacking the *PTEN* phosphatase enzyme contain elevated amounts of the Akt activator, $PI(3,4,5)P_3$, and this causes the prostate cells to proliferate abnormally, progressing over time to form a cancer. In this way, the process of metastasis could be followed in the same animal over the course of a year. The light intensity in these experiments is proportional to the number of prostate-cell descendants, increasing from light blue to green, to yellow, to red in this representation. (Adapted from C.-P. Liao et al., *Cancer Res.* 67:7525–7533, 2007. With permission from the American Association for Cancer Research.)

gene, one can make a transgenic mouse that overexpresses it or a knockout mouse that lacks it. Using the techniques described in Chapter 8, one can engineer mice in which the misexpression or deletion of the gene is restricted to a specific set of cells, or in which expression of the gene can be switched on at will at a chosen point in time, or both, to see whether and how tumors develop. Moreover, to follow the growth of tumors from day to day in the living mouse, the cells of interest can be genetically marked and made visible by expression of a fluorescent or luminescent reporter (Figure 20-32). In these ways, one can begin to clarify the part that each cancer-critical gene plays in cancer initiation or progression.

Transgenic mouse studies confirm, for example, that a single oncogene is generally not enough to turn a normal cell into a cancer cell. Thus, in mice engineered to express a *Myc* or *Ras* oncogenic transgene, some of the tissues that express the oncogene may show enhanced cell proliferation, and, over time, occasional cells will undergo further changes to give rise to cancers. Most cells expressing the oncogene, however, do not give rise to cancers. Nevertheless, from the point of view of the whole animal, the inherited oncogene is a serious menace because it creates a high risk that a cancer will arise somewhere in the body. Mice that express both *Myc* and *Ras* oncogenes develop cancers earlier and at a much higher rate than mice that express either gene alone (Figure 20-33); but, again, the cancers originate as scattered, isolated tumors among noncancerous

Figure 20-33 Oncogene collaboration in transgenic mice. The graphs show the incidence of tumors in three types of transgenic mouse strains, one carrying a *Myc* oncogene, one carrying a *Ras* oncogene, and one carrying both oncogenes. For these experiments, two lines of transgenic mice were first generated. One carries an inserted copy of an oncogene created by fusing the proto-oncogene *Myc* with the mouse mammary tumor virus regulatory DNA (which then drives *Myc* overexpression in the mammary gland). The other line carries an inserted copy of the *Ras* oncogene under the control of the same regulatory element. Both strains of mice develop tumors much more frequently than normal, most often in the mammary or salivary glands. Mice that carry both oncogenes together are obtained by crossing the two strains. These hybrids develop tumors at a far higher rate still, much greater than the sum of the rates for the two oncogenes separately. Nevertheless, the tumors arise only after a delay and only from a small proportion of the cells in the tissues where the two genes are expressed. Further accidental changes, in addition to the two oncogenes, are apparently required for the development of cancer. (After E. Sinn et al., *Cell* 49:465–475, 1987.)



cells. Thus, even cells expressing these two oncogenes must undergo further, randomly generated changes to become cancerous. This strongly suggests that multiple mutations are required for tumorigenesis, as supported by a great deal of other evidence discussed earlier. Experiments using mice with deletions of tumor suppressor genes lead to similar conclusions.

Cancers Become More and More Heterogeneous as They Progress

Although useful to study the function and interaction of cancer driver genes *in vivo*, there are important limitations to mouse models of human cancer. Whereas a mouse tumor is small and arises over a period of months or even days, a human tumor may be the size of a mouse or larger and will often have been growing for decades. From simple histology looking at stained tissue sections, it is clear that some tumors contain distinct sectors, all clearly cancerous, but differing in appearance because they differ genetically or epigenetically: the cancer cell population is heterogeneous. Evidently, within the initial clone of cancerous cells, additional mutations have arisen and conferred selective advantages, such as increased growth rates, creating diverse subclones. Today, the ability to analyze cancer genomes lets us look much deeper into the process. Comparison of samples from different regions of a tumor and the metastases it has spawned reveal a classic picture of Darwinian evolution, occurring on a time scale of months or years rather than millions of years, but governed by the same rules of natural selection.

One approach to examine the development of tumor heterogeneity takes advantage of human cells grown outside the body, termed “organoids” (described in Chapter 22). Unlike the immortalized cancer cell lines described previously that have adapted to an artificial, two-dimensional lifestyle (see Figure 20–15), organoids are grown from adult stem cells under more physiological conditions within a three-dimensional matrix. In this environment, a single stem cell can proliferate and differentiate into a self-organizing structure that resembles a miniature, simplified version of the organ from which it was taken, with realistic microanatomy. One study applied this organoid system to examine the genetic heterogeneity within individual colorectal tumors (Figure 20–34A). Organoids were grown from single cells isolated from different regions of a tumor as well as from neighboring, normal tissue. Genome analysis revealed detailed patterns of mutations in each organoid that indicated how closely it was related to the others, and from these data one could draw up a family tree. Those organoids derived from the same tumor site bore the most similarities, and all possessed the same set of cancer-driving mutations that arose in their common ancestor within the trunk of the tree, corresponding to the early stages of tumor growth.

Clearly, cancer cells are constantly mutating, multiplying, competing, evolving, and diversifying as they exploit new ecological niches and react to the treatments that are used against them (Figure 20–34B). Diversification accelerates as they metastasize and colonize new territories, where they encounter new selection pressures. The longer the evolutionary process continues, the harder it becomes to catch them all in the same net and kill them.

Colorectal Cancers Evolve Slowly Via a Succession of Visible Changes

At the beginning of this chapter, we saw that most cancers develop gradually from a single aberrant cell, progressing from benign to malignant tumors by the accumulation of a number of independent genetic and epigenetic changes. We have discussed what some of these changes are in molecular terms and have seen how they contribute to cancerous behavior. We now examine one of the common human cancers more closely, using it to illustrate and enlarge upon some of the general principles and molecular mechanisms we have introduced. We take **colorectal cancer** as our example.

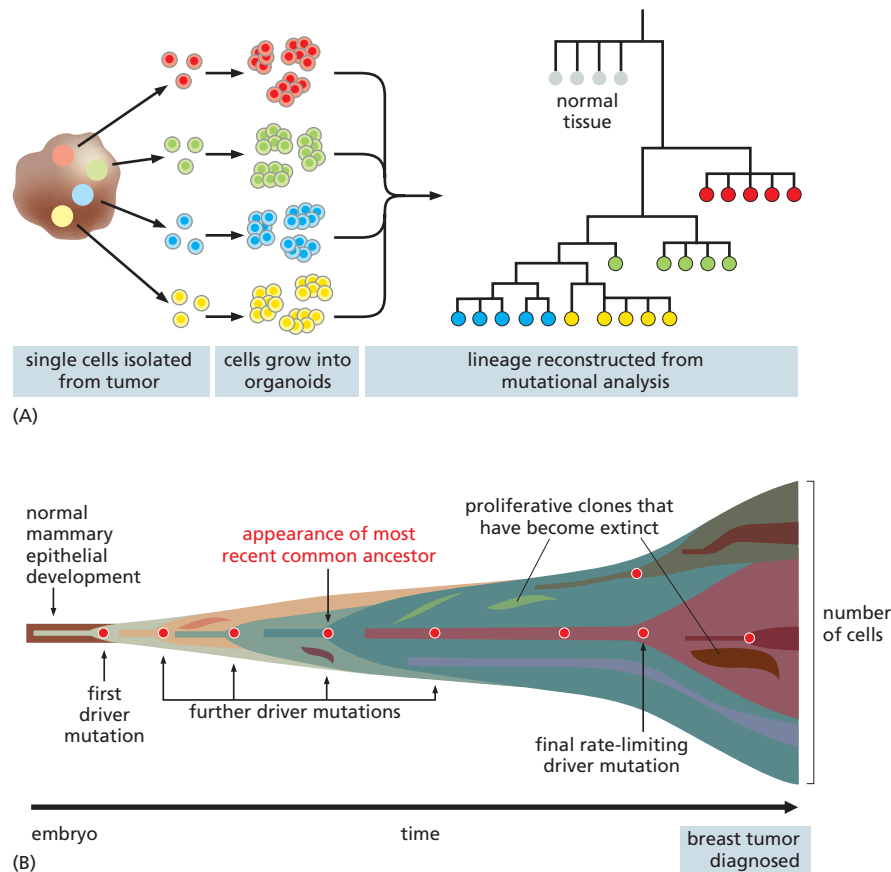


Figure 20-34 How cancers progress as a series of subclones. (A) Cancer analysis using organoids. Single cells were isolated from four different regions of a human colorectal tumor (color coded) as well as from nearby healthy tissue (not shown). These cells were then used to produce organoids, allowing large amounts of material to be obtained for analysis. DNA sequencing was used to determine the mutations in each organoid. These data then enabled construction of a phylogenetic tree that reveals the order in which the mutations emerged in the original tumor. (B) A depiction of how driver mutations are thought to cause cancer progression over long periods of time, before producing a large enough clone of proliferating cells to be detected as a tumor. The data indicate that driver mutations occur only rarely in a background of long-lived subclones of cells that continually accumulate passenger mutations without gaining a growth advantage. (A, adapted from C.J. Kuo and C. Curtis, *Nature* 556:441–442, 2016. With permission from Springer Nature. B, adapted from S. Nik-Zainal et al., *Cell* 149:994–1007, 2012. This article is distributed under the terms of the Creative Commons Attribution License.)

Colorectal cancers arise from the epithelium lining the colon (the large intestine) and rectum (the terminal segment of the gut). The organization of this tissue is broadly similar to that of the small intestine, discussed in Chapter 22 (pp. 1281–1282). For both the small and large intestine, the epithelium is renewed at an extraordinarily rapid rate, taking about a week to completely replace most of the epithelial sheet. In both regions, the renewal depends on stem cells that lie in deep pockets of the epithelium, called intestinal crypts. The signals that maintain the stem cells and control the normal organization and renewal of the epithelium are beginning to be quite well understood, as explained in Chapter 22. Mutations that disrupt these signals begin the process of tumor progression for most colorectal cancers ([Movie 20.7](#)).

Colorectal cancers are common, currently causing nearly 60,000 deaths a year in the United States, or about 10% of total deaths from cancer. Like most cancers, they are not usually diagnosed until late in life (90% occur after the age of 55). However, routine examination of normal adults with a colonoscope (a fiberoptic device for viewing the interior of the colon and rectum) often reveals a small benign tumor, or adenoma, of the gut epithelium in the form of a protruding mass of tissue called a *polyp*. These adenomatous polyps are believed to be the

TABLE 20–1 Some Genetic Abnormalities Detected in Colorectal Cancer Cells

Gene	Class	Pathway affected	Human colon cancers (%)
<i>K-Ras</i>	Oncogene	Receptor tyrosine kinase signaling	40
<i>β-Catenin</i> ¹	Oncogene	Wnt signaling	5–10
<i>Apc</i> ¹	Tumor suppressor	Wnt signaling	>80
<i>p53</i>	Tumor suppressor	Response to stress and DNA damage	60
<i>TGFβ receptor II</i> ²	Tumor suppressor	TGFβ signaling	10
<i>Smad4</i> ²	Tumor suppressor	TGFβ signaling	30
<i>MLH1</i> and other DNA mismatch repair genes (often silenced by DNA methylation)	Tumor suppressor (genetic stability)	DNA mismatch repair	15

^{1,2}The genes with the same superscript numeral act in the same pathway, and therefore only one of the components is mutated in an individual cancer.

precursors of a large proportion of colorectal cancers. Because the progression of the disease is usually very slow, there is typically a period of about 10 years in which the slowly growing tumor is detectable but has not yet turned malignant. Thus, when people are screened by colonoscopy in their fifties and the polyps are removed through the colonoscope—a quick and easy procedure—the subsequent incidence of colorectal cancer is much lower: according to some studies, less than a quarter of what it would be otherwise.

In microscopic sections of polyps smaller than 1 cm in diameter, the cells and their arrangement in the epithelium usually appear almost normal. The larger the polyp, the more likely it is to contain cells that look aberrant and are abnormally organized. Sometimes, two or more distinct areas can be distinguished within a single polyp, with the cells in one area appearing relatively normal and those in the other appearing clearly cancerous, as though they have arisen as a mutant subclone within the original clone of adenomatous cells. At later stages in the disease, some tumor cells become invasive in a small fraction of the polyps, first breaking through the epithelial basal lamina, then spreading through the layer of muscle that surrounds the gut, and finally metastasizing to lymph nodes via lymphatic vessels and to liver, lung, and other organs via blood vessels.

A Few Key Genetic Lesions Are Common to a Large Fraction of Colorectal Cancers

What are the mutations that accumulate with time to produce this chain of events? Of those genes so far discovered to be involved in colorectal cancer, three stand out as most frequently mutated: the proto-oncogene *K-Ras* (a member of the *Ras* gene family), in about 40% of cases; *p53*, in about 60% of cases; and the tumor suppressor gene *Apc* (discussed below), in more than 80% of cases. Others are involved in smaller numbers of colon cancers, and some of these are listed in [Table 20–1](#).

The role of *Apc* first came to light through study of certain families showing a rare type of hereditary predisposition to colorectal cancer, called *familial adenomatous polyposis coli* (FAP). In this syndrome, hundreds or thousands of polyps develop along the length of the colon ([Figure 20–35](#)). These polyps start

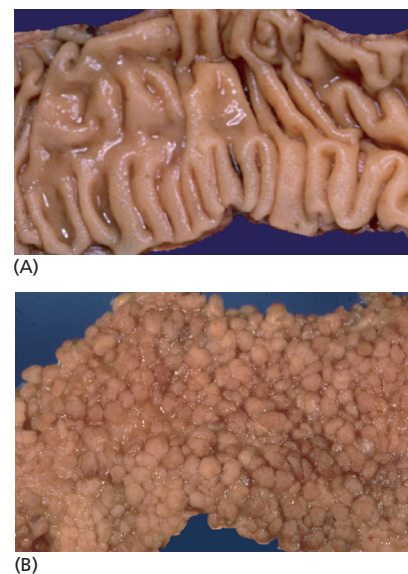


Figure 20–35 Colon of individual with familial adenomatous polyposis coli compared to normal colon. (A) The normal colon wall is a gently undulating but smooth surface. (B) The polyposis colon is completely covered by hundreds of projecting polyps, each resembling a tiny cauliflower when viewed with the naked eye. (Courtesy of Mark Arends.)

to appear in early adult life, and if they are not removed, one or more will almost always progress to become malignant; the average time from the first detection of polyps to the diagnosis of cancer is 12 years. The disease can be traced to a deletion or inactivation of the tumor suppressor gene *Apc*, named after the syndrome. Individuals with FAP have inactivating mutations or deletions of one copy of the *Apc* gene in all their cells and show loss of heterozygosity in tumors, meaning that both copies have been lost or inactivated, even in the benign polyps. Most individuals with colorectal cancer do not have the hereditary condition. Nevertheless, in more than 80% of the cases, their cancer cells (but not their normal cells) have inactivated both copies of the *Apc* gene through mutations acquired during the individual's lifetime. Thus, by a route similar to what we discussed for retinoblastoma, mutation of the *Apc* gene was identified as one of the central ingredients of colorectal cancer.

The APC protein is an inhibitory component of the *Wnt signaling pathway* (discussed in Chapter 15). It binds to the β -catenin protein, another component of the Wnt pathway, and helps to induce the protein's degradation. By inhibiting β -catenin in this way, APC prevents it from localizing to the nucleus, where it would act as a transcriptional regulator to drive cell proliferation and maintain the stem-cell state (see Figure 15-61). Loss of APC results in an excess of free β -catenin and thus leads to an uncontrolled expansion of the stem-cell population. This causes massive increase in the number and size of the intestinal crypts (see Figure 22-4).

When the β -catenin gene was sequenced in a collection of colorectal tumors, it was discovered that many of the tumors that did not have *Apc* mutations had activating mutations in the β -catenin protein instead. Thus, it is excessive activity in the Wnt signaling pathway that is critical for the initiation of this cancer, rather than any single oncogene or tumor suppressor gene that the pathway contains.

This being so, why is the *Apc* gene in particular so often the most common culprit in colorectal cancer? The APC protein is large and it interacts not only with β -catenin but also with various other cell components, including microtubules. Loss of APC appears to increase the frequency of mitotic spindle defects, leading to chromosome abnormalities when cells divide. This additional, independent cancer-promoting effect could explain why *Apc* mutations feature so prominently in the causation of colorectal cancer.

Some Colorectal Cancers Have Defects in DNA Mismatch Repair

In addition to the hereditary disease (FAP) associated with *Apc* mutations, there is a second, more common kind of hereditary predisposition to colon carcinoma in which the course of events differs from the one we have described for FAP. In this more common condition, called *hereditary nonpolyposis colorectal cancer* (HNPCC), or *Lynch syndrome*, the probability of colon cancer is increased without any increase in the number of colorectal polyps (adenomas). Moreover, the cancer cells are unusual, in that they have a normal (or almost normal) karyotype. The majority of colorectal tumors in non-HNPCC individuals, in contrast, have gross chromosomal abnormalities, with multiple translocations, deletions, and other aberrations, and have many more chromosomes than normal (see Figure 20-10).

The mutations that predispose HNPCC individuals to colorectal cancer occur in one of several genes that code for central components of the DNA *mismatch repair system*. These genes are homologous in structure and function to the *MutL* and *MutS* genes in bacteria and yeast (see Figure 5-20). Only one of the two copies of the involved gene is defective, so the repair system is still able to remove the inevitable DNA replication errors that occur in the individual's cells. However, as discussed previously, these individuals are at risk because the accidental loss or inactivation of the remaining good gene copy will immediately elevate the spontaneous mutation rate by a hundredfold or more (discussed in

Chapter 5). These genetically unstable cells can then speed through the standard processes of mutation and natural selection that allow clones of cells to progress to malignancy.

This particular type of genetic instability produces invisible changes in the chromosomes—most notably changes in individual nucleotides and short expansions and contractions of mononucleotide and dinucleotide repeats such as AAAA... or CACACA.... Once the defect in HNPCC individuals was recognized, the epigenetic silencing or mutation of mismatch repair genes was found in about 15% of the colorectal cancers occurring in people with no inherited predisposing mutation.

Thus, the genetic instability found in many colorectal cancers can be acquired in at least two ways. The majority of the cancers display a form of chromosomal instability that leads to visibly altered chromosomes, whereas in the others the instability occurs on a much smaller scale and reflects a defect in DNA mismatch repair. Indeed, many carcinomas show either chromosomal instability or defective mismatch repair—but rarely both. These findings clearly demonstrate that genetic instability is not an accidental by-product of malignant behavior but a contributory cause—and that cancer cells can acquire this instability in multiple ways.

The Steps of Tumor Progression Can Often Be Correlated with Specific Mutations

In what order do *K-Ras*, *p53*, *Apc*, and the other identified colorectal cancer-critical genes mutate, and what contribution does each of them make to the asocial behavior of the cancer cell? There is no single answer because colorectal cancer can arise by more than one route: thus, we know that in some cases, the first mutation can be in a DNA mismatch repair gene; in others, it can be in a gene regulating cell proliferation. Moreover, as previously discussed, a general feature such as genetic instability or a tendency to proliferate abnormally can arise in a variety of ways, through mutations in different genes.

Nevertheless, certain sets of mutations are particularly common in colorectal cancer, and they occur in a characteristic order. Thus, in most cases, mutations inactivating the *Apc* gene appear to be the first or, at least, a very early step, as they are detected at the same high frequency in small benign polyps as in large malignant tumors. Changes that lead to genetic and epigenetic instability are likely also to arise early in tumor progression, as they are needed to drive the later steps.

Activating mutations in the *K-Ras* gene occur later, as they are rare in small polyps but common in larger ones that show disturbances in cell differentiation and histological pattern.

Inactivating mutations in *p53* are thought to come later still, as they are rare in polyps but common in carcinomas (Figure 20-36). We have seen that loss of *p53* function allows cancer cells to endure stress and to avoid apoptosis and cell-cycle arrest. Additionally, loss of *p53* is related to the heightened activation of oncogenes such as *Ras*. Experiments in mice show that an initial low level of oncogene

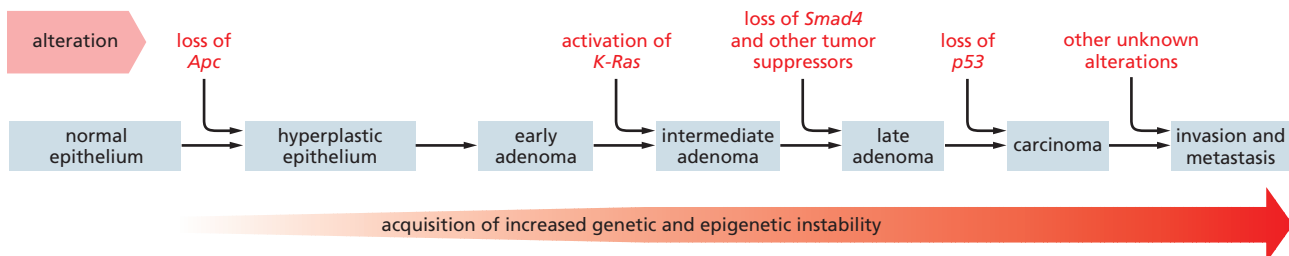


Figure 20-36 Suggested typical sequence of genetic changes underlying the development of a colorectal carcinoma. This oversimplified diagram provides a general idea of the way mutation and tumor development are related. But many other mutations are generally involved, and different colon cancers can progress through different sequences of mutations (and/or epigenetic changes).

activation can give rise to a slowly growing tumor even while *p53* is functional: genes such as *Ras* are, after all, part of the normal machinery of growth control, and moderate activation is not stressful for a cell and does not call the *p53* protein into play. Progression of a tumor from slow to rapid, malignant growth, however, involves activation of oncogenes beyond normal physiological limits to a higher, stressful level. If the *p53* protein is present and functional, this should lead to cell-cycle arrest or death. Only by losing *p53* function can the cancer cells with hyperactive oncogenes survive and progress.

The steps we have just described are only part of the picture. It is important to emphasize that each case of colorectal cancer is different, with its own detailed combination of mutations, and that even for the mutations that are commonly shared, the sequence of occurrence may vary.

The Changes in Tumor Cells That Lead to Metastasis Are Still Largely a Mystery

Perhaps the most significant gap in our understanding of cancer concerns invasiveness and metastasis (see Figure 20–20). More than 25 years after the multistep progression of colorectal cancer was first described, no genetic mutations have been identified that are characteristically associated with metastatic disease. To date, even large-scale genomic sequencing efforts have yet to uncover recurrent mutations that adequately explain the escape of cancer cells from a primary tumor and their dissemination and colonization of a distant tissue. One possibility is that metastasis can be initiated by epigenetic changes that alter patterns of gene expression. These changes could allow cancer cells to take on new traits in the absence of further genetic changes, thereby reprogramming their behavior to promote invasiveness and motility or to generate cancer stem cells. Importantly, changes in cellular programming and behavior operate normally in a variety of contexts during cellular differentiation. The example most relevant to metastasis is the *epithelial-to-mesenchymal transition* (EMT), a highly regulated and poorly understood process that allows epithelial cells to lose their characteristic polarity and adhesiveness and take on a mesenchymal phenotype, which includes enhanced migratory behavior. An EMT program operates at several stages of embryogenesis as well as during wound healing, and its activation in cancer cells could explain how they escape from a primary tumor and invade a new tissue. However, EMT is unlikely to explain the major puzzle of how disseminated cells acquire the ability to survive in the microenvironment of that new tissue, with its unfamiliar mix of growth factors and extracellular matrix components. Because the EMT program (like almost all cellular differentiation programs) involves changes in gene expression, it does not alter DNA sequences and would not be detected by sequencing the genomes of metastasized cancer cells.

Observations of the many stages and varied features of tumorigenesis highlight that cancer cells do not invent new biological phenomena, but instead deploy existing mechanisms and pathways at inappropriate times and places. A better understanding of the normal, underlying cell biology, and of the molecular changes that act to subvert it in cancer, will provide promising leads for cancer treatment, as we discuss in the final section of this chapter.

Summary

The molecular analysis of cancer cells reveals two classes of cancer-critical genes: oncogenes and tumor suppressor genes. A set of these genes becomes altered by a combination of genetic and epigenetic accidents to drive tumor progression. Many cancer-critical genes code for components of the social control pathways that regulate when cells grow, divide, differentiate, or die. In addition, a subclass of tumor suppressors can be categorized as genome maintenance genes, because their normal role is to help maintain genome integrity.

*The inactivation of the *p53* pathway, which occurs in nearly all human cancers, allows genetically damaged cells to escape apoptosis and continue to proliferate.*

Inactivation of the Rb pathway also occurs in most human cancers, illustrating how fundamental each of these pathways is for protecting us against cancer. More generally, the extensive sequencing of cancer cell genomes reveals that the development of a cancer requires the acquisition of heritable disturbances in each of three types of normal controls—those in the cell cycle, cell proliferation, and cell survival pathways.

The sequencing of cancer cell genomes indicates that—except for the cancers of childhood—many cancers acquire multiple driver mutations over the long course of tumor progression, along with a considerably larger number of passenger mutations of no consequence. The same methods reveal how subclones of cells arise and die out as a tumor ages. Tumors thus contain a heterogeneous mixture of cells, some—the so-called cancer stem cells—being much more dangerous than others.

We can often correlate the steps of tumor progression with mutations that activate specific oncogenes and inactivate specific tumor suppressor genes, and colon cancer provides a good example. But different combinations of mutations and epigenetic changes are found in different types of cancer, and even in different individuals with the same type of cancer, reflecting the random way in which these inherited changes arise. Nevertheless, many of the same changes are encountered repeatedly, suggesting that there are a limited number of ways to breach our defenses against cancer. However, the molecular basis of the final and most deadly step in cancer, metastasis, remains poorly understood.

CANCER PREVENTION AND TREATMENT: PRESENT AND FUTURE

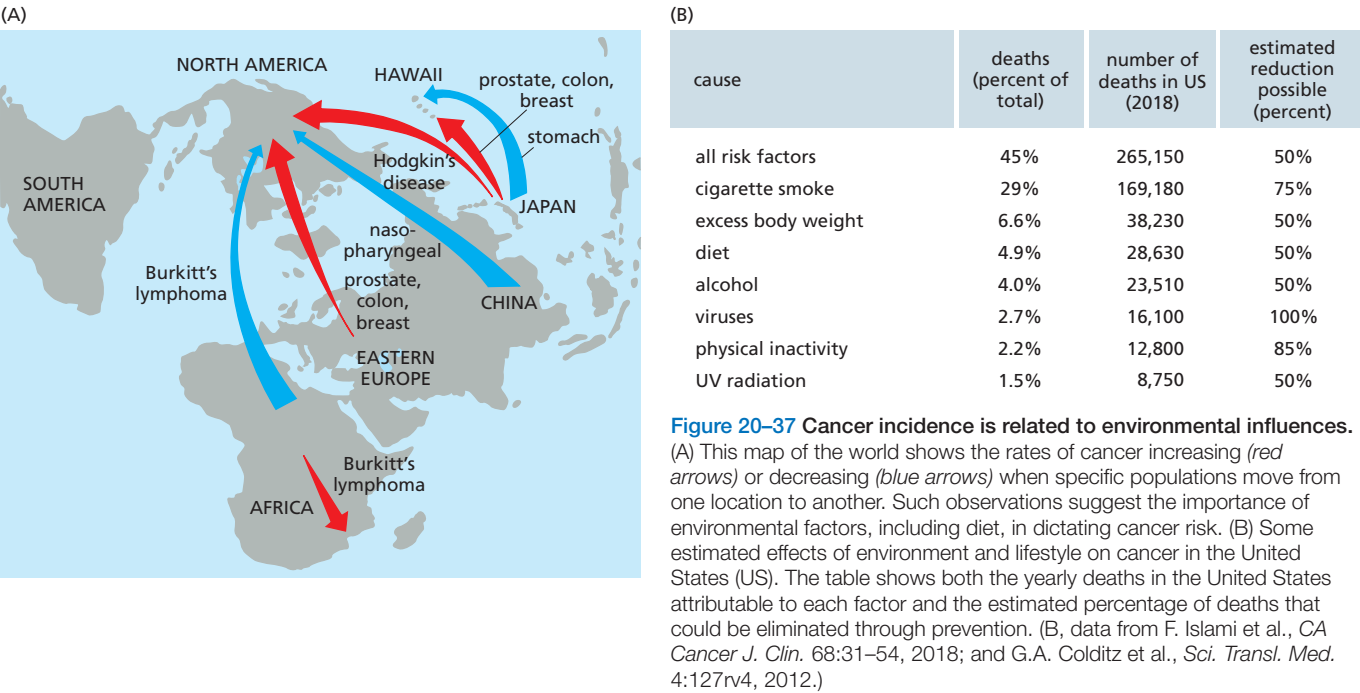
We can apply the growing understanding of the molecular biology of cancer to sharpen our attack on the disease at three levels: prevention, diagnosis, and treatment. Prevention is always better than cure, and indeed many cancers can be prevented, especially by avoiding smoking. Highly sensitive molecular assays promise new opportunities for earlier and more precise diagnosis, with the aim of detecting primary tumors while they are still small and have not yet metastasized. Cancers caught at these early stages can often be nipped in the bud by surgery or radiotherapy, as we saw for colorectal polyps. Nevertheless, full-blown malignant disease will continue to be common for many years to come, and cancer treatments will continue to be needed.

In this section, we first examine the preventable causes of cancer and then consider how advances in our understanding at a molecular level are beginning to transform the treatment of the disease.

Epidemiology Reveals That Many Cases of Cancer Are Preventable

A certain irreducible background incidence of cancer is to be expected regardless of circumstances. As discussed in Chapter 5, mutations can never be absolutely avoided because they are an inescapable consequence of fundamental limitations on the accuracy of DNA replication and repair. If a person could live long enough, it is inevitable that at least one of his or her cells would eventually accumulate a set of mutations sufficient for cancer to develop.

Nevertheless, environmental factors seem to play a large part in determining the risk for cancer. This is demonstrated most clearly by a comparison of cancer incidence in different countries: for almost every cancer that is common in one country, there is another country where the incidence is much lower. Because migrant populations tend to adopt the pattern of cancer incidence typical of their new host country, the differences are thought to be due mostly to environmental, not genetic, factors (**Figure 20-37A**). From epidemiologic studies compiled by the American Cancer Society, it is estimated that at least 45% of cancer deaths can be attributed to modifiable risk factors, indicating that more than half of all cancers should be avoidable (**Figure 20-37B**).



Unfortunately, different cancers have different environmental risk factors, and a population that escapes one such danger is usually exposed to another. This is not, however, inevitable. There are some human subgroups whose way of life substantially reduces the total cancer death rate among individuals of a given age. Under the current conditions in the United States and Europe, approximately one in five people will die of cancer. But the incidence of cancer among strict Mormons in Utah—who avoid alcohol, coffee, cigarettes, drugs, and unsafe sex—is only about half the incidence for nonpracticing members of the same family or for Americans in general.

Although such observations on human populations indicate that cancer can often be avoided, it has been difficult in most cases—with cigarette smoking as a striking exception—to pinpoint the specific environmental factors responsible for these large population differences or to establish how they act. Nevertheless, several important classes of environmental cancer risk factors have been identified (Figure 20–37B). But there are also many other influences—including the chemicals in our environment, the hormones that circulate in our bodies, and the irritations, infections, and damage to which we expose our tissues—that are no less important and favor development of the disease in other ways.

Sensitive Assays Can Detect Those Cancer-causing Agents That Damage DNA

Many quite disparate chemicals are carcinogenic when they are fed to experimental animals or painted repeatedly on their skin. Examples include a range of aromatic hydrocarbons and derivatives of them such as aromatic amines, nitro-samines, and alkylating agents such as mustard gas. Although these **chemical carcinogens** are diverse in structure, a large proportion of them have at least one shared property—they cause mutations. In one common test for mutagenicity (the *Ames test*), the carcinogen is mixed with an activating extract prepared from rat liver cells (to mimic the biochemical processing that occurs in an intact animal). The mixture is then added to a culture of specially designed test bacteria and the bacterial mutation rate measured. Most of the compounds scored as mutagenic by this rapid and convenient assay in bacteria also cause mutations or chromosome aberrations when tested on mammalian cells.

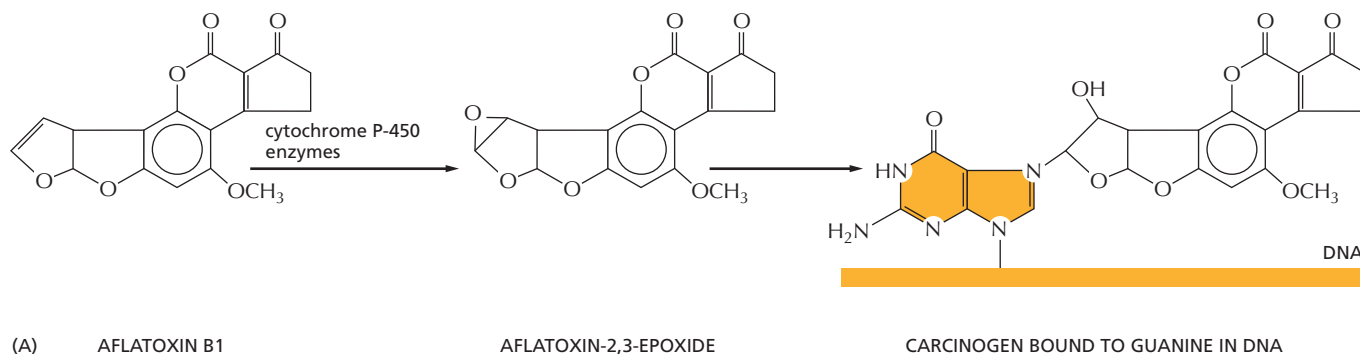


Figure 20–38 Some known carcinogens. (A) Carcinogen activation. A metabolic transformation that occurs in the liver must activate many chemical carcinogens before they will cause mutations by reacting with DNA (labeled *orange*). The compound illustrated here is *aflatoxin B1*, a toxin from a mold (*Aspergillus flavus*) that grows on grain and peanuts when they are stored under humid tropical conditions. Aflatoxin is an important cause of liver cancer in the tropics. (B) Different carcinogens cause different types of cancer. (B, data from Institute of Medicine, Cancer and the Environment: Gene-Environment Interactions. Washington, DC: National Academies Press, 2002.)

A few of these carcinogens act directly on DNA. But generally the more potent ones are relatively inert chemically; these chemicals become damaging only after they have been converted to a more reactive molecule by metabolic processes in the liver, catalyzed by a set of intracellular enzymes known as the *cytochrome P-450 oxidases*. These enzymes normally help to convert ingested toxins into harmless and easily excreted compounds. Unhappily, their activity on certain chemicals generates products that are highly mutagenic. Examples of carcinogens activated in this way include *benzo[a]pyrene*, a cancer-causing polycyclic aromatic hydrocarbon present in coal tar, tobacco smoke, and the fungal toxin *aflatoxin B1* (Figure 20–38).

Fifty Percent of Cancers Could Be Prevented by Changes in Lifestyle

Tobacco smoke is the most important carcinogen in the world today. Even though many other risk factors have been identified, none of these appear to be responsible for anything like the same numbers of human cancer deaths attributable to tobacco smoke. It is sometimes thought that the main environmental causes of cancer are the products of a highly industrialized way of life—the rise in pollution, the enhanced use of food additives, and so on—but there is little evidence to support this view. The idea may have come in part from the identification of some highly carcinogenic materials used in industry, such as 2-naphthylamine and asbestos. Except for the increase in cancers caused by smoking, however, age-adjusted death rates for most common human cancers have stayed much the same over the past half-century or, in some cases, have declined significantly (Figure 20–39).

Most of the carcinogenic factors that are known to be significant are by no means specific to the modern world. The most potent known carcinogen, by certain assays at least, is aflatoxin B1 (see Figure 20–38). It is produced by fungi that naturally contaminate foods such as tropical peanuts and is an important cause of liver cancer in Africa and Asia.

Except for tobacco, chemical toxins and mutagens are of lesser importance as contributory causes of cancer than other factors that are a matter of human behavior. For example, currently nearly three-fourths of adults and one-third of children and adolescents in the United States are overweight or obese. The combination of four risk factors—excess body weight, alcohol intake, poor diet, and physical inactivity—now accounts for the highest proportion of all cancer cases in women and is second only to smoking in men. Thus, it is estimated that as many

- **VINYL CHLORIDE:**
liver angiosarcoma
- **BENZENE:**
acute leukemias
- **ARSENIC:**
skin carcinomas, bladder cancer
- **ASBESTOS:**
mesothelioma
- **RADIUM:**
osteosarcoma

(B)

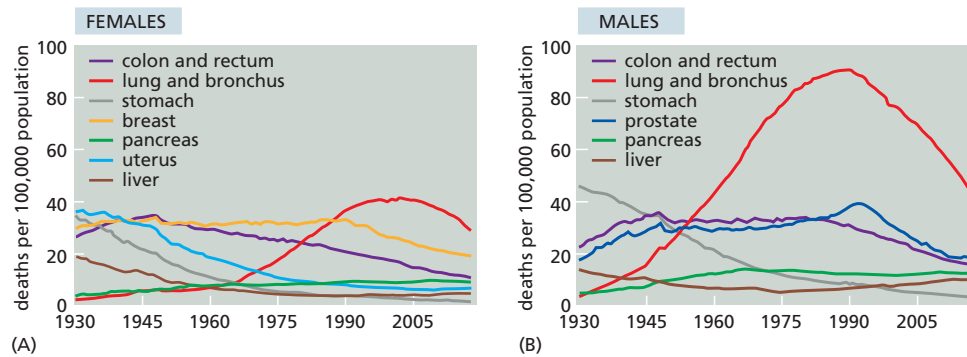


Figure 20-39 Age-adjusted cancer death rates, United States, 1930–2018. Selected death rates, adjusted to the age distribution of the US population, are plotted for (A) females and (B) males. Note the dramatic rise in lung cancer for both sexes, following the pattern of tobacco smoking, and the fall in deaths from stomach cancer, thought to be related to a fall in rates of infection with *Helicobacter pylori*. Recent reductions in other cancer death rates may correspond to improvements in detection and treatment. Adjustment of the data for the age of individuals is needed to compensate for the inevitable increase in cancer as people live longer, on average. (Adapted from American Cancer Society. Cancer Facts and Figures, 2021. Atlanta: American Cancer Society, Inc. Data from US Mortality Volumes 1930 to 1959, US Mortality Data 1960 to 2016, National Center for Health Statistics, Centers for Disease Control and Prevention. Note that uterus includes uterine cervix and uterine corpus combined due to limitations in coding prior to 1970. With permission from American Cancer Society.)

as 50% of all cancers could be avoided by identifiable changes in lifestyle (see Figure 20-37B).

Viruses and Other Infections Contribute to a Significant Proportion of Human Cancers

Human cancers are not contagious, but a significant minority of cancers are related directly or indirectly to infections with viruses and, less frequently, bacteria or parasites. Such infections account for approximately 15% of cancers worldwide and are more common in developing countries. Evidence for their involvement comes partly from the detection of viruses in individuals with cancer and partly from epidemiology. Thus, cancer of the uterine cervix is associated with infection with a papillomavirus, while liver cancer is very common in parts of the world (Africa and Southeast Asia) where hepatitis-B viral infections are common. Chronic infection with hepatitis-C virus, which has infected 170 million people worldwide, is also clearly associated with the development of liver cancer.

The main culprits, as shown in Table 20-2, are the DNA viruses. The **DNA tumor viruses** cause cancer by the most direct route—by interfering with controls of the cell cycle and apoptosis. To understand this type of viral carcinogenesis, it is important to review the life history of viruses. Many DNA viruses use the host cell's DNA replication machinery to replicate their own genomes. However, to produce a large number of infectious virus particles within a single host cell, the DNA virus has to commandeer this machinery and drive it hard, breaking through the normal constraints on DNA replication and usually killing the host cell in the process. Many DNA viruses reproduce only in this way. But some have a second option: they can propagate their genome as a quiet, well-behaved passenger in the host cell, replicating in parallel with the host cell's DNA (either integrated into the host genome or as an extrachromosomal plasmid) in the course of ordinary cell-division cycles. These viruses will switch between two modes of existence according to circumstances, remaining latent and harmless for a long time, but then proliferating in occasional cells in a process that kills the host cell and generates large numbers of infectious particles.

Neither of these conditions converts the host cell to a cancerous character, nor is it in the interest of the virus to do so. But for viruses with a latent phase, accidents can occur that prematurely activate some of the viral proteins that the

TABLE 20–2 Viruses Associated with Human Cancers

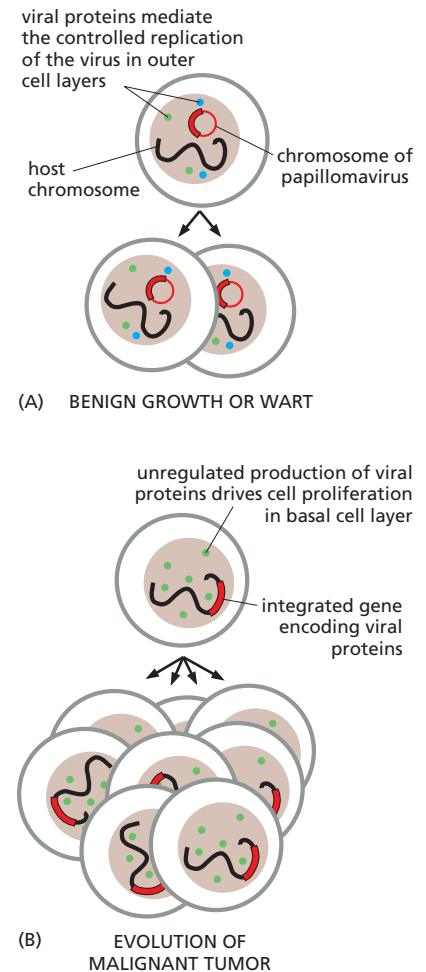
Virus	Associated cancer	Areas of high incidence
DNA viruses		
<i>Papovavirus family</i>		
Papillomavirus (many distinct strains)	Warts (benign)	Worldwide
Papillomavirus (many distinct strains)	Carcinoma of the uterine cervix	Worldwide
<i>Hepadnavirus family</i>		
Hepatitis-B virus	Liver cancer (hepatocellular carcinoma)	Southeast Asia, tropical Africa
<i>Herpesvirus family</i>		
Epstein–Barr virus	Burkitt's lymphoma (cancer of B lymphocytes)	West Africa, Papua New Guinea
Epstein–Barr virus	Nasopharyngeal carcinoma	Southern China, Greenland
Human herpesvirus 8	Kaposi's sarcoma	Central and southern Africa
RNA viruses		
<i>Retrovirus family</i>		
Human T-cell leukemia virus type I (HTLV-1)	Adult T-cell leukemia/lymphoma	Japan, West Indies
Human immunodeficiency virus (HIV, the AIDS virus)	Kaposi's sarcoma (via human herpesvirus 8)	Central and southern Africa
<i>Flavivirus family</i>		
Hepatitis-C virus	Liver cancer (hepatocellular carcinoma)	Worldwide
For all these viruses, the number of people infected is much larger than the number who develop cancer: the viruses must act in conjunction with other factors. As described in the text, different viruses contribute to cancer in different ways.		

virus would normally use in its replicative phase to allow the viral DNA to replicate independently of the cell cycle. As in the case of papillomavirus described below, this type of accident can switch on the persistent proliferation of the host cell itself, leading to cancer.

Cancers of the Uterine Cervix Can Be Prevented by Vaccination Against Human Papillomavirus

The **papillomaviruses** are a prime example of DNA tumor viruses. They are responsible for human warts and are especially important as a cause of carcinoma of the uterine cervix: this is the second most common cancer of women in the world as a whole, representing about 6% of all human cancers. Human papillomaviruses (HPVs) infect the cervical epithelium and maintain themselves in a latent phase in the basal layer of cells as extrachromosomal plasmids, which replicate in step with the chromosomes. Infectious virus particles are generated through a switch to a replicative phase in the outer epithelial layers, as progeny of these cells begin to differentiate before being sloughed from the surface. Here, cell division should normally stop, but the virus interferes with this cell-cycle arrest so

Figure 20–40 How certain papillomaviruses are thought to give rise to cancer of the uterine cervix. Papillomaviruses have double-stranded circular DNA chromosomes of about 8000 nucleotide pairs. These chromosomes are normally stably maintained in the basal cells of the epithelium as plasmids (red circles), whose replication is regulated so as to keep step with the chromosomes of the host. (A) Normally, the virus perturbs the host cell cycle only when the virus is programmed to produce infectious progeny, in the outer layers of an epithelium. This is relatively harmless. (B) Rare accidents can cause the integration of a fragment of such a plasmid into a chromosome of the host, altering the environment of the viral genes in the basal cells of an epithelium. This can disrupt the normal control of viral gene expression. The unregulated production of certain viral proteins (named E6 and E7) interferes with the control of cell division in the basal cells, thereby helping to generate a cancer (bottom).



as to allow replication of its own genome. Usually, the effect is restricted to the outer layers of cells and is relatively harmless, as in a wart. Occasionally, however, a genetic accident causes the viral genes that encode the proteins that prevent cell-cycle arrest to integrate into the host chromosome and become active in the basal layer, where the stem cells of the epithelium reside (see Figure 22–6). This can lead to cancer, with the viral genes acting as oncogenes (Figure 20–40).

The whole process, from initial infection to invasive cancer, is slow, taking many years. It involves a long intermediate stage when the affected patch of cervical epithelium is visibly disordered but the cells have not yet begun to invade the underlying connective tissue—a phenomenon called *intraepithelial neoplasia*. Many such lesions regress spontaneously. Moreover, at this stage, it is still easy to cure the condition by destroying or surgically removing the abnormal tissue. Fortunately, the presence of such lesions can be detected by scraping off a sample of cells from the surface of the cervix and viewing it under the microscope (the *Pap smear* technique).

Better still, a vaccine has now been developed that protects against infection with the relevant strains of human papillomavirus. This vaccine, if given to girls before puberty and thus before they become sexually active, has been shown to greatly reduce their risk of ever developing cervical cancer. Because the virus spreads through sexual activity, it is now recommended that both young males and young females be routinely vaccinated. Mass immunization programs have begun in several countries.

Infectious Agents Can Cause Cancer in a Variety of Ways

In papillomaviruses, the viral genes that are mainly to blame are called *E6* and *E7*. The protein products of these viral oncogenes interact with many host-cell proteins, but, in particular, they bind to two key tumor suppressor proteins of the host cell, putting them both out of action and so permitting the cell to replicate its DNA and divide in an uncontrolled way. One of these host proteins is Rb; the other is p53. Other DNA tumor viruses use similar mechanisms to inhibit Rb and p53, underlining the central importance of altering both the cell-cycle control pathway and the cell survival pathway if a cell is to escape the normal constraints on proliferation, as we have previously discussed.

In other cancers, viruses have indirect tumor-promoting actions. The hepatitis-B and hepatitis-C viruses, for example, favor the development of liver cancer by causing chronic inflammation (hepatitis), which stimulates extensive cell division in the liver that promotes the eventual evolution of tumor cells. In AIDS, the human immunodeficiency virus (HIV) promotes development of an otherwise rare cancer called Kaposi's sarcoma by destroying the immune system, thereby permitting a secondary infection with a human herpesvirus (HHV-8) that has a direct carcinogenic action. By causing severe inflammation, chronic infection with parasites and bacteria can also promote the development of some cancers. For example, chronic infection of the stomach with the bacterium *Helicobacter pylori*, which causes ulcers, appears to be a major cause of stomach cancer; dramatic falls in the incidence of stomach cancer over the past half-century (see Figure 20–39) correlate with a decline in the incidence of *Helicobacter* infections.

The Search for Cancer Cures Is Difficult but Not Hopeless

The difficulty of curing a cancer is similar to the difficulty of getting rid of weeds. Cancer cells can be removed surgically or destroyed with toxic chemicals or radiation, but it is hard to eradicate every single one of them. Surgery can rarely ferret out every metastasis, and treatments that kill cancer cells are generally toxic to normal cells as well. Moreover, unlike normal cells, cancer cells can mutate rapidly and will often evolve resistance to the poisons and irradiation used against them.

In spite of these difficulties, effective cures using anticancer drugs (alone or in combination with other treatments) have already been found for some formerly highly lethal cancers, including Hodgkin's lymphoma, testicular cancer, choriocarcinoma, and some leukemias and other cancers of childhood. Even for types of cancer where a cure at present seems beyond our reach, there are treatments that will prolong life or at least relieve distress. But what prospect is there of doing better and finding cures for the most common forms of cancer, which still cause great suffering and so many deaths?

Traditional Therapies Exploit the Genetic Instability and Loss of Cell-Cycle Checkpoint Responses in Cancer Cells

Anticancer therapies need to take advantage of some molecular peculiarity of cancer cells that distinguishes them from normal cells. One such property is genetic instability, reflecting deficiencies in chromosome maintenance, cell-cycle checkpoints, and/or DNA repair. Remarkably, the most widely used cancer therapies seem to work by exploiting these abnormalities, although this was not known by the scientists who first developed the treatments. Ionizing radiation and most anticancer drugs damage DNA or interfere with chromosome segregation at mitosis, and they preferentially kill cancer cells because cancer cells have a diminished ability to survive the damage. Normal cells treated with radiation, for example, arrest their cell cycle until they have repaired the damage to their DNA, thanks to the cell-cycle checkpoint responses discussed in Chapter 17. Because cancer cells generally have defects in their checkpoint responses, they may continue to divide after irradiation, only to die after a few days because the genetic damage remains unrepaired. More generally, most cancer cells are physiologically deranged to a stressful degree: they live dangerously. Even though the cells in a tumor have evolved to be unusually tolerant of minor DNA damage, they are hypersensitive to the much greater amount of damage that can be created by radiation and by DNA-damaging drugs. A small increase of genetic damage can be enough to tip the balance between proliferation and death. Unfortunately, however, DNA-damaging therapies typically cause terrible side effects on normal, rapidly dividing cells in the digestive system, bone marrow, and mucous membranes. Furthermore, these treatments are themselves carcinogenic and can lead to second cancers.

Furthermore, while the molecular abnormalities present in cancer cells often enhance their sensitivity to cytotoxic agents, they can also increase their resistance. For example, where a normal cell might die by apoptosis in response to DNA damage, thanks to the stress response mediated by p53, a cancer cell may escape apoptosis because it lacks p53. Cancers vary widely in their sensitivity to cytotoxic treatments, some responding to one drug, some to another, probably reflecting the particular kinds of defects that a particular cancer has in its DNA repair processes, cell-cycle checkpoints, and control of apoptosis.

New Drugs Can Kill Cancer Cells Selectively by Targeting Specific Mutations

Radiotherapy and traditional cytotoxic drugs are rather weakly selective: they hurt normal cells as well as the cancer cells, and the safety margin is narrow. The dose often cannot be raised high enough to kill all the cancer cells, because

this would kill the patient, and curative treatments, where achievable, generally require a combination of several cytotoxic agents. The side effects can be harsh and hard to endure. How can we do better?

An ideal treatment is one that is cell-lethal in combination with some lesion that is present in the cancer cells, but harmless to cells where this lesion is absent. Such a treatment is said to be *synthetic-lethal* (from the original sense of the word *synthesis*, meaning “putting together”): it kills only in partnership with the cancer-specific mutation. As we become increasingly able to pinpoint the specific molecular alterations in cancer cells that make them different from their normal neighbors, new opportunities for such precisely targeted treatments are coming into view. We end this chapter with some examples of new treatments that are already being put into practice.

PARP Inhibitors Kill Cancer Cells That Have Defects in *Brca1* or *Brca2* Genes

As we have emphasized, the genetic instability of cancer cells makes the cells both dangerous and vulnerable—dangerous because of their enhanced ability to evolve and proliferate, and vulnerable because treatment that leads to still more extreme genetic disruption can take them over the brink and kill them. In some cancers, genetic instability results from an identified fault in one of the many devices on which normal cells depend for DNA repair and maintenance. In this case, a drug tailored to block a complementary part of the DNA repair machinery can lead to such severe genetic damage that the cancer cells die.

Detailed studies of the mechanisms for DNA maintenance discussed in Chapter 5 reveal a surprising amount of apparent redundancy. Thus, knocking out a particular pathway for DNA repair is generally less disastrous than one might expect, because alternate repair pathways exist. For example, stalled DNA replication forks can arise when the fork encounters a single-strand break in a template strand, but cells can avoid the disaster that would otherwise result either by directly repairing these single-strand breaks or, if that fails, by using homologous recombination to repair the broken fork (see Figure 5–49). Suppose that the cells in a particular cancer have become genetically unstable by acquiring a mutation that reduces their ability to repair broken replication forks by homologous recombination. Might it be possible to eradicate that cancer by treating it with a drug that inhibits the repair of single-strand breaks, thereby greatly increasing the number of forks that break? The consequences of such drug treatment might be expected to be relatively harmless for normal cells, but lethal for the cancer.

This strategy appears to work to kill the cells in at least one class of cancers—those that have inactivated both copies of either their *Brca1* or their *Brca2* tumor suppressor genes. As described in Chapter 5, *Brca2* is an accessory protein that interacts with the Rad51 protein (the RecA analog in humans) in the repair of DNA double-strand breaks by homologous recombination. *Brca1* is another protein that is also required for this repair process. Like *Rb*, the *Brca1* and *Brca2* genes were discovered as mutations that predispose humans to cancer—in this case, chiefly cancers of the breast and ovaries (though unlike *Rb*, they seem to be involved in only a small proportion of such cancers). Individuals who inherit one mutant copy of *Brca1* or *Brca2* develop tumors that have inactivated the second copy of the same gene, presumably because this change makes the cells genetically unstable and speeds tumor progression.

While *Brca1* and *Brca2* are needed for the repair of DNA double-strand breaks, single-strand breaks are repaired by other machinery, involving an enzyme called PARP (polyADP-ribose polymerase). This understanding of the basic mechanisms of DNA repair led to a striking discovery: drugs that block PARP activity kill *Brca*-deficient cells with extraordinary selectivity. At the same time, PARP inhibition has very little effect on normal cells; in fact, mice that have been engineered to lack PARP1—the major PARP family member involved in DNA repair—remain healthy under laboratory conditions. This result suggests that,

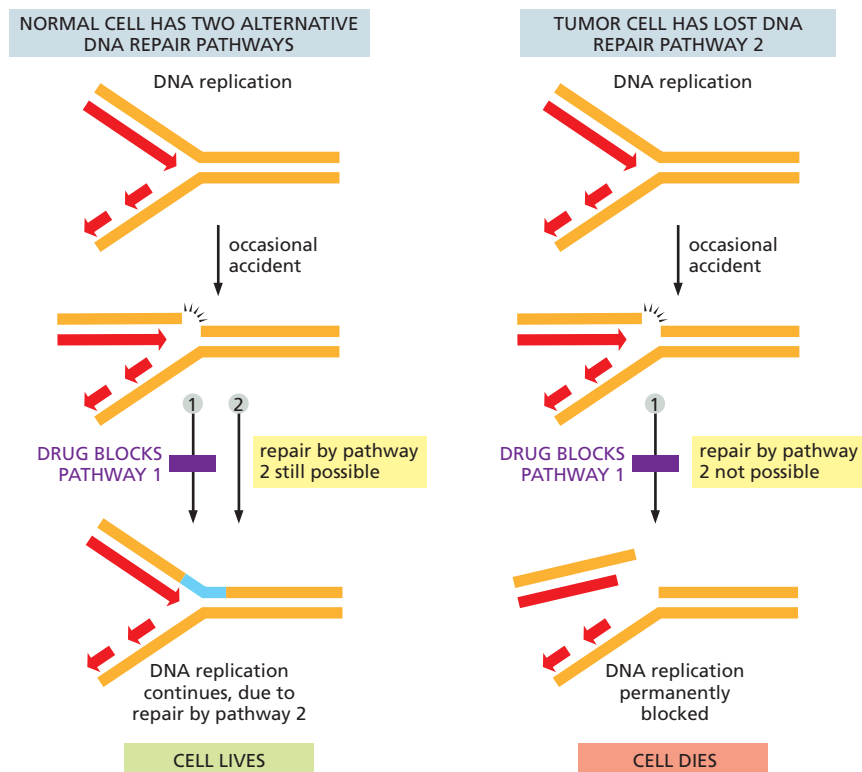


Figure 20–41 How a tumor’s genetic instability can be exploited for cancer therapy. As explained in Chapter 5, the maintenance of DNA sequences is so critical for life that cells have evolved multiple pathways for repairing DNA damage and reducing DNA replication errors. As illustrated, a DNA replication fork will stall whenever it encounters a break in a DNA template strand. In this example, normal cells have two different repair pathways that help them to avoid the problem, pathways 1 and 2. They are therefore not harmed by treatment with a drug that blocks repair pathway 1. But, because the inactivation of repair pathway 2 was selected for during the evolution of the tumor cell, the tumor cells are killed by the same drug treatment.

In the actual case that underlies this example, the function of repair pathway 1 (requiring the PARP protein discussed in the text) is to remove persistent, accidental breaks in a DNA single strand before they are encountered by a moving replication fork. Pathway 2 is the recombination-dependent process (requiring the Brca2 and Brca1 proteins) for repairing stalled replication forks illustrated in Figure 5–49. PARP inhibitors are often used for treating cancers with defective *Brca2* or *Brca1* tumor suppressor genes.

while the repair pathway requiring PARP provides a first line of defense against persistent breaks in a DNA strand, these breaks can be repaired efficiently by a genetic recombination pathway in normal cells. In contrast, tumor cells that have acquired their genetic instability by the loss of *Brca1* or *Brca2* have lost this second line of defense, and they are therefore uniquely sensitive to PARP inhibitors (Figure 20–41).

PARP inhibitors have produced some striking results, causing tumors to regress in many *Brca*-deficient patients and delaying progression of their disease, with relatively few disagreeable side effects. These drugs also appear to be applicable to cancers with other mutations that cause defects in the cell’s homologous recombination machinery—a small, though significant, proportion of cancer cases.

PARP inhibition provides an example of the type of rational, highly selective approach to cancer therapy that is beginning to be possible. Along with other new treatments to be discussed below, it raises high hopes for treating many other cancers.

Small Molecules Can Be Designed to Inhibit Specific Oncogenic Proteins

An obvious tactic for treating cancer is to attack a tumor expressing an oncogene with a drug designed to specifically block the function of the protein that the oncogene produces. But how can such a treatment avoid hurting the normal cells that depend on the function of the proto-oncogene from which the oncogene has evolved, and why should the drug kill the cancer cells, rather than simply calm them down? One answer may lie in the phenomenon of *oncogene dependence*. Once a cancer cell has undergone an oncogenic mutation, it will often undergo further mutations, epigenetic changes, or physiological adaptations that make it reliant on the hyperactivity of the initial oncogene, just as drug addicts become reliant on high doses of their drug. Blocking the activity of the oncogenic protein may then kill the cancer cell without significantly harming its normal neighbors. Some remarkable successes have been achieved in this way.

As we saw earlier, chronic myelogenous leukemia (CML) is usually associated with a particular chromosomal translocation, visible as the Philadelphia chromosome (see Figure 20–5). This results from chromosome breakage and rejoining at the sites of two specific genes, *Abl* and *Bcr*. The fusion of these genes creates a hybrid gene, called *Bcr-Abl*, that codes for a chimeric protein consisting of the N-terminal fragment of Bcr fused to the C-terminal portion of Abl (Figure 20–42). Abl is a tyrosine kinase involved in cell signaling. The substitution of the Bcr fragment for the normal N-terminus of Abl makes it hyperactive, so that it stimulates inappropriate proliferation of the hemopoietic precursor cells that contain it and prevents these cells from dying by apoptosis—which many of them would normally do. As a result, excessive numbers of white blood cells accumulate in the bloodstream, producing CML.

The chimeric Bcr-Abl protein is an obvious target for therapeutic attack. Searches for synthetic drug molecules that can inhibit the activity of tyrosine kinases discovered one, called *imatinib* (trade name Gleevec), that blocks Bcr-Abl (Figure 20–43). When the drug was first given to patients with CML, nearly

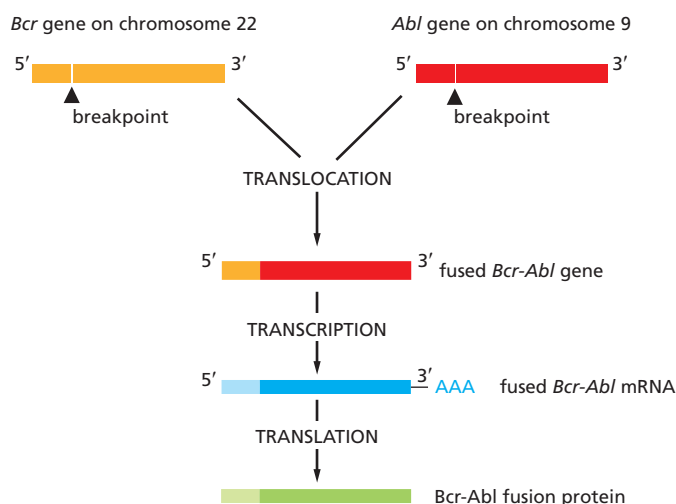


Figure 20–42 The conversion of the *Abl* proto-oncogene into an oncogene in individuals with chronic myelogenous leukemia. The chromosome translocation responsible joins the *Bcr* gene on chromosome 22 to the *Abl* gene from chromosome 9, thereby generating a Philadelphia chromosome (see Figure 20–5). The resulting fusion protein has the N-terminus of the Bcr protein joined to the C-terminus of the Abl tyrosine protein kinase; in consequence, the Abl kinase domain becomes inappropriately active, driving excessive proliferation of a clone of hemopoietic cells in the bone marrow.

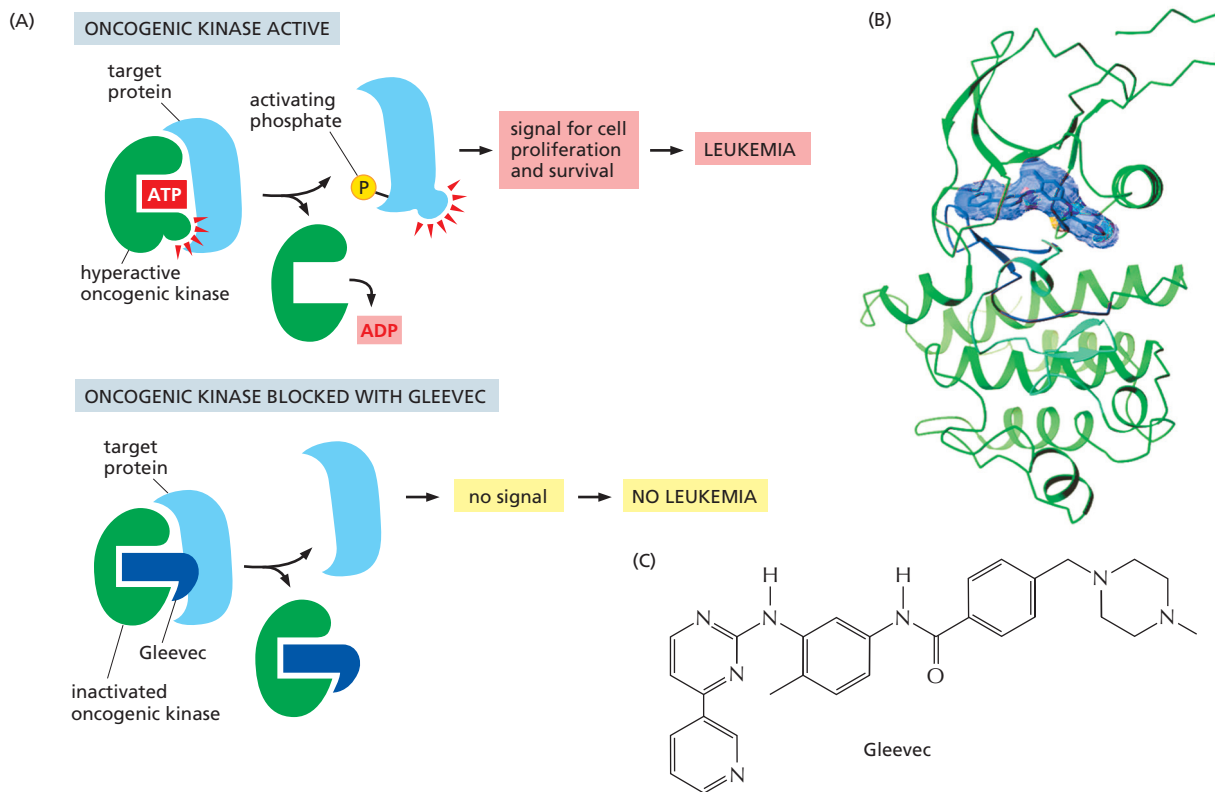


Figure 20-43 How imatinib (Gleevec) blocks the activity of Bcr-Abl protein and halts chronic myelogenous leukemia.

(A) Imatinib sits in the ATP-binding pocket of the tyrosine kinase domain of Bcr-Abl and thereby prevents Bcr-Abl from transferring a phosphate group from ATP onto a tyrosine residue in a substrate protein. This blocks transmission of a signal for cell proliferation and survival. (B) The structure of the complex of imatinib (solid *blue* object) with the tyrosine kinase domain of the Abl protein (ribbon diagram), as determined by x-ray crystallography. (C) The chemical structure of the drug. It can be administered orally; it has side effects, but they are usually quite tolerable. (B, from T. Schindler et al., *Science* 289:1938–1942, 2000. With permission from AAAS. PDB code: 3K5V.)

all of them showed a dramatic response, with an apparent disappearance of the cells carrying the Philadelphia chromosome in more than 80% of patients. The response appears relatively durable: after years of continual treatment, many patients have not progressed to later stages of the disease—although imatinib-resistant cancers emerge with a probability of about 5% per year during the early years.

Results are not so good for those patients who have already progressed to the more acute phase of myeloid leukemia, known as blast crisis, where genetic instability has set in and the march of the disease is far more rapid. These patients show a response at first and then relapse because the cancer cells develop a resistance to imatinib. This resistance is usually associated with secondary mutations in the part of the *Bcr-Abl* gene that encodes the kinase domain, disrupting the ability of imatinib to bind to Bcr-Abl kinase. Second-generation inhibitors that function effectively against a whole range of imatinib-resistant mutants have now been developed. By combining one or more of these new inhibitors with imatinib as the initial therapy, it seems that CML—at least in the chronic (early) stage—may be on its way to becoming a curable disease.

Despite the complications with resistance, the extraordinary success of imatinib is enough to drive home an important principle: once we understand precisely what genetic lesions have occurred in a cancer, we can begin to design effective rational methods to treat it. This success story has fueled efforts to identify small-molecule inhibitors for other oncogenic protein kinases and to use them to attack the appropriate cancer cells. Increasing numbers are being developed. These include molecules that target the EGF receptor and are currently approved

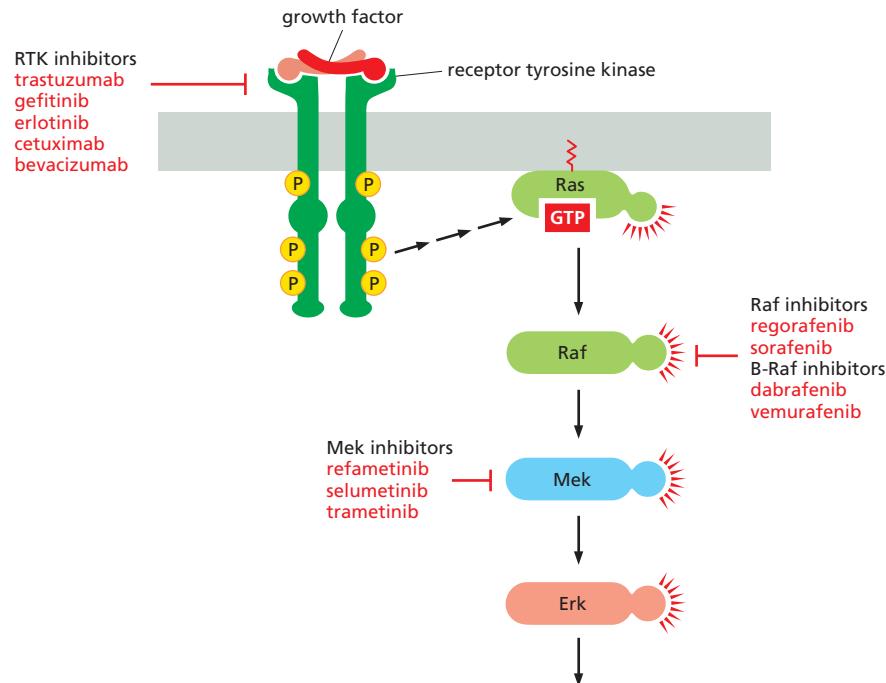


Figure 20–44 Some anticancer drugs and drug targets in the Ras–MAP-kinase signaling pathway. This Ras–MAP-kinase signaling pathway is triggered by a variety of receptor tyrosine kinases (RTKs), including the EGF receptor (see Figures 15–48 and 15–50). Raf kinase inhibitors include several that specifically target the oncoprotein B-Raf. By convention, those drugs that are antibodies end in “mab,” while those that are small molecules end in “nib.” (Adapted from B. Vogelstein et al., *Science* 339:1546–1558, 2013.)

for the treatment of some lung cancers, as well as drugs that specifically target the B-Raf oncoprotein in melanomas.

Protein kinases have been relatively easy to inhibit with small molecules such as imatinib, and many kinase inhibitors are being produced by pharmaceutical companies in the hope that they can be effective as drugs for some forms of cancer. Many cancers lack an oncogenic mutation in a protein kinase. But most tumors contain inappropriately activated signaling pathways, for which a target somewhere in the pathway can hopefully be found (**Movie 20.8**). As an example, **Figure 20–44** displays some of the anticancer drugs and drug targets that are currently being tested for a pathway frequently activated in cancers.

Many Cancers May Be Treatable by Enhancing Immune Responses

Cancers have complex interactions with the immune system, and its various components may sometimes help as well as hinder tumor progression. But for more than a century it has been a dream of cancer researchers to somehow harness our immune systems in a controlled and efficient way to exterminate cancer cells, just as it exterminates pathogenic microorganisms. There are finally signs that this dream may one day be realized, at least for some forms of cancer.

The simplest type of immunological therapy, conceptually at least, is to inject the patient with antibodies that target the cancer cells. This approach has had some successes. About 25% of breast cancers, for example, express unusually high levels of the Her2 (human epidermal growth factor receptor 2) protein, a receptor tyrosine kinase related to the EGF receptor that plays a part in the normal development of mammary epithelium. A monoclonal antibody called *trastuzumab* (trade name Herceptin), which binds to Her2 and inhibits its function, slows the growth of human breast cancers that overexpress Her2, and it is now a standard therapy for these cancers (see Figure 20–44). A related approach uses antibodies to deliver poisons to the cancer cells. Antibodies against proteins that are abundant on the surface of a particular type of cancer cell but rare on normal cells can be coupled to a toxin that kills the cells that the antibody binds to.

A great deal of current excitement centers around a different type of immunological approach, which is based on a class of lymphocytes called T cells. As

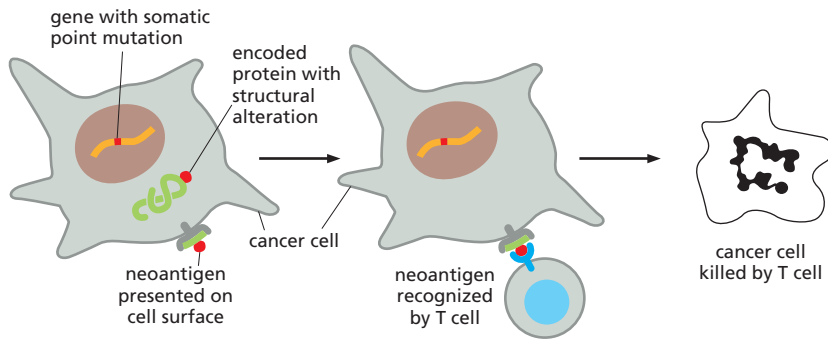


Figure 20–45 Tumor-specific antigens are recognized by the immune system.

Due to somatic mutation, cells in tumors will produce many different mutant proteins. As described in Chapter 24, peptides from these proteins will be displayed as neoantigens on the tumor cell surface and have the potential to activate a T cell response that kills the cell (see Figure 24–42).

described in detail in Chapter 24, cytotoxic T cells can kill a host cell that displays foreign peptide antigens on its surface. Because the peptide is usually derived from an invading pathogen, the response helps terminate the infection. The challenge is to find ways of recruiting cytotoxic T cells to attack cancers with similar efficiency and specificity, presuming that the cancer cells express tumor-specific antigens. Recall that, from the thousands of tumor genome sequences thus far determined, we know that a typical cancer cell contains approximately 50 proteins with a mutation that alters an amino acid sequence, most of these being passenger mutations, as previously explained (see p. 1186). The amino acid changes resulting from either passenger or driver mutations have the potential to produce cell-surface *neoantigens* that can be recognized as foreign by T cells, resulting in cancer cell death (Figure 20–45). However, because cytotoxic T cells kill infected host cells, they are under tight control to keep their activity within safe bounds. Furthermore, cancer cells typically lack some of the components present in pathogen-infected cells that are necessary to initiate a robust T cell response. Thus, very few tumor-reactive T cells exist, and therapies to boost their numbers are at the forefront of immunotherapy.

One approach to augment the T cell response to cancer cells is to collect a patient's own T cells from his or her blood, expand the cancer-reactive population, and then re-infuse them into the patient. This method of introducing a large number of T cells that already recognize the tumor cells bearing cancer-specific antigens has been very effective in metastatic melanoma patients, achieving complete remission rates as high as 20%. In an even more elaborate variation of this strategy, the patient's T cells can be genetically engineered before their numbers are increased, in a procedure called chimeric antigen receptor T cell (CAR T) therapy (Movie 20.9). The engineered CAR T cells not only recognize the patient's tumor-specific antigen but also possess co-stimulatory activities that boost the response of the T cells. This therapy is extremely effective against certain blood cancers, but, unfortunately, it also has severe side effects due to immune system hyperactivation.

Immunosuppression Is a Major Hurdle for Cancer Immunotherapy

If cancer cell neoantigens are recognized as foreign, why doesn't the immune system work better to eliminate the cancer in the first place? As discussed previously (see pp. 1175–1176), cancer cells manipulate their surrounding stroma, including immune cells, to create a microenvironment that simultaneously promotes tumor growth and suppresses immune responses. Investigation of how tumors escape immune destruction led to the discovery of crucial immunosuppressive mechanisms—the expression on the cancer cell surface of one or more proteins that bind to inhibitory receptors on various immune cells, including activated T cells. As a result, even if cytotoxic T cells recognize tumor antigens, they are prevented from killing the tumor cells. The inhibitory receptors expressed on the surface of T cells are part of the *immune checkpoint* that plays an important normal function in preventing excessive, tissue-damaging immune responses during infections. But, in the context of cancer, immune

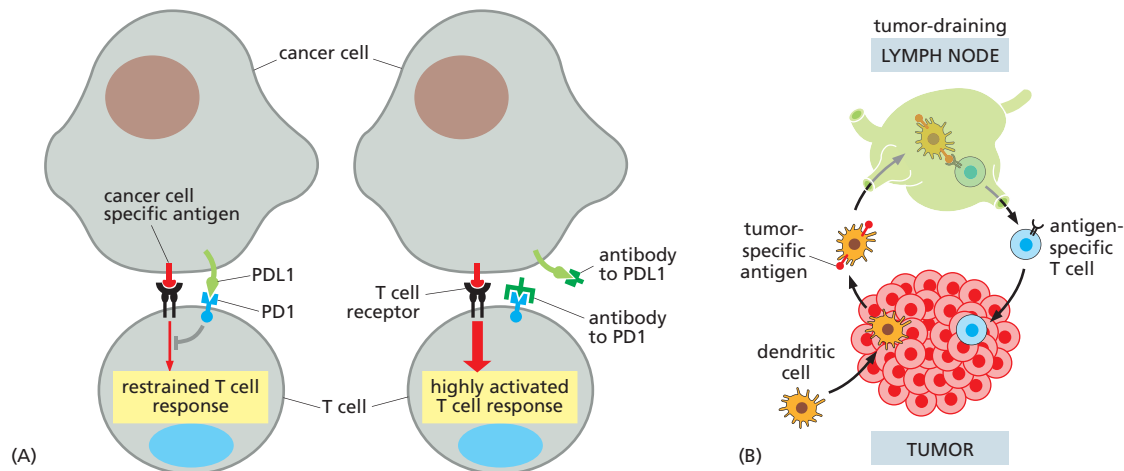


Figure 20-46 How the immune system interacts with cancer. (A) An antibody therapy for overcoming the immunosuppressive microenvironment in tumors. Cancer cells often protect themselves from immune attack by expressing proteins on their surface that bind to inhibitory receptors on T cells. In the example shown, the cancer cell expresses PDL1, which binds the PD1 receptor on a T cell and interferes with T cell activation. This makes the tumor susceptible to antibodies that unleash T cell attack. (B) Dendritic cells play a key role in activating T cells by acquiring tumor cell antigens from tumors and presenting them in lymph nodes, as illustrated, and some cancers have evolved mechanisms that interfere with this process (not shown).

checkpoint mechanisms can prevent an individual from killing the cancer cells that are threatening his or her survival.

To overcome the immunosuppressive environment and provide a strong T cell activating signal, a promising new type of anticancer therapy focuses on developing *immune checkpoint inhibitors* such as antibodies that prevent the tumor cells from engaging with the inhibitory receptors on T cells. As illustrated in [Figure 20-46A](#), targeting inhibitory receptors or their ligands with antibodies can unleash an immune attack on the cancer cells. Importantly, multiple neoantigens are recognized as foreign, so that cancer cells cannot escape an immune attack through the mutational loss of a single neoantigen, making it difficult for the tumor to become resistant to the antibody treatment. A substantial fraction of metastatic melanoma patients injected repeatedly with monoclonal antibodies that bind to either one such receptor, PD1 (programmed cell death 1), or its ligand on the surface of cancer cells, PDL1, respond in a dramatic way, with their cancer being driven into remission for years. Unfortunately, the treatment fails to help others with the same type of cancer.

An important clue as to why immunotherapies are effective in some patients but not others comes from the extensive analysis of cancer genomes described in the second part of this chapter. A fortuitous property of tumor samples is that they contain not only the cancer cells but also the cells of the associated stroma, including fibroblasts and endothelial and various immune cells. Because each cell type expresses a signature set of RNAs, the presence and proportion of each cell type in the tumor can be assessed, including the T cells. A comparison of the immune cell profiles of thousands of different tumors has revealed interesting correlations. For example, as one might predict, cancers with genomes that contain large numbers of point mutations are frequently found to possess a higher proportion of T cells compared to cancers with genomes that contain fewer mutations, presumably due to an increase in neoantigens. In turn, this tumor profile correlates with better treatment outcomes after treatment with immune checkpoint inhibitors.

Some cancers appear to avoid immune destruction by preventing T cells from infiltrating the tumor in the first place. The cellular basis of this phenomenon is poorly understood and is an important ongoing area of investigation. It is known that T cell activation requires that T cells physically interact with antigen-presenting cells called dendritic cells. Dendritic cells are highly migratory and pick up pathogens or their products at sites of infection and deliver them to lymph

nodes or other lymphoid organs, where they present them to T cells. In the case of cancer, dendritic cells present neoantigens to T cells, so that they can track down the cells expressing the antigens and destroy them ([Figure 20–46B](#)). Some cancer cells inhibit this process by preventing the dendritic cells from interacting with them and acquiring their neoantigens.

These observations illustrate how the immune system is subject to many levels of control and how cancer cells have been selected to take advantage of them. Importantly, research into the cell biological mechanisms behind these phenomena not only helps to develop new cancer treatments but will also teach us basic principles of how the system operates in the first place, as described in Chapter 24.

Cancers Evolve Resistance to Therapies

High hopes have to be tempered with sobering realities. We have seen that genetic instability can provide an Achilles' heel that cancer therapies can exploit, but at the same time it can make eradicating the disease more difficult by allowing the cancer cells to evolve resistance to therapeutic drugs, often at an alarming rate. This applies even to the drugs that target genetic instability itself. Thus, PARP inhibitors give valuable remission of illness, but in the long term the disease generally comes back. For example, *Brca*-deficient cancers can sometimes develop resistance to PARP inhibitors by undergoing a second mutation in an affected *Brca* gene that restores its function. By then, the cancer is already out of control and it may be too late to affect the course of the disease with additional treatments.

There are many different strategies by which cancers can evolve resistance to anticancer drugs. Often, a cancer will be dramatically reduced in size by an initial drug treatment, with all of the detectable tumor cells seeming to disappear. But months or years later the cancer will reappear in an altered form that is resistant to the drug that was at first so successful. In such cases, the initial drug treatment has evidently failed to destroy some tiny fraction of cells in the original tumor-cell population. These cells may have escaped death because they carry a protective mutation or epigenetic change or perhaps simply because they were lurking in a protected environment. Alternatively, and most insidiously, a phenotypically distinct subpopulation of tumor cells may function as cancer stem cells. The surviving cells eventually regenerate the cancer by continuing to proliferate, mutating and evolving still further as they do so. Combination therapies, in which the right two agents are used simultaneously to target the same cancer cells, should in principle help greatly with such problems.

However, in some cases, cells that are exposed to one anticancer drug evolve a resistance not only to that drug but also to other drugs to which they have never been exposed. This phenomenon of **multidrug resistance** frequently correlates with amplification of a part of the genome that contains a gene called *Mdr1* or *Abcb1*. This gene encodes a plasma membrane-bound transport ATPase of the ABC transporter superfamily (discussed in Chapter 11), which pumps lipophilic drugs out of the cell (see Movie 11.5). The overproduction of this protein (or some of its other family members) by a cancer cell can prevent the intracellular accumulation of many cytotoxic drugs, making the cell insensitive to them.

In the to-and-fro struggle between advanced metastatic cancer and the therapist, as current practice stands, the cancer usually wins in the end. Does it have to be so? As we discuss below, there is reason to think that by attacking a cancer with many weapons at once—instead of using them one after another, each until it fails—it may be possible to do much better.

We Now Have the Tools to Devise Combination Therapies Tailored to the Individual

Nowadays, cancers caught at an early stage can often be cured by surgery, radiation, or drugs. For most cancers that have progressed and metastasized widely, however, cure is still frequently beyond us. Treatments such as those described

earlier can give valuable remissions, but sooner or later these are typically followed by relapse.

Nevertheless, for some forms of advanced cancer, curative therapies are being developed that utilize multiple strategies; for example, combining chemotherapy and radiation therapy with immunotherapy. Ideally, the choice of drugs to be given in combination should be tailored to the individual patient. Cancers evolve by a fundamentally random process, and each individual is different, but modern methods of genome analysis now let us characterize the cells from a tumor biopsy in exhaustive detail so as to discover which cancer-critical genes are affected in a particular case. Admittedly, this is not straightforward: the tumor cells in an individual are heterogeneous and do not all contain the same genetic lesions. With increased understanding of the pathways of cancer evolution, however, and with the experience gained from many different cases, it should become possible to make informed decisions about the optimal therapies to use.

Improvements in tissue sampling, genomics, and biostatistics have enabled the direct characterization of primary human tumors. However, these analyses do not take into account the contributions of the entire body. Advances in cancer therapy are also coming from the development of more sophisticated models of the disease, which can be used to investigate the factors involved in tumor growth and metastasis, as well as to examine response to therapy. In one approach, mice can be genetically engineered to introduce particular combinations of human mutations (see Figure 20–33). A second approach is the human tumor xenograft. In this model, human cancer cells are transplanted—either under the skin or into the organ type in which the cancer originated—into immunocompromised mice that do not reject human cells. In an approach that bypasses the use of mice, organoids can be grown in 3D culture from patient-derived healthy and tumor tissues (see Figure 20–34A). These systems facilitate patient-specific drug testing and the development of individualized treatment regimens.

From the perspective of the patient, the pace of advance in cancer research can seem frustratingly slow. Each new drug has to be tested in the clinic, first for safety and then for efficacy, before it can be released for general use. And if the drug is to be used in combination with others, the combination therapy must then go through the same long process. Strict ethical rules constrain the conduct of trials, which means that they take time—typically several years. But slow and cautious steps, taken systematically in the right direction, can lead to great advances. There is still far to go, but the examples that we have discussed provide proof of principle and grounds for optimism.

From the cancer research effort, we have learned a great deal of what we know about the molecular biology of the normal cell. Now, more and more, we are discovering how to put that knowledge to use in the battle with cancer itself.

Summary

Our growing understanding of the cell biology of cancers has already begun to lead to better ways of preventing, diagnosing, and treating these diseases. Anticancer therapies can be designed to destroy cancer cells preferentially by exploiting the properties that distinguish cancer cells from normal cells, including the cancer cells' dependence on oncogenic proteins and the defects they harbor in their DNA repair mechanisms. We now have good evidence that, by increasing our understanding of normal cell control mechanisms and exactly how they are subverted in specific cancers, we can eventually devise drugs to kill cancers precisely by attacking specific molecules critical for the proliferation and survival of the cancer cells. In addition, great progress has recently been made through sophisticated immunological approaches to cancer therapy. And, as we become better able to determine which genes are altered in the cells of any given tumor, we can begin to tailor treatments more accurately to each individual.

PROBLEMS

Which statements are true? Explain why or why not.

20-1 Cancer therapies directed solely at killing the rapidly dividing cells that make up the bulk of a tumor are unlikely to eliminate the cancer from many patients.

20-2 In the regulatory pathways that control cell growth and proliferation, the products of oncogenes are stimulatory components and the products of tumor suppressor genes are inhibitory components.

20-3 The chemical carcinogen dimethylbenz[*a*]anthracene (DMBA) must be an extraordinarily specific mutagen because 90% of the skin tumors it causes have an A-to-T alteration at exactly the same site in the mutant *Ras* gene.

20-4 The main environmental causes of cancer are the products of our highly industrialized way of life such as pollution and food additives.

Discuss the following problems.

20-5 What is the term for a cancer arising from epithelial cells?

20-6 In contrast to colon cancer, the incidence of which increases dramatically with age, the incidence of osteosarcoma—a tumor that occurs most commonly in the long bones—peaks during adolescence. Osteosarcomas are relatively rare in young children (up to age 9) and in adults (over age 20). Why do you suppose that the incidence of osteosarcoma does not show the same sort of age dependence as colon cancer?

20-7 Mortality due to lung cancer was followed in groups of males in the United Kingdom for 50 years. Figure Q20-1 shows the cumulative risk of dying from lung cancer as a function of age and smoking habits for four groups of males: those who never smoked, those who stopped at age 30, those who stopped at age 50, and those who continued to smoke. These data show clearly that individuals can substantially reduce their cumulative risk of dying from lung cancer by stopping smoking. What do you suppose is the biological basis for this observation?

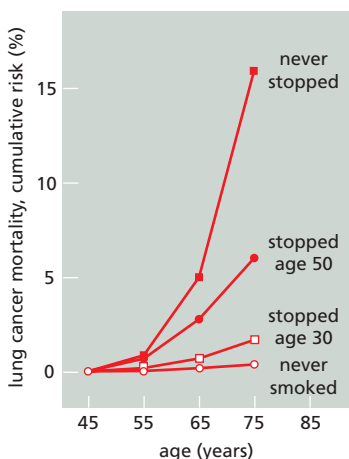


Figure Q20-1 Cumulative risk of lung cancer mortality for nonsmokers, smokers, and former smokers (Problem 20-7). Cumulative risk is the running total of deaths, as a percentage, for each group. Thus, for continuing smokers, 1% died of lung cancer between ages 45 and 55; an additional 4% died between 55 and 65 (giving a cumulative risk of 5%); and 11% more died between 65 and 75 (for a cumulative risk of 16%).

20-8 A small fraction—2 to 3%—of all cancers, across many subtypes, displays a quite remarkable phenomenon: tens to hundreds of rearrangements that primarily involve a single chromosome or chromosomal region. The breakpoints can be tightly clustered, with several in a few kilobases, and the junctions of the rearrangements often involve segments of DNA that were not originally close together on the chromosome. The copy number of various segments within the rearranged chromosome is found to be 0, indicating deletion, or 1, indicating retention.

You can imagine two ways in which such multiple, localized rearrangements might happen: a progressive rearrangements model with ongoing inversions, deletions, and duplications involving a localized area, or a catastrophic model in which the chromosome is shattered into fragments that are stitched back together in random order by a double-strand break repair process such as nonhomologous end joining (Figure Q20-2).

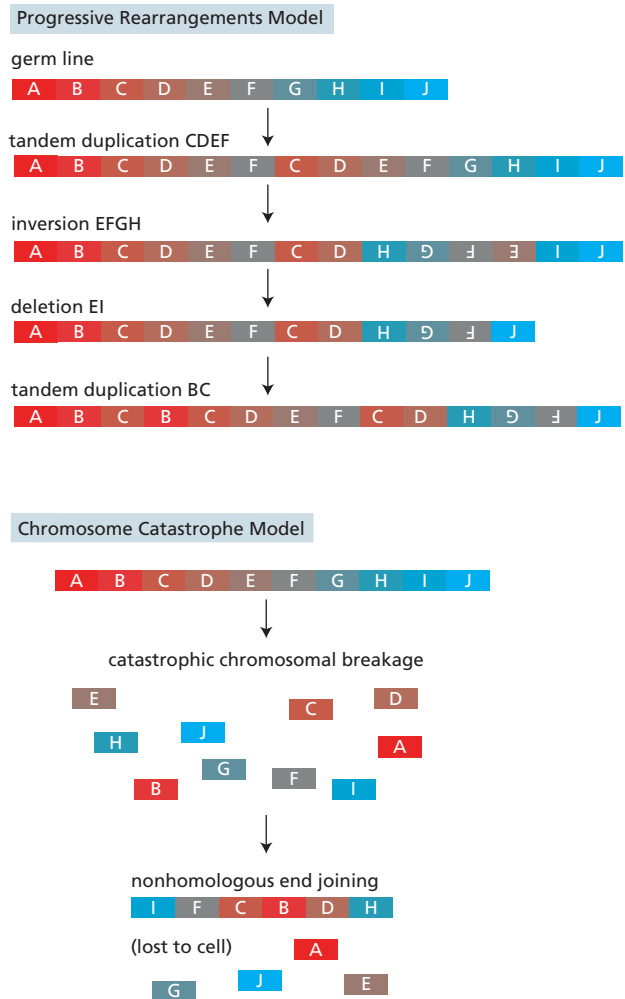


Figure Q20-2 Two models to explain the multiple, localized chromosome rearrangements found in some cancers (Problem 20-8). The progressive rearrangements model shows a sequence of rearrangements that disrupts the chromosome, generating increasingly complex chromosomal configurations. The chromosome catastrophe model shows the chromosome being fragmented and then reassembled randomly, with some pieces left out.

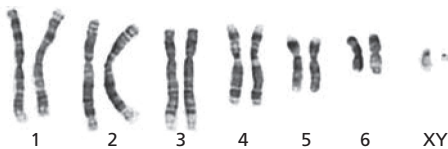
Which of the two models in Figure Q20-2 accounts more readily for the features of these highly rearranged chromosomes? Explain your reasoning.

20-9 The Tasmanian devil (**Figure Q20-3A**) is a carnivorous Australian marsupial that is threatened with extinction by the spread of a fatal disease in which a malignant oral-facial tumor interferes with the animal's ability to feed. You have been called in to analyze the source of this unusual cancer. It seems clear to you that the cancer is somehow spread from animal to animal, very likely by their frequent fighting, which is accompanied by biting around the face and mouth. To uncover the source of the cancer, you isolate tumors from 11 Tasmanian devils captured in widely separated regions and examine them. The karyotypes of nontumor cells from these Tasmanian devils are normal (**Figure Q20-3B**). As expected, the tumor cells are highly rearranged relative to the normal karyotype, but, surprisingly, the karyotypes from all 11 tumor samples are very similar (**Figure Q20-3C**). Moreover, one of the Tasmanian devils has an inversion on chromosome 5 that is not present in its facial tumor. How do you suppose this cancer is transmitted from animal to animal? Is it likely to arise as a consequence of an infection by a virus or microorganism? Explain your reasoning.

(A) Tasmanian devil (*Sarcophilus harrisii*)



(B) normal karyotype



(C) tumor karyotype

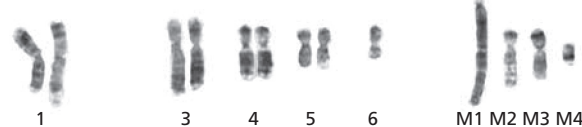


Figure Q20-3 Karyotypes of cells from Tasmanian devils (Problem 20-9). (A) A Tasmanian devil. (B) Normal karyotype for a male Tasmanian devil. The karyotype has 14 chromosomes, including XY. (C) Karyotype of cancer cells found in each of the 11 facial tumors studied. The karyotype has 13 chromosomes, no sex chromosomes, no chromosome 2 pair, one chromosome 6, two chromosomes 1 with deleted long arms, and four highly rearranged marker chromosomes (M1–M4). (A, reproduced courtesy of Museum Victoria; B and C, from A.-M. Pearse and K. Swift, *Nature* 439:549, 2006. Reproduced with permission from SNCSC.)

20-10 Virtually all cancer treatments are designed to kill cancer cells. However, one particular cancer—acute promyelocytic leukemia (APL), which is caused by too many immature blood-forming cells (promyelocytes) in the blood—has been successfully treated with all-*trans*-retinoic acid, which causes the promyelocytes to differentiate into neutrophils. How might a change in the state of differentiation of APL cancer cells help the patient?

20-11 PolyADP-ribose polymerase (PARP) plays a key role in the repair of DNA single-strand breaks. In the presence of the PARP inhibitor olaparib, single-strand breaks accumulate. When a replication fork encounters a single-strand break, it converts it to a double-strand break, which in normal cells is then repaired by homologous recombination. In cells defective for homologous recombination, however, inhibition of PARP triggers cell death.

Individuals who have only one functional copy of the *Brca1* gene, which is required for homologous recombination, are at much higher risk for cancer of the breast and ovary. Cancers that arise in these tissues in these individuals can be treated successfully with olaparib. Explain how it is that treatment with olaparib kills the cancer cells in these individuals but does not harm their normal cells.

20-12 One major goal of modern cancer therapy is to identify small molecules—anticancer drugs—that can be used to inhibit the products of specific cancer-critical genes. If you were searching for such molecules, would you design inhibitors for the products of oncogenes or the products of tumor suppressor genes? Explain why you would (or would not) select each type of gene.

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